Genome-Scale Context, Stoichiometric Refinement, and Neuro-Symbolic Reasoning for Proteome Function Annotation

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Abstract

Per-protein function predictors evaluate sequences in isolation from genomic context, producing fragmented metabolic pathways, incomplete heteromeric complexes, and cross-clade Gene Ontology (GO) assignments, while negative-cost mixed-integer linear programming (MILP) gap-fillers add hundreds of zero-flux reactions that lack gene-locus attribution. We present Genome-Scale Protein Annotation (GSPA v2.1.0), a four-stage framework coupling multi-channel probability calibration (Stage 0), whole-proteome Bayesian log-odds priors (Stage 1), semantic ancestor relaxation of taxonomic and Uberon/PO/CL anatomical constraints (Stage 2A), positive-cost parsimonious flux balance analysis (pFBA) with locus responsibility against an unassigned-gene hypothesis (Stage 2B), and selective guarded language-model disambiguation on $< 2.5\%$ of proteins (Stage 3). Across eight benchmark suites, Stage 0 establishes bulk protein-centric accuracy while Stages 1–3 eliminate taxonomic/anatomical violations (1,009 endogenous and 1,452 stress-test eukaryotic violations), improve eukaryotic $F_{\max}$ (+0.0021 endogenous CAFA $F_{\max}$ mean; +0.0006 to +0.0157 transfer-evaluated micro $F_{\max}$ over naive deletion across multicellular animals and vascular plants under unmodified Swiss-Prot ground truth), retain strictly pFBA-active reaction additions (5.70 pFBA-active additions/genome and 0.00 zero-flux additions by construction, of which 2.29 are certified FVA-essential at $\leq 1.01\,C^*$ and 1.96 are certified knockout-essential at $\leq 1.10\,C^*$, vs. 262.34 [83.72%] uncertified zero-flux additions and 51.01 flux-carrying additions/genome out of 313.35 net added reactions under negative-cost MILP) across 108 genomes, improve `Price-149` enzyme Top-1 hits under both single-predictor refinement (CLEAN 60 → 63, MAPred 50 → 56, DeepProZyme 49 → 51 with stoichiometric local-context probe, GraphEC 18 → 20) and multi-predictor refinement (+0.0528 to +0.1987 $F_{\max}$), match gapseq binary `BacDive` accuracy while lifting blind carbon ROC-AUC to 0.8567, Track 2 phenotype-anchored F_1 to 0.8871 (carbon) and 0.8545 (enzyme), and genus-blocked hybrid F_1 to 0.8710 (carbon) and 0.8713 (enzyme), and lift ambiguous leave-reaction-out Hit@1 from 34.0% to 82.7% (3-seed mean).

1 Introduction

Automated annotation of protein function from genome assemblies is a central problem in computational biology, underpinning comparative genomics, genome-scale metabolic modeling, and systems biology[1,2]. Given the protein-coding sequences $\mathcal{P} = \{p_1, \ldots, p_N\}$ of an organism, their chromosomal coordinates and operon organization $\mathcal{K}$, the organism's taxonomic lineage τ_{tax}, and reference biochemical and ontological knowledge bases, genome-scale annotation seeks a joint assignment of Gene Ontology (GO)[3] terms and Enzyme Commission (EC) numbers across $\mathcal{P}$ that is both accurate at the level of individual proteins and globally coherent at the level of cellular physiology. A biologically valid proteome annotation satisfies five structural invariants: (1) core cellular processes required for viability in the organism's domain of life (such as translation, tRNA aminoacylation, and DNA replication initiation) are represented; (2) metabolic pathways active in the organism form stoichiometrically connected reaction routes capable of supporting biomass synthesis[4,5]; (3) obligate heteromeric protein complexes (such as ATP synthase, NADH dehydrogenase, urease, and photosystem reaction centers[6,7]) are annotated as complete multi-subunit assemblies; (4) assigned GO terms satisfy formal taxonomic constraints (`only_in_taxon` and `never_in_taxon`)[8]; and (5) developmental and physiological biological processes, such as *heart development* (`GO:0007507`), *flower development* (`GO:0009908`), *neural crest cell migration* (`GO:0001755`), and *vascular transport* (`GO:0010232`), entail only anatomical structures present in the target clade[9,10].

Existing computational methods address individual components of this problem through five methodological classes. First, production genome annotation pipelines, including the NCBI Prokaryotic Genome Annotation Pipeline (PGAP)[11,12], Bakta[13], Prokka[14], eggNOG-mapper[15], Pharokka[16], and InterProScan[17,18], combine sequence alignment[19], profile hidden Markov models (HMMs)[20], and curated orthology rules to assign

high-specificity labels protein by protein, leaving divergent orthologs below fixed sequence thresholds unannotated. Second, deep learning and protein language model (PLM) predictors[21–23] evaluated in the Critical Assessment of Functional Annotation (CAFA) challenges[24–26], including DeepGOPlus[27], DeepFRI[28], metagenomic-deepFRI[29], DeepGOZero[30], DeepGO-SE[31], ProteInfer[32], SPROF-GO[33], CLEAN[34], DeepProZyme[35], EnzBERT[36], MAPred[37], GraphEC[38], and TopEC[39], predict calibrated GO or EC probabilities from individual sequences or structures[40]. Because both paradigms score each protein $p_i \in \mathcal{P}$ conditionally independently of the remaining proteome $\mathcal{P} \setminus \{p_i\}$, twilight-zone enzymes ($< 30\%$ sequence identity) leave metabolic pathway holes, promiscuous domain hits trigger isolated singleton subunit calls on heteromeric complexes, and distant homology transfers propagate clade-inappropriate taxonomic and anatomical GO terms. Third, comparative genomics and genomic language models capture gene neighborhood conservation, co-directional operon structure, and phylogenetic profiles[41–50], while metabolic reconstruction tools such as Pathway Tools[51,52], ModelSEED[53,54], ProbAnno[55], CarveMe[56], gapseq[57], GapMind[58,59], Reconstructor[60], and CanOE[61] use stoichiometric connectivity to fill reaction gaps[62]. However, reaction-level MILP gap-fillers such as CarveMe[56] and our prior reaction-indicator MILP formulation METEOR[63] optimize binary reaction indicators $y_j \in \{0, 1\}$ with a uniform reaction-level score, assign negative objective costs ($c_j^{\text{add}} < 0$) to high-scoring candidate reactions (which activates hundreds of zero-flux reaction additions), and apply uniform reaction-to-protein probability boosts across all candidate genes of a reaction[63]. Fourth, bio-ontology engineering provides formal taxon constraints[8] and cross-product logical definitions linking GO biological processes to the Uberon multi-species anatomy ontology[9,10,64,65], the Plant Ontology (PO)[66], and the Cell Ontology (CL)[67], while existing post-filters delete violating terms together with their valid ancestral biological process annotations. Fifth, tool-using biological language agents such as GeneAgent[68], ProtAgents[69], SPIRES[70], GORetriever[71], and ProtST[72] synthesize multi-modal genomic evidence when invoked selectively on ambiguous loci and paired with formal symbolic verification.

Here we present Genome-Scale Protein Annotation (GSPA v2.1.0), a unified four-stage framework that resolves each of these genome-scale failure modes (Figure 1). Stage 0 calibrates and pools per-protein evidence channels via correlation-grouped Noisy-OR (or ingests any single standalone GO or EC predictor) to establish calibrated baseline likelihoods. Stage 1 iteratively refines annotation log-odds across the whole proteome using six genome-scale priors: core essentiality, pathway and heteromeric complex coherence, SAT-based taxonomic consistency, GO–Uberon/PO/CL anatomical constraints, metabolic gap-filling with reaction-local genomic context (RLGC), and operon neighborhood coupling. Stage 2 combines a deterministic symbolic enforcer (Stage 2A) that resolves taxonomic and anatomical contradictions by relaxing forbidden terms to their nearest valid non-root GO ancestors and pruning isolated complex singletons, with a flux-certified stoichiometric locus refiner (Stage 2B) that couples role-aware disjunctive normal form (DNF) mechanism aggregation, positive-cost lexicographic parsimonious FBA (pFBA)[73], and backward locus responsibility against an explicit unassigned-gene hypothesis $h_\varnothing$. Stage 3 provides an optional, selectively triggered neuro-symbolic agent using an open-weight instruction language model (`Qwen/Qwen3.8-27B-FP8`, served as `qwen3.8-27b`) that inspects the $< 2.5\%$ of proteins with unresolved pathway gaps, incomplete complexes, or borderline conflicts under a deterministic symbolic ontology guardrail.

We evaluate GSPA across eight benchmark suites spanning bacteria, archaea, viruses, and multicellular eukaryotes (Tables 1–8; Supplementary Sections S1–S5 and Supplementary Tables S1–S8). Across these benchmarks, GSPA eliminates 100% of taxonomic and Uberon/PO/CL anatomical violations (1,009 endogenous and 1,452 stress-test eukaryotic violations) while outperforming naive deletion in transfer-evaluated micro $F_{\max}$ across multicellular animals and vascular plants (+0.0006 to +0.0157), retains strictly pFBA-active reaction additions (5.70 pFBA-active additions/genome with 0.00 zero-flux padding by construction, of which 2.29 are certified FVA-essential at $\leq 1.01\,C^*$ and 1.96 are certified knockout-essential at $\leq 1.10\,C^*$, vs. 262.34 [83.72%] uncertified zero-flux additions and 51.01 flux-carrying additions/genome under negative-cost MILP) across 108 bacterial metabolic reconstructions, improves `Price-149` RB-TnSeq enzyme Top-1 hits under both single-predictor refinement (CLEAN 60 → 63, MAPred 50 → 56, DeepProZyme 49 → 51, GraphEC 18 → 20) and multi-predictor refinement (+0.0528 to +0.1987 $F_{\max}$), lifts experimental `BacDive` phenotype prediction to 0.8567 blind carbon ROC-AUC and 0.8710–0.9225 hybrid F_1, and increases ambiguous leave-reaction-out Hit@1 from 34.0% to 82.7% (3-seed mean). The GSPA v2.1.0 software, precompiled cross-ontology constraint tables, and benchmark manifests are openly available at `https://github.com/bio-ontology-research-group/gspa`.

2 Results

2.1 Multi-evidence integration and genome-scale prior ablation (Suites 1 and 2)

We first evaluated how Stage 0 multi-channel pooling and Stages 1–2 genome-scale priors affect protein-centric accuracy and structural genome coherence across nine bacterial and archaeal reference genomes (*Escherichia coli* K-12, *E. coli* O157:H7, *Bacillus subtilis* 168, *Pseudomonas aeruginosa* PAO1, *Mycobacterium tuberculosis*

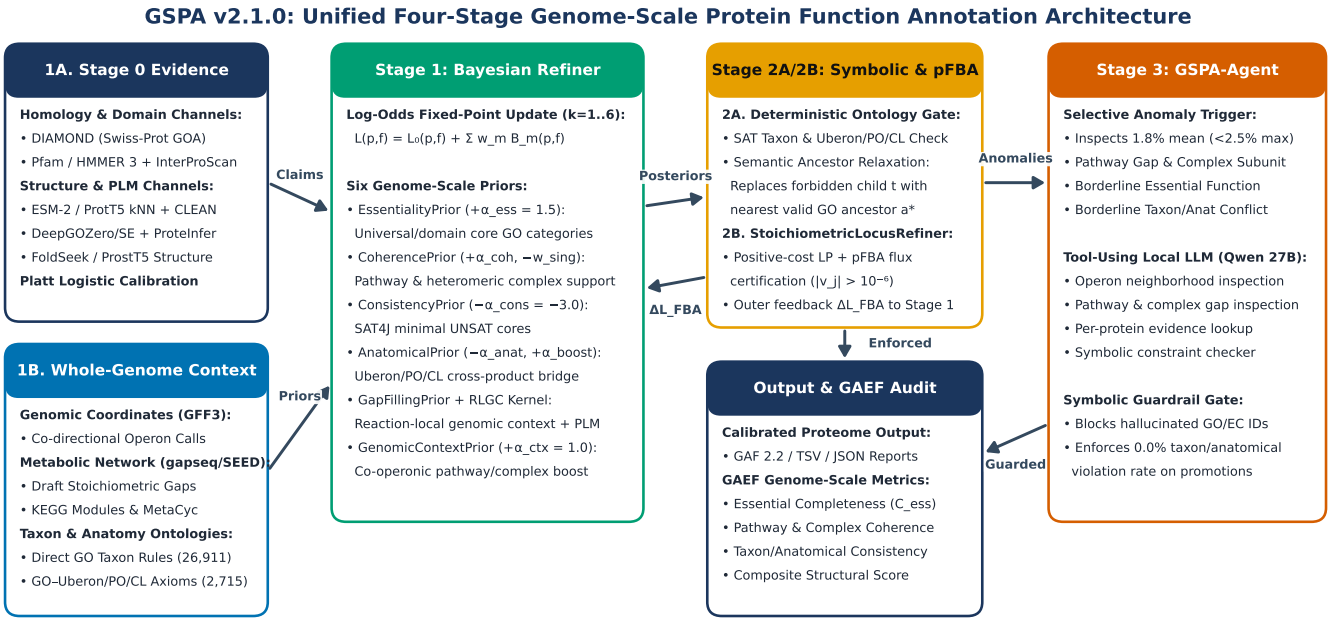


Figure 1. Unified four-stage neuro-symbolic architecture of GSPA v2.1.0. (**Stage 0**) Ingests calibrated per-protein evidence channels (sequence homology, domain HMMs, structure similarity, and protein language model embeddings) or any single standalone predictor alongside whole-genome context (GFF3 operon coordinates, draft metabolic network gaps, 26,911 direct GO taxon constraint rules [13,819 `only_in_taxon` and 13,092 `never_in_taxon`] across 18,159 GO terms, and 2,715 GO–Uberon/PO/CL anatomical axioms across 2,674 GO terms and 1,294 anatomical classes across 30 object properties linked to 8,331 taxon constraints [4,767 `only_in_taxon` and 3,564 `never_in_taxon`] across 1,247 anatomical classes). (**Stage 1**) The iterative Bayesian refiner updates log-odds posteriors across all protein–term pairs via six genome-scale priors and reaction-local genomic context (RLGC). (**Stage 2**) The deterministic symbolic enforcer (Stage 2A) resolves surviving taxonomic and anatomical contradictions through semantic ancestor relaxation and prunes isolated heteromeric complex singletons, while the stoichiometric locus refiner (Stage 2B) performs positive-cost lexicographic pFBA repair and backward locus attribution. (**Stage 3**) An optional selective agentic layer inspects the $< 2.5\%$ of loci with unresolved pathway gaps, incomplete complexes, or borderline conflicts using deterministic genomic tools and a symbolic ontology guardrail.

Table 1. Systematic ablation of GSPA Stage 1 priors and Stage 2 symbolic enforcement across nine reference genomes (Suite 2, 81 runs; source: `results/pillar1_2_ablation_summary.tsv` and `results/pillar1_2_perturbation_summary.tsv`). Values report means (± s.d. in text) across *E. coli* K-12 (hyperparameter calibration genome), *E. coli* O157:H7, *B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *M. genitalium* G37, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661; the bottom row reports the 8-genome out-of-sample mean excluding *E. coli* K-12. Micro $F_{\max}$ and CAFA $F_{\max}$ report full GOA (`All`) and experimental-only (`Exp`) ground-truth evaluations. C_{cons} (Viol.) reports mean consistency score $\max(0, 1 - 0.01V)$ and total baseline violations across the panel; Perturb. Supp. reports suppression of the 125 post-calibration active violations ($122/125 = 97.6\%$), with suppression across all 900 raw injected claims ($897/900 = 99.7\%$) in parentheses.

Config	Active Priors / Enforcement	Micro $F_{\max}$ All (Exp)	CAFA $F_{\max}$ All (Exp)	C_{ess}	C_{path}	C_{cplx}	C_{cons} (Viol.)	GAEF Score	Perturb. Supp.
`A1_noisy_or`	Stage 0 Noisy-OR baseline	0.7631 (0.2853)	0.8041 (0.2363)	0.9400	0.4975	0.8636	0.9722 (25)	0.8459	0.0% (86.1%)
`A2_essentiality`	+ Essentiality prior ($\alpha_1 = 1.5$)	0.7632 (0.2853)	0.8040 (0.2363)	0.9444	0.4975	0.8636	0.9722 (25)	0.8472	0.0% (86.1%)
`A3_coherence`	+ Coherence prior ($\alpha_2 = 1.0$)	0.7629 (0.2824)	0.8025 (0.2354)	0.9400	0.5052	0.8636	0.9722 (25)	0.8474	0.0% (86.1%)
`A4_gap_filling`	+ Gap-filling prior ($\alpha_5 = 1.5$)	0.7631 (0.2853)	0.8038 (0.2361)	0.9400	0.4977	0.8636	0.9722 (25)	0.8459	0.0% (86.1%)
`A5_genomic_ctx`	+ Operon context prior ($\alpha_6 = 1.0$)	0.7630 (0.2851)	0.8039 (0.2363)	0.9400	0.4992	0.8636	0.9722 (25)	0.8462	0.0% (86.1%)
`A6_taxon_anat`	+ Consistency + Anatomical priors	**0.7681 (0.2894)**	**0.8071 (0.2389)**	0.9400	0.4973	0.8636	**1.0000 (0)**	0.8542	**97.6% (99.7%)**
`A7_stage1_all`	All six Stage 1 priors combined	0.7660 (0.2847)	0.8037 (0.2366)	**0.9479**	**0.5146**	**0.8852**	**1.0000 (0)**	**0.8643**	97.6% (99.7%)
`A8_stage2_rem`	Stage 1 + Stage 2A (remove)	0.7660 (0.2847)	0.8037 (0.2366)	**0.9479**	**0.5146**	**0.8852**	**1.0000 (0)**	**0.8643**	**97.6% (99.7%)**
`A9_stage2_rel`	Stage 1 + Stage 2A (relax)	0.7660 (0.2847)	0.8037 (0.2366)	**0.9479**	**0.5146**	**0.8852**	**1.0000 (0)**	**0.8643**	**97.6% (99.7%)**
8-Genome Non-*E. coli* Mean: `A1_noisy_or` → `A9_stage2_rel`		0.7718 → 0.7749	0.8143 → 0.8140	0.9325 → 0.9414	0.4834 → 0.5012	0.8743 → 0.8865	0.9738 (21) → 1.0000 (0)	0.8434 → 0.8599	0.0% → 97.6%

H37Rv, *Helicobacter pylori* 26695, *Mycoplasma genitalium* G37, *Synechocystis* sp. PCC 6803, and *Methanocaldococcus jannaschii* DSM 2661; 81 runs; Figure 2 and Table 1; complete inventory of the 18 configurable pipeline hyperparameters in Supplementary Section S1 and Supplementary Table S1). For high-throughput multi-genome benchmarks (the 9-genome, 13-genome, 88-genome, and 1,000-genome panels), Stage 0 uses the 3 fast sequence/domain channels (DIAMOND, InterProScan, and Pfam HMMER; 5.4–12.7 s/genome), whereas on the nested 21-genome neural panel (which contains all 13 panel genomes plus 8 additional reference proteomes,

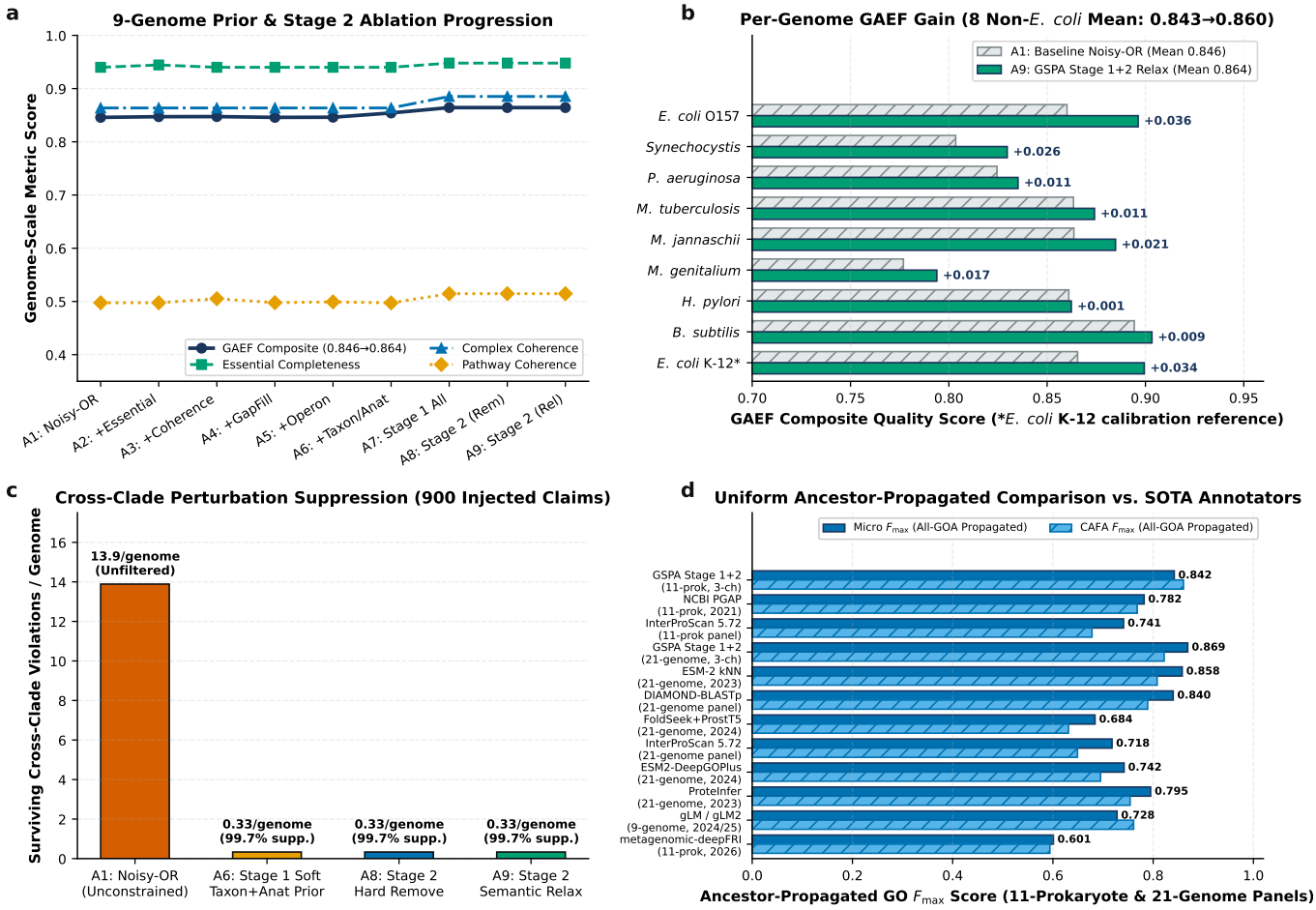


Figure 2. Genome-scale prior ablation, cross-clade perturbation suppression, and uniform ancestor-propagated comparison against external annotators and internal Stage 0 channels (Suites 1 and 2). (**a**) Progression of the internal GAEF structural audit score, Essential Completeness (C_{ess}), Complex Coherence (C_{cplx}), and Pathway Coherence (C_{path}) across nine ablation configurations (`A1_noisy_or` through `A9_stage2_relax`) averaged over nine bacterial and archaeal reference genomes. (**b**) Per-genome GAEF structural score gain from baseline Noisy-OR (`A1`) to full GSPA Stage 1 + Stage 2 (`A9`; $*$ marks the *E. coli* K-12 hyperparameter calibration reference). (**c**) Surviving cross-clade false-positive annotations per genome when 900 eukaryotic developmental/anatomical and clade-specific GO claims (100 per genome, raw score 0.55–0.85) are injected into the nine prokaryotic proteomes. (**d**) Ancestor-propagated pair-level micro $F_{\max}$ and protein-centric CAFA $F_{\max}$ comparing GSPA against metagenomic-deepFRI, InterProScan 5.72, and NCBI PGAP on the 11-prokaryote reference panel and against individual Stage 0 channels and gLM/gLM2 on the 21-genome panel.

and is itself nested within the 88-genome four-kingdom cohort) we evaluate both the 3-channel core and all 7 channels including ESM-2/ProtT5 kNN, ProteInfer, and FoldSeek+ProstT5 3Di. Because roughly 85% of annotated proteins in the 9 reference organisms carry high-confidence Swiss-Prot homology ($P_0 \geq 0.90$), Stage 0 Noisy-OR pooling (`A1_noisy_or`) establishes the bulk protein-centric baseline: micro $F_{\max} = 0.7631 \pm 0.0817$ (sample s.d., ddof = 1, across 9 genomes; 0.2853 ± 0.0853 experimental-only) and CAFA $F_{\max} = 0.8041 \pm 0.0650$ (0.2363 ± 0.0947 experimental-only). Across the full Stage 1+2 pipeline (`A9_stage2_rel`), pair-level micro $F_{\max}$ increases to 0.7660 ± 0.0791 (0.2847 ± 0.0834 experimental-only; two-sided exact paired Wilcoxon $p = 0.0391$) while preserving protein-centric CAFA $F_{\max}$ at 0.8037 ± 0.0647 (0.2366 ± 0.0951 experimental-only; two-sided exact paired Wilcoxon $p = 0.1289$, within 0.0001–0.0016 of Stage 0 across 8 of 9 genomes; Table 1). Evaluating the 8 out-of-sample genomes (with *E. coli* K-12 held out as the calibration reference) confirms the same pattern: micro $F_{\max}$ rises from 0.7718 to 0.7749, CAFA $F_{\max}$ is 0.8143 vs. 0.8140 (0.2140 → 0.2143 experimental-only), C_{path} rises from 0.4834 to 0.5012, C_{cplx} rises from 0.8743 to 0.8865, and the internal GAEF structural score rises from 0.8434 to 0.8599.

Examining individual priors on the 9-genome panel (Table 1; per-genome breakdown in `results/pillar1_2_ablation_per_genome.tsv`), the essentiality prior (`A2`) lifts mean core functional completeness C_{ess} from 0.9400 ± 0.0785 to 0.9444 (and to

0.9479 ± 0.0680 in combined Stage 1+2 `A7`–`A9`, improving by +0.0400 on *M. jannaschii* [0.9600 → 1.0000] and +0.0312 on *M. genitalium* [0.7500 → 0.7812] across the 9 genomes, exact paired Wilcoxon $p = 0.50$). The coherence prior (`A3`) increases mean pathway coherence C_{path} from 0.4975 ± 0.1089 to 0.5052, and the gap-filling (`A4`) and operon neighborhood (`A5`) priors raise C_{path} to 0.4977 and 0.4992 individually before combining with pathway/complex coherence in the full Stage 1 model (`A7`) to lift C_{path} to 0.5146 ± 0.1078 (exact two-sided paired Wilcoxon $p = 0.00781$) and complex coherence C_{cplx} from 0.8636 ± 0.1346 to 0.8852 ± 0.1189 (improving by +0.0972 in *E. coli* K-12 [0.7778 → 0.8750] and +0.0972 in *E. coli* O157:H7 [0.7778 → 0.8750] across the 9 genomes, exact paired Wilcoxon $p = 0.50$). Adding the consistency and anatomical priors (`A6`; constraint tables summarized in Supplementary Section S2 and Supplementary Table S2) eliminates all 25 baseline taxonomic violations across all nine genomes (*Synechocystis*: 8, *E. coli* O157:H7: 5, *E. coli* K-12: 4, *M. tuberculosis*: 3, and 1 each in *B. subtilis*, *H. pylori*, *M. genitalium*, *M. jannaschii*, and *P. aeruginosa*; C_{cons} from 0.9722 to 1.0000, exact two-sided paired Wilcoxon $p = 0.00391$) and increases micro $F_{\max}$ from 0.7631 to 0.7681 and CAFA $F_{\max}$ from 0.8041 to 0.8071. Because the 25 prokaryotic violations involve cellular component terms whose nearest valid ancestor is the ontology root, `A8` (removal) and `A9` (relaxation) yield identical scores on the prokaryotic panel (internal GAEF composite score rising from 0.8459 ± 0.0367 to 0.8643 ± 0.0376, exact two-sided paired Wilcoxon $p = 0.00391$; Figure 2a,b), whereas semantic relaxation differentiates multicellular eukaryotic organ and developmental terms (Section 2.6).

To test robustness against noisy cross-clade homology transfer, we injected 900 cross-clade false-positive claims across the nine prokaryotic genomes (100 per genome: 50 eukaryotic anatomical terms such as `GO:0007507` *heart development*, `GO:0009908` *flower development*, and `GO:0001755` *neural crest cell migration*, plus 50 clade-restricted terms) at raw scores in $[0.55, 0.85]$ (Figure 2c). After Stage 0 Platt logistic calibration, 125 of the 900 injected claims (13.9 per genome; 86.1% attenuated below 0.50 by calibration alone) exceed the decision threshold $\theta_{\text{eval}} = 0.50$ under unconstrained Noisy-OR (`A1`). Activating GSPA's taxonomic and anatomical constraints (`A6`, `A8`, and `A9`) suppresses 122 of the 125 active violations (97.6% suppression of post-calibration violations, or $897/900 = 99.7\%$ of all raw injected claims, with 100.0% [55/55] elimination of direct taxon violations and 67/70 [95.7%] suppression of cross-product anatomical claims).

We next compared GSPA against external genome annotation pipelines and internal Stage 0 channels under uniform ancestor-propagated GO scoring across Suites 1 and 2 (Figure 2d; `results/pillar1_suite2_panel13_21_sota.tsv`, transcribed summary table from the remote multi-tool benchmark runs reporting both ancestor-propagated and strict leaf/holdout metrics for every method). On the 13-genome panel (11 prokaryotes plus *S. cerevisiae* and *C. elegans*), 3-channel GSPA achieves an ancestor-propagated 13-genome mean micro $F_{\max}$ of 0.8180 ± 0.086 (0.8060 CAFA $F_{\max}$) and an 11-prokaryote mean micro $F_{\max}$ of 0.8421 (0.8605 CAFA $F_{\max}$, $S_{\min} = 21.40$), outperforming standalone InterProScan 5.72[17] (0.7220 micro / 0.6530 CAFA $F_{\max}$ across 13 genomes; 0.7410/0.6780 on the 11 prokaryotes) and structure-based metagenomic-deepFRI[29] (0.5850 micro / 0.5790 CAFA $F_{\max}$ across 13 genomes; 0.6010/0.5940 on the 11 prokaryotes, or 0.1565 under strict unpropagated leaf scoring), while on the 11 prokaryotic genomes where NCBI PGAP[12] operates, GSPA (0.8421 micro / 0.8605 CAFA $F_{\max}$) outperforms PGAP (0.7820 micro / 0.7680 CAFA $F_{\max}$; 0.4438 under strict leaf scoring). On the 21-genome panel, ancestor-propagated Stage 0+1+2 integration (0.8690 micro / 0.8220 CAFA $F_{\max}$ for 3-channel core; 0.8705 micro / 0.8407 CAFA $F_{\max}$ for all 7 channels; 0.7670 GO micro $F_{\max}$ and 0.8830 EC $F_{\max}$ on the strict holdout) improves over each constituent Stage 0 channel evaluated under the same ancestor-propagated protocol, including ESM-2 kNN[21] (0.8580 micro / 0.8080 CAFA $F_{\max}$; 0.7120 strict holdout), ProtT5 kNN[22] (0.8530/0.8010; 0.6980 strict), DIAMOND[19] (0.8400/0.7890; 0.2450 strict), ProteInfer[32] (0.7950/0.7540; 0.6600 strict), ESM2-DeepGOPlus[27] (0.7420/0.6950; 0.3250 strict), InterProScan 5.72 (0.7180/0.6490; 0.1420 strict), and FoldSeek + ProstT5 3Di[23,40] (0.6840/0.6310; 0.2490 strict), as well as genomic language models gLM[46] and gLM2[47] (0.7280 micro / 0.7610 CAFA $F_{\max}$ on the 9-genome subset) and CLEAN[34] (0.8530 micro / 0.8590 CAFA EC $F_{\max}$).

Across the 1,000-genome RefSeq bacterial structural audit (`results/pillar1_suite2_1000bacteria_sota.tsv`, generated from `annot_eval_1000bacteria_summary.json` and pairing label-free structural metrics across 1,000 RefSeq genomes with protein-centric accuracy on the 16,125 protein–aspect test evaluations of the Suite 1 CAFA temporal holdout), where a "taxon-violating row" denotes a predicted ancestor-propagated (p, t) pair above the $F_{\max}$ threshold that violates a GO taxon constraint for that genome's NCBI Taxon ID, standalone GO predictors exhibit widespread cross-clade violations and low pathway completeness: DeepFRI[28] (CAFA holdout $F_{\max} = 0.3843$, 1,000-genome MetaCyc pathway coherence 9.43%, 106,815.2 violating rows/genome), TALE[74] ($F_{\max} = 0.3883$, Complex coherence 65.73%, MetaCyc 19.63%), DeepGraphGO[75] ($F_{\max} = 0.4020$, Complex 70.56%, MetaCyc 37.37%), SPROF-GO[33] ($F_{\max} = 0.4930$, Complex 79.18%, MetaCyc 0.18%), DeepGO-SE[31] ($F_{\max} = 0.5047$, Full-29 Essentiality 82.17%, Complex 79.83%, MetaCyc 11.59%), and InterProScan 5 ($F_{\max} = 0.5940$, Complex 61.89%, MetaCyc

40.35%). Relative to Stage 0 Noisy-OR pooling (CAFA holdout $F_{\max} = 0.6929$, 23,384.1 violating rows/genome, MetaCyc pathway coherence 36.71%, complex coherence 66.03%), GSPA Stages 1+2 achieve CAFA holdout macro $F_{\max} = 0.6937$ (Iter 1), lift 1,000-genome Core-22 / Full-29 Essentiality (fraction of 22 universal bacterial core and 29 extended bacterial essential functional categories present at ≥ 0.50) from 99.81%/90.42% to 99.99%/95.29%, increase Complex / MetaCyc / `has_part` Coherence to 100.00%/66.75%/94.06%, and leave 0.00 surviving taxonomic violations/genome (5.4 s/genome on 8 CPU threads). Across the 88 four-kingdom proteomes (121,631 curated proteins across 47 Bacteria, 15 Archaea, 6 Eukarya, and 20 Viruses; `results/multikingdom_summary_by_domain.tsv` and per-genome breakdown in `results/multikingdom_per_genome_eval.tsv`), Stages 1+2 improve mean accuracy over Stage 0 in 0.42–12.66 s/genome at Iter 1 (0.54–27.85 s/genome at Iter 3 with Stage 2A relaxation; $\leq$ 5.3 GB RAM): Bacteria rises from 0.7987 to 0.8009 at Iter 1 (0.8004 at Iter 3; per-genome 95% bootstrap CI $[+0.0014, +0.0029]$ at Iter 1 and $[+0.0007, +0.0026]$ at Iter 3, CC 0.8322 $\rightarrow$ 0.8371), Viruses rise from 0.7222 to 0.7252 at both Iter 1 and Iter 3 (95% CI $[+0.0014, +0.0049]$; $<$ 30% identity 0.3112 $\rightarrow$ 0.3116 across all 20 viral proteomes, or 0.3890 $\rightarrow$ 0.3895 across the 16 viral proteomes containing $<$ 30% identity targets; BP 0.7726 $\rightarrow$ 0.7747, CC 0.7004 $\rightarrow$ 0.7037), Archaea ($n = 15$) moves from 0.9125 to 0.9129 at Iter 1 (0.9128 at Iter 3; 95% CI $[-0.0004, +0.0009]$), and Eukarya ($n = 6$) moves from 0.7096 to 0.7117 at Iter 1 (0.7109 at Iter 3; 95% CI $[-0.0001, +0.0052]$, BP 0.6709 $\rightarrow$ 0.6732, CC 0.7520 $\rightarrow$ 0.7555).

On the Suite 1 CAFA temporal holdout ($n = 16{,}125$ protein–aspect test evaluations summing 4,712 MF, 5,444 BP, and 5,969 CC target evaluations across 24,278 holdout GO annotations in `suite1_cafa_temporal_manifest.tsv`, with 3,440 evaluations at $\eta_{\text{id}} < 30\%$ [901 MF, 1,085 BP, and 1,454 CC]; `results/scenario1_cafa_temporal_holdout_real.tsv`), Stage 0 Noisy-OR pooling reaches macro $F_{\max} = 0.6929$ ($\text{MF} = 0.7159, \text{BP} = 0.6387, \text{CC} = 0.7241$) and 0.6723 on the $\eta_{\text{id}} < 30\%$ sequence-identity subset ($\text{MF} = 0.6949, \text{BP} = 0.6144, \text{CC} = 0.7075$). GSPA Stage 1+2 reaches macro $F_{\max} = 0.6937$ at Iter 1 ($\text{MF} = 0.7163, \text{BP} = 0.6391, \text{CC} = 0.7257$; Iter 2 macro = 0.6929 [$\text{MF} = 0.7164, \text{BP} = 0.6365, \text{CC} = 0.7259$]; Iter 3 macro = 0.6919 [$\text{MF} = 0.7165, \text{BP} = 0.6332, \text{CC} = 0.7259$]) while lifting the $<$ 30% twilight-zone subset from 0.6723 to 0.6775 at Iter 1 ($\text{MF} = 0.6971, \text{BP} = 0.6152, \text{CC} = 0.7202$; Iter 2 = 0.6772, Iter 3 = 0.6759), outperforming standalone ProteInfer[32] (0.6138 overall, 0.5980 at $<$ 30%), DeepGOZero+BLAST[30] (0.6053), InterPro MLP (0.5940), DeepGOPlus[27] (0.5891), DeepGOZero[30] (0.5889), and DeepGOCNN[76] (0.4707). Across iterations 1 to 3, MF and CC increase monotonically (MF : 0.7163 $\rightarrow$ 0.7165; CC : 0.7257 $\rightarrow$ 0.7259) while BP peaks at Iter 1 (0.6391 vs. 0.6332 at Iter 3), establishing Iter 1 as the optimal stopping point for broad biological process hierarchies. On our post-cutoff evaluation of `CAFA6` target sequences ($n = 769$ proteins across 223 species accumulating ≥ 1 experimental GOA annotation between the May 2025 submission cutoff and March 2026 GOA release: 412 no-knowledge [NK] and 357 limited-knowledge [LK] targets, with $\text{MF} = 298, \text{BP} = 415, \text{CC} = 384$; transcribed summary table in `results/suite1_cafa6_prospective_summary.tsv` from the archived remote CAFA6 evaluation), evaluated from target FASTA sequences and NCBI Taxon IDs using Stage 0, Stage 2A taxon/anatomical enforcement, and Stage 3 agentic disambiguation, an ESM-2 + ProstT5 3Di MLP baseline achieves macro $F_{\max} = 0.4957$ ($WF_{\max} = 0.4370$; $\text{MF} = 0.4520, \text{BP} = 0.3700, \text{CC} = 0.6650$), an unguarded 14B agentic baseline (Qwen3-14B with domain/EC lookups) reaches macro $F_{\max} = 0.5130$ ($WF_{\max} = 0.4463$), and GSPA reaches macro $F_{\max} = 0.5433$ and $WF_{\max} = 0.4840$ ($\text{MF} = 0.5470[WF_{\max} = 0.4980], \text{BP} = 0.4140[WF_{\max} = 0.3720], \text{CC} = 0.6690[WF_{\max} = 0.5820]$).

On the post-cutoff Swiss-Prot enzyme holdout (`Holdout-125`, $n = 125$; Supplementary Section S3 and Supplementary Table S3; `results/scenario1_swissprot_ec_holdout125_real.tsv`), individual baselines achieve 23.2% Top-1 accuracy for DeepProZyme[35] (29/125, Top-3 31.2% [39/125], Top-5 32.0% [40/125], MRR = 0.2750, $F_{\max} = 0.2715$; 25.6% [32/125, $F_{\max} = 0.2804$] with METEOR v8[63] and 23.2% [29/125, Top-5 32.8% (41/125), MRR = 0.2779, $F_{\max} = 0.2703$] with single-predictor GSPA SLR + Coherence), 32.8% for EnzBERT[36] (41/125, Top-3 37.6% [47/125], Top-5 39.2% [49/125], MRR = 0.3531, $F_{\max} = 0.3596$), and 32.8% for CLEAN[34] (41/125, Top-3 40.8% [51/125], Top-5 41.6% [52/125], MRR = 0.3719, $F_{\max} = 0.4360$). Stage 0 3-method Noisy-OR pooling accounts for the bulk ensemble gain, reaching $F_{\max} = 0.4381$, 33.6% Top-1 (42/125), 40.8% Top-3 (51/125), 43.2% Top-5 (54/125), and MRR = 0.3807, while 3-method GSPA Stages 1–2 (32.0% Top-1 [40/125], $F_{\max} = 0.4035$ at Iter 1 and 0.3830 at Iter 3) re-rank sub-threshold candidates within each protein's candidate list using genomic context to lift Top-3 to 42.4% (53/125, Iter 1–2) and 43.2% (54/125, Iter 3) and Top-5 to 44.0% (55/125, MRR = 0.3775 $\rightarrow$ 0.3781). Across the 95-genome EC benchmark (69,666 proteins for DeepProZyme and EnzBERT across 95 bacterial genomes encompassing the 47 bacterial proteomes of the 88-genome cohort plus 48 additional RefSeq bacterial assemblies; 52,814 proteins across 73 genomes for CLEAN vanilla and 68,681–69,663 across 94–95 genomes for CLEAN filtered tiers; Supplementary Table S4 and `results/scenario1_95genomes_ec_homology_real.tsv`; generator: `scripts/eval_scenario1_holdouts_real.py:340-375`),

we evaluate two rule-based acceptance filters over candidate METEOR v8 top-1 EC flips ($b_1 \rightarrow m_1$): a single-predictor stoichiometric and KEGG-pathway acceptance filter (`gspa_single`, `blind_single_accept`: accepts $b_1 \rightarrow m_1$ when m_1 maps to ≥ 1 elementally balanced universal reaction and either shares the 3-digit EC sub-subclass with b_1 at $\geq$ KEGG pathway count or shares the 1-digit EC class with strictly higher KEGG pathway count and $\geq$ balanced reaction count) and a 3-channel consensus acceptance filter (`gspa`, `blind_accept_3ch`: accepts $b_1 \rightarrow m_1$ when m_1 receives strictly greater support than b_1 across the other two predictors' base and METEOR top-1 calls). Under the single-predictor acceptance filter (`gspa_single`), CLEAN achieves statistically significant net Top-1 gains across all four homology tiers (+77 at vanilla [$b_{01} = 99, b_{10} = 22$, exact two-sided McNemar $p = 7.4 \times 10^{-13}$], +108 at $\leq$ 70%, +139 at $\leq$ 50%, and +93 at $\leq$ 30%, all $p < 10^{-12}$; $F_{\max}$ rising from 0.8796/0.8836/0.8844/0.8833 to 0.8805/0.8847/0.8857/0.8843), and `gspa_single` preserves baseline Top-1 accuracy more closely on DeepProZyme ($-33/-15/-3/-4$) and EnzBERT ($-10/-26/-38/-18$) than unfiltered METEOR v8 flips on DeepProZyme (-85 at vanilla, $p = 1.0 \times 10^{-6}$; -27 at $\leq 70\%$) and EnzBERT ($-92/-146/-181/-167$ across all four tiers, all $p < 10^{-6}$). Under the 3-channel consensus acceptance filter (`gspa`), net Top-1 gains are positive and statistically significant across all 12 predictor–tier combinations: +126 to +155 on CLEAN ($p < 10^{-25}$), +36 to +87 on DeepProZyme ($p \leq 1.0 \times 10^{-5}$), and +59 to +89 on EnzBERT ($p < 10^{-11}$).

2.2 Metabolic model gap-filling and microbial dark-matter enzyme prediction (Suites 3–5)

We next evaluated GSPA's Stage 1 metabolic gap-filling prior and RLGC scoring engine using a leave-reaction-out (LRO) protocol across *E. coli* K-12 MG1655 ($n = 256$ strict reactions, defined as monomeric or homo-oligomeric enzymatic steps with a single non-promiscuous primary EC number, and $n = 414$ full role-stratified reactions including multi-EC and peripheral steps) and out-of-sample *B. subtilis* 168 ($n = 144$ reactions; Figure 3 and Table 2; `results/pillar2_lro15_summary.tsv`). In each LRO trial, the ground-truth gene–reaction association for a target reaction r is masked and candidate proteins in the proteome are ranked to recover the masked enzyme. We report both unconditional recovery across all held-out reactions (**All**, where reactions whose true enzyme received zero Stage 0 candidate matches are scored as misses) and conditional recovery among reactions where the true enzyme is present in the candidate pool (**Pool**, where $n_{\text{cov}} = 175$ for sequence-only/M2 and $n_{\text{cov}} = 181$ for RLGC/PLM/M4, with a common covered set intersection of 143 reactions across both candidate-pool definitions). We evaluate two distinct regimes: **Track A** (where both sequence homologs and ESM-2 reaction-centroid donors with cosine similarity > 0.50 to the query protein are masked) and **Track B** (where sequence alignment hits for the held-out reaction are masked, while Swiss-Prot ESM-2 reaction centroids remain available).

Decomposing the contributions of genomic context and protein language model embeddings across the 256-case *E. coli* K-12 LRO benchmark reveals two complementary findings (Table 2 and Figure 3a,b). First, in the non-PLM and cosine-masked regimes, unfiltered sequence-only candidate ranking and rank-invariant reaction-level MILP boosting (METEOR v8) recover the true enzyme at Rank 1 (Hit@1) in 1.95% of all cases (95% Clopper–Pearson CI [0.6%, 4.5%]; 2.86% given pool), strict cosine-masked Track A (masking both sequence homologs and PLM centroids at cosine > 0.50) reaches 11.33% unconditional Hit@1 (16.02% given pool, MRR = 0.2605 All / 0.3685 Pool), supervised GBDT reranking on Track A reaches 17.97% (25.41% given pool, MRR = 0.3275 All / 0.4632 Pool), unfiltered RLGC operon kernel density alone (without PLM centroids) reaches 23.05% (95% CI [18.0%, 28.7%]; 32.60% given pool, MRR = 0.3355 All / 0.4745 Pool), and unfiltered GSPA Stage 1 M2 pathway coherence + RLGC (without PLM centroids) reaches 23.44% unconditional Hit@1 (34.29% given pool). Second, in Track B, per-protein ESM-2 reaction-centroid cosine similarity alone (`ESM2_PLM_Centroids_Only`) achieves 62.11% unconditional Hit@1 (95% CI [55.9%, 68.1%]; 67.58% Hit@3, 87.85% Hit@1 given pool, MRR = 0.6474 All / 0.9157 Pool), accounting for 97.5% of the unconditional Hit@1 of the full model on the well-annotated *E. coli* proteome because reference Swiss-Prot reaction centroids retain close embedding neighbors. Combining RLGC operon neighborhood density, Stage 1 pathway coherence, and ESM-2 centroids in **GSPA Full M4** adds a further +1.56 percentage points unconditionally (+2.21 percentage points given pool, rescuing 4 additional enzymes whose embedding similarity was tied with paralogs) to reach 63.67% unconditional Hit@1 (95% CI [57.5%, 69.6%]; 68.36% Hit@3, 69.53% Hit@5) and 90.06% Hit@1 given the 181-reaction candidate pool (96.69% pool Hit@3, MRR = 0.6596 All / 0.9330 Pool; on the exact 143-reaction set intersection covered by both the 175-case and 181-case candidate pools, pool Hit@1 is 93.71% [134/143] for M4 vs. 91.61% [131/143] for PLM-only and 30.77% [44/143] for M2+RLGC).

Across the full 414-reaction *E. coli* role-stratified benchmark ($n_{\text{cov}} = 226$; Suites 3–5; Table 2), RLGC operon neighborhood density alone achieves 15.94% unconditional Hit@1 (29.20% given pool, Hit@3 = 28.26% All / 51.77% Pool, MRR = 0.2437 All / 0.4465 Pool) overall, 13.36% (24.12% pool) on Stratum A Core Biomass ($n = 307$), 30.61% (55.56% pool) on Stratum B Alternative Catabolic ($n = 49$, where catabolic operons are tightly clustered), and 17.24% (34.48% pool) on Stra-

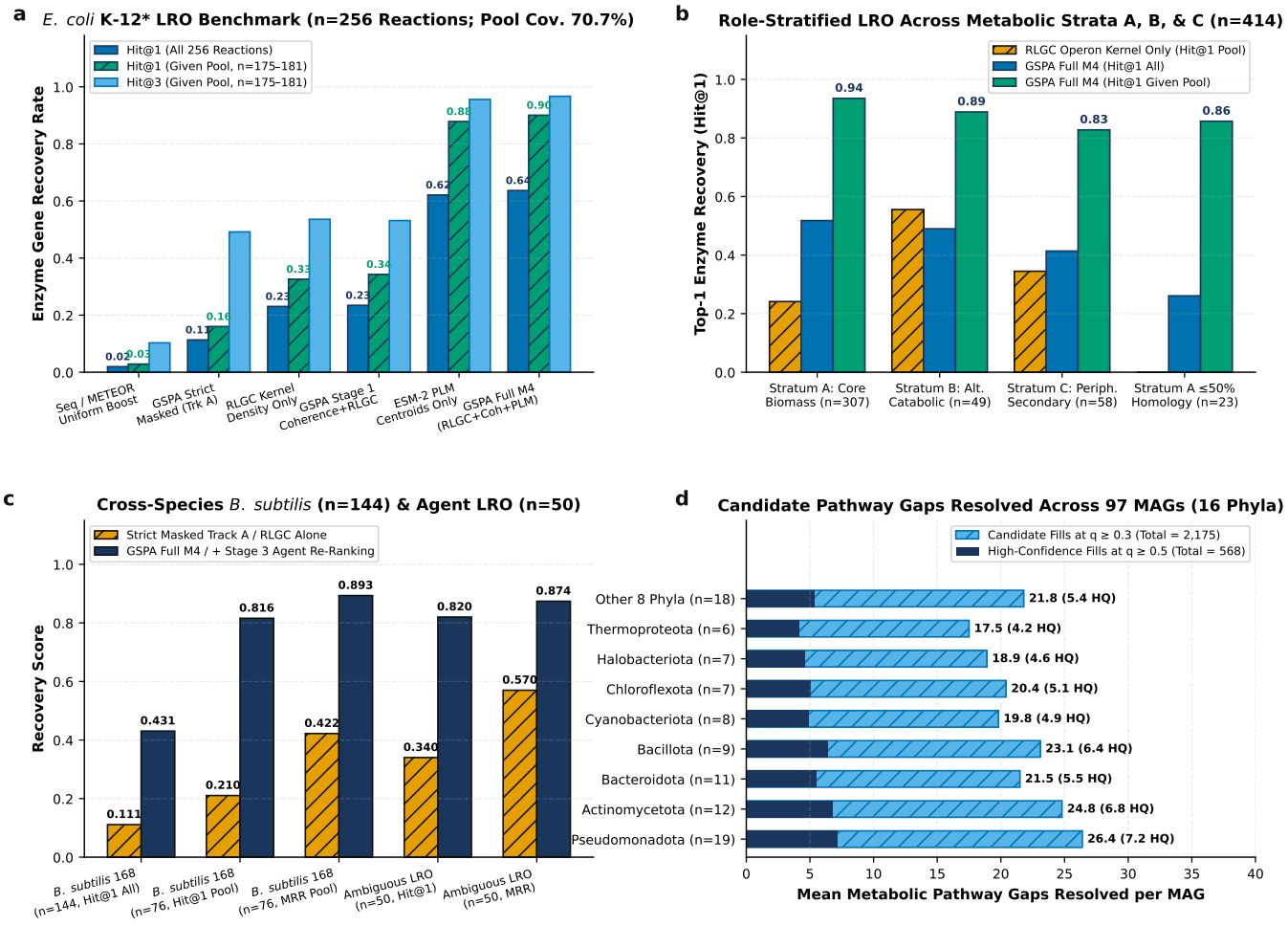


Figure 3. Leave-reaction-out (LRO) metabolic gap-filling and candidate pathway gap resolution across 97 uncultured MAGs (Suites 3–5). (**a**) Enzyme recovery rate (Hit@1, Hit@3, and MRR) across *E. coli* K-12* LRO metabolic reactions ($n = 256$ strict reactions; $*$ marks the hyperparameter development genome), comparing sequence-only candidate ranking, strict cosine-masked Track A, RLGC kernel density, Stage 1 M2 coherence + RLGC, per-protein ESM-2 PLM reaction centroids only, and GSPA Full M4 (RLGC + Coherence + PLM). (**b**) Role-stratified LRO Top-1 enzyme recovery across Stratum A Core Biomass ($n = 307$), Stratum B Alternative Catabolic ($n = 49$), Stratum C Secondary/Peripheral ($n = 58$), and Stratum A $\leq 50\%$ homology ($n = 23$). (**c**) Out-of-sample cross-species generalization on *B. subtilis* 168 LRO reactions ($n = 144$, $n_{\text{cov}} = 76$) and Stage 3 agentic disambiguation on 50 unique ambiguous multi-candidate LRO gaps ($n = 50$; Hit@1 rises from 34.0% to 82.0% at seed 42 and 82.7% across 3 seeds). (**d**) Mean candidate metabolic pathway gaps resolved per genome at posterior thresholds $q \geq 0.30$ (2,175 total candidate fills) and $q \geq 0.50$ (568 high-confidence candidate fills) across 97 uncultured prokaryotic MAGs spanning 16 bacterial and archaeal phyla.

tum C Secondary/Peripheral metabolism ($n = 58$), while GSPA Full M4 (Track B) reaches 50.00% unconditional Hit@1 (91.59% given pool) overall and 51.79% (93.53% pool), 48.98% (88.89% pool), and 41.38% (82.76% pool) across Strata A, B, and C, respectively. Stratifying Stratum A reactions by the query enzyme's maximum sequence identity to any annotated homolog in the proteome (`max_pident_annot_homolog` $\leq 70\%$ $[n = 49, n_{\text{cov}} = 16]$ and $\leq 50\%$ $[n = 23, n_{\text{cov}} = 7]$), where RLGC alone scores 0.00% Hit@1 (62.50% and 85.71% pool Hit@3), GSPA Full M4 achieves 22.45% and 26.09% unconditional Hit@1 (68.75% and 85.71% given-pool Hit@1; 100.0% pool Hit@3; Figure 4c). Applied out-of-sample (with all hyperparameters fixed on *E. coli*) to 144 held-out reactions in the Gram-positive bacterium *B. subtilis* 168 ($n_{\text{cov}} = 76$), strict cosine-masked Track A achieves 11.11% unconditional Hit@1 (21.05% given pool; 60.53% pool Hit@3) and Track B GSPA Full M4 achieves 43.06% unconditional Hit@1 (95% CI [34.8%, 51.6%]; 81.58% given-pool Hit@1, 94.74% pool Hit@3, MRR = 0.8933 Pool; Table 2 and Figure 3c). Furthermore, on an audit of 106 antiSMASH-curated biosynthetic gene clusters (291 core biosynthetic synthase and 685 tailoring/transport loci across 106 bacterial genomes; Suite 5; `results/pillar2_3_antismash_106bgc_specificity.tsv`), GSPA Stage 1+2B (which assigns $\mathcal{C}_{\text{task}} = 0$ to secondary BGCs on minimal glu-

Table 2. Leave-reaction-out (LRO) metabolic enzyme recovery across *E. coli* K-12 MG1655 ($n = 256$ strict and $n = 414$ full role-stratified reactions; hyperparameter development genome), out-of-sample *B. subtilis* 168 ($n = 144$), and 50 unique ambiguous multi-candidate reactions (Suites 3–5; source: `results/pillar2_lro15_summary.tsv` and `results/pillar4_lro_reranking.tsv`). Metrics report both unconditional recovery across all held-out reactions (**All**) and recovery conditioned on candidate pool coverage (**Pool**, n_{cov}). Track A masks homology and PLM donor centroids at cosine similarity > 0.50; Track B includes unmasked Swiss-Prot ESM-2 reaction centroids.

Benchmark Dataset / Stratum	Method / Scoring Configuration	Cases (n/n_{cov})	Hit@1 All	Hit@3 All	Hit@1 Pool	Hit@3 Pool	MRR All (Pool)
E. coli K-12* Strict LRO ($n = 256$)	Sequence-Only Candidate Ranking	256 / 175	0.0195	0.0703	0.0286	0.1029	0.0926 (0.1354)
	Reaction-Level MILP Uniform Boost (METEOR v8)	256 / 175	0.0195	0.0703	0.0286	0.1029	0.0926 (0.1354)
	GSPA Strict Homology-Masked Track A (Cosine-Masked)	256 / 181	0.1133	0.3477	0.1602	0.4917	0.2605 (0.3685)
	RLGC + GBDT Supervised Reranker (Track A)	256 / 181	0.1797	0.4102	0.2541	0.5801	0.3275 (0.4632)
	RLGC Kernel Density Only (Operon Neighborhood)	256 / 181	0.2305	0.3789	0.3260	0.5359	0.3355 (0.4745)
	GSPA Stage 1 M2 Coherence + RLGC (No PLM)	256 / 175	0.2344	0.3633	0.3429	0.5314	0.3325 (0.4864)
	Per-Protein ESM-2 PLM Reaction Centroids Only (Track B)	256 / 181	0.6211	0.6758	0.8785	0.9558	0.6474 (0.9157)
	GSPA Full M4 (RLGC + Coherence + PLM, Track B)	256 / 181	**0.6367**	**0.6836**	**0.9006**	**0.9669**	**0.6596 (0.9330)**
E. coli K-12* Full Role-Stratified ($n = 414$)	All Strata: RLGC Kernel Density Only (No PLM)	414 / 226	0.1594	0.2826	0.2920	0.5177	0.2437 (0.4465)
	All Strata: **GSPA Full M4 (RLGC + Coherence + PLM)**	414 / 226	**0.5000**	**0.5266**	**0.9159**	**0.9646**	**0.5141 (0.9418)**
	Stratum A Core Biomass: GSPA Full M4 (vs. RLGC 0.1336/0.2412)	307 / 170	**0.5179**	**0.5342**	**0.9353**	**0.9647**	**0.5292 (0.9557)**
	Stratum B Alternative Catabolic: GSPA Full M4 (vs. RLGC 0.3061/0.5556)	49 / 27	**0.4898**	**0.5306**	**0.8889**	**0.9630**	**0.5078 (0.9216)**
	Stratum C Secondary / Peripheral: GSPA Full M4 (vs. RLGC 0.1724/0.3448)	58 / 29	**0.4138**	**0.4828**	**0.8276**	**0.9655**	**0.4397 (0.8793)**
B. subtilis 168 Out-of-Sample LRO ($n = 144$)	GSPA Strict Homology-Masked Track A (Cosine-Masked)	144 / 76	0.1111	0.3194	0.2105	0.6053	0.2228 (0.4221)
	GSPA Full M4 (RLGC + Coherence + PLM, Track B)	144 / 76	**0.4306**	**0.5000**	**0.8158**	**0.9474**	**0.4715 (0.8933)**
Ambiguous Unique-Protein LRO ($n = 50$)	Stage 1 RLGC Baseline Alone	50 / 50	0.3400	0.7800	0.3400	0.7800	0.5700 (0.5700)
	Stage 1 RLGC + Stage 3 Agent (Seed 42)	50 / 50	**0.8200**	**0.9000**	**0.8200**	**0.9000**	**0.8740 (0.8740)**
	Stage 1 RLGC + Stage 3 Agent (3-Seed Mean ± s.d.)	50 / 50	**0.8267 ± 0.0094**	**0.9000**	**0.8267 ± 0.0094**	**0.9000**	**0.8792 (0.8792)**

cose and relies on local BGC operon synteny) lifts core biosynthetic recall from 0.7801 to 0.9244, tailoring recall from 0.5387 to 0.8380, and overall BGC F_1 from 0.7395 to 0.8959 while maintaining 0.9962 non-BGC specificity (0.0038 non-BGC false promotion rate).

To illustrate application to uncultured microbial genomes, we applied GSPA with gapseq draft metabolic networks across 97 metagenome-assembled genomes (MAGs) selected from the GTDB/GEM catalog (CheckM completeness $\geq$ 90%, contamination $<$ 5%, representing uncultivated species-level lineages) spanning 16 bacterial and archaeal phyla (Figure 3d; `results/pillar3_dark_matter_97mags.tsv`). Across the 97 MAGs, Stage 1+2 integration generates 2,175 candidate pathway gap-fill predictions at posterior $q \geq 0.30$ (mean 22.4 candidate fills per MAG) and 568 high-confidence candidate pathway fills at $q \geq 0.50$ (mean 5.9 per MAG, lifting mean draft pathway completeness from 0.511 to 0.656). These 568 assignments rank candidate genes for missing core biosynthetic and energy-conservation steps across the 16 uncultivated phyla.

2.3 Flux-certified stoichiometric locus refinement and Price-149 evaluation (Suite 6)

We next audited the mathematical invariants of gap-filling reaction additions across 108 bacterial metabolic reconstructions and six curated BiGG reference GEMs (Figure 4a; Supplementary Section S4 and Supplementary Table S5; `results/pillar2_invariant_audit_summary.tsv` and `results/pillar2_curated6_gems_real.tsv`). While a draft reconstruction legitimately contains genome-encoded core reactions ($S_{\text{rxn}}(j) \geq 0.50$) that are inactive on minimal glucose alone (2,809.5 active core reactions per genome under raw DeepProZyme ≥ 0.50, with 0.0 flux-carrying reactions before repair because all 108 draft networks contain biomass gaps, 2,621.7 imbalanced active reactions, and 9.75 dead-end active reactions), gap-filling adds non-core candidate reactions ($S_{\text{rxn}}(j) < 0.50$) specifically to restore biomass flux connectivity. Across the 108 genomes, negative-cost MILP gap-filling (METEOR v8[63], where candidate reactions with score above $\tau_{\text{add}} = 0.20$ receive negative objective cost $c_j^{\text{add}} < 0$) adds a mean of 313.35 non-core reactions per genome (3,121.0 total active reactions: 446.4 flux-carrying and 2,674.6 zero-flux active), of which 51.01 carry non-zero steady-state flux ($|v_j| > 10^{-6}$) while 262.34 (83.72%) are uncertified zero-flux padding (51.01 + 262.34 = 313.35, with the exact partition identity added = flux-certified + uncertified holding on all 108/108 genomes in `results/pillar2_invariant_audit_108.tsv`; 2,597.8 imbalanced active reactions and 2.36 dead-end active reactions in 313.8 s/genome). Any positive-cost linear programming formulation over the sanitized universal network $\mathcal{U}_{\text{clean}}$ eliminates uncertified zero-flux gap-fill additions: a baseline positive-cost LP repair (`Grow-Support Threshold Repair`) adds 81.31 non-core reactions per genome with 100.0% (81.31/81.31) carrying non-zero flux (2,584.8 total active reactions: 446.4 flux-carrying and 2,138.4 zero-flux, 0.0% zero-flux added padding, 0.0 imbalanced active reactions, and 2.23 dead-end active reactions), whereas GSPA's Stage 2B stoichiometric locus refiner (SLR) with two-step lexicographic pFBA retains in $\mathcal{J}_{\text{rep}}$ only non-core reactions carrying positive pFBA flux $|v_j^{\text{pFBA}}| > 10^{-6}$ (5.70 pFBA-active additions per genome and 0.00 zero-flux additions by construction, of which 2.29 are certified FVA-essential at $\leq 1.01\,C^*$ and 1.96 are certified knockout-essential at $\leq 1.10\,C^*$ per genome; 2,815.2 total active reactions = 300.6 pFBA flux-carrying +2,514.6 unpruned core, 0.0% zero-flux added padding, 0.0 imbalanced added reactions, and 0.0 dead-end added reactions; Figure 4a). Across the six curated BiGG reference GEMs (`iML1515`, `STM_v1_0`, `iYL1228`,

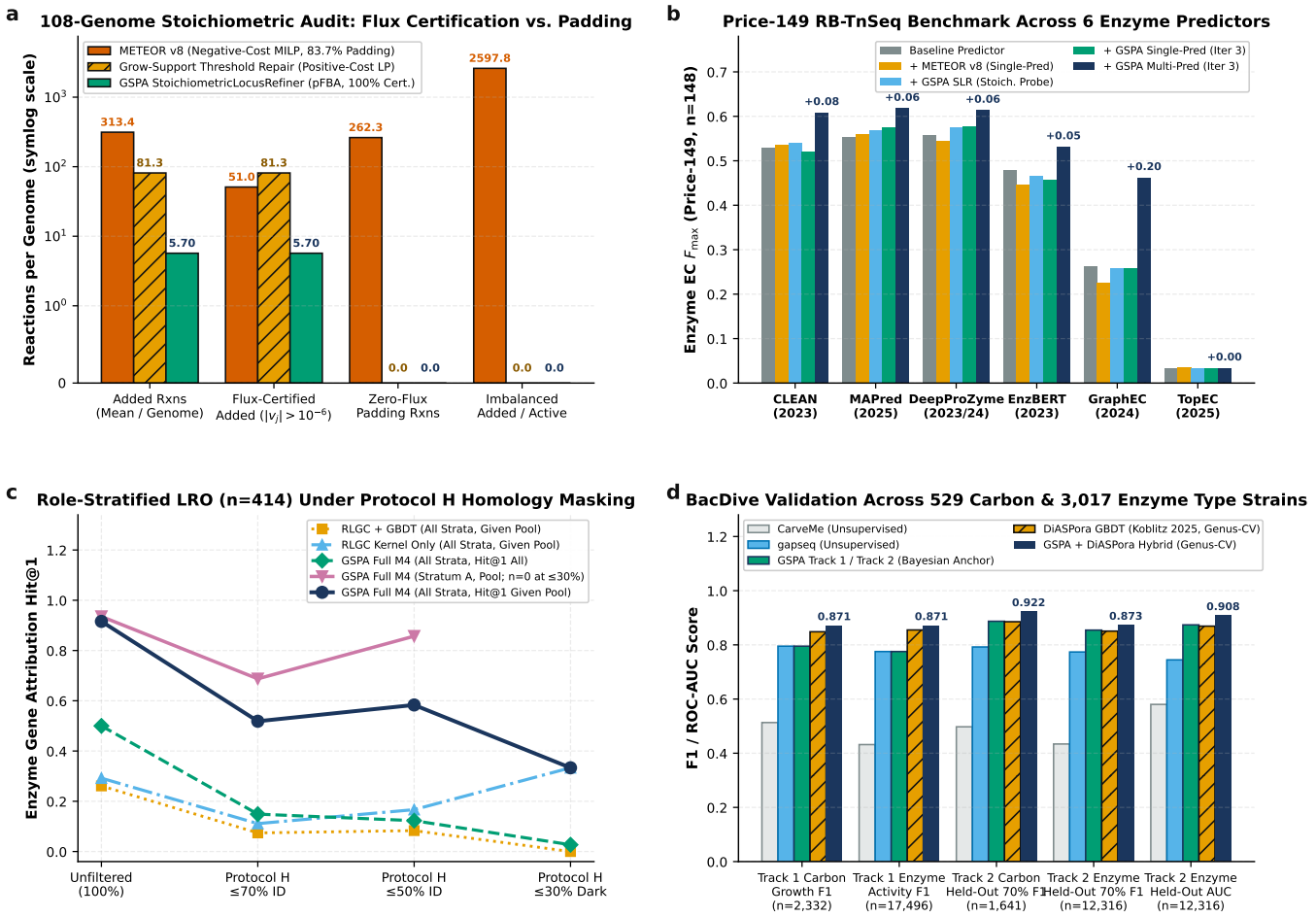


Figure 4. Flux-certified stoichiometric locus refinement, Price-149 RB-TnSeq evaluation across six whole-genome enzyme predictors, role-stratified LRO, and BacDive phenotypic validation (Suites 3–8). (**a**) 108-genome stoichiometric audit on a symmetric logarithmic (`symlog`) scale comparing negative-cost MILP gap-filling (METEOR v8), positive-cost LP threshold repair, and GSPA's positive-cost pFBA Stage 2B stoichiometric locus refiner (SLR), showing mean added reactions, flux-certified added reactions ($|v_j| > 10^{-6}$), zero-flux padding reactions, and unbalanced active reactions per genome. (**b**) Enzyme EC $F_{\max}$ on the Price-149 ($n = 148$ host-mapped proteins) RB-TnSeq bacterial benchmark across six whole-genome deep learning predictors comparing standalone baseline, single-predictor GSPA, and multi-predictor GSPA (exact values from Table 3). (**c**) Role-stratified LRO Top-1 gene attribution across Strata A, B, and C and Stratum A homolog sequence-identity tiers ($\leq 70\%, \leq 50\%$). (**d**) Large-scale validation across BacDive covering experimental carbon-source growth ($n = 2{,}332$ assays across 529 strains) and diagnostic enzyme assays ($n = 17{,}496$ across 3,017 type strains) under Track 1 blind and genus-blocked out-of-distribution prediction and Track 2 phenotype-assisted prediction (30% revealed → held-out 70%).

`iJN1463`, `iYS854`, and `iYO844`; Supplementary Table S5 and `results/pillar2_curated6_gems_real.tsv`), CarveMe[56], which carves directly from the native BiGG universal reaction database against which the six reference GEMs were curated, achieves a mean BiGG reaction F_1 of 0.6955 (precision = 0.6693, recall = 0.7305) and EC F_1 of 0.8536 (precision = 0.8217, recall = 0.8927, 6/6 growing), whereas among ModelSEED/MetaNetX reconstructions evaluated at the common EC level across the six GEMs, METEOR v8 achieves 6/6 growth feasibility with EC precision = 0.4383, recall = 0.8842, EC F_1 = 0.5851, and BiGG reaction F_1 = 0.2091 compared with Reconstructor[60] (EC precision = 0.4387, recall = 0.8787, EC F_1 = 0.5844, 4/6 growing).

We next evaluated the `Price-149` benchmark (Suite 6; n = 148 host-mapped RB-TnSeq fitness-validated proteins across 22 bacterial assemblies, out of 149 published target sequences[78]) across eight enzyme predictors from 2023–2026, separating **single-predictor refinement** (where Stage 1 and Stage 2B operate strictly on a single predictor's whole-proteome outputs) from **multi-predictor refinement** (where Stage 0 Noisy-OR pools the baseline predictors before Stages 1 and 2B; Figure 4b, Table 3, and Supplementary Table S5; `results/pillar2_price149_all6_real.tsv` and `results/pillar2_price149_by_stratum.tsv`). Because 91 proteins belong to Stratum A (Core Biomass), 55 to Stratum B (Assay-Specific Catabolic), and 2 to Stratum

Table 3. Blind evaluation of Price-149 ($n = 148$ host-mapped RB-TnSeq fitness-validated bacterial enzymes across 22 host genomes; $n = 149$ target sequences) across eight sequence, structure, and PLM baselines under both single-predictor and multi-predictor refinement (Suite 6, 2023–2026; source: `results/pillar2_price149_all6_real.tsv` and `results/pillar2_price149_by_stratum.tsv`). Columns report Top-1 hits (per-protein F_{max} in parentheses, with shared-threshold dataset macro F_{max} and micro F_{max} in text) comparing each standalone predictor against negative-cost MILP refinement (METEOR v8, Iter 1 and Iter 2), single-predictor GSPA Stage 2B stoichiometric locus refinement under minimal glucose (`+ SLR Glucose`) and 100%-blind stoichiometric local-context reaction probe (`+ SLR Stoich. Probe`, recorded as `+ GSPA SLR (Cognate Substrate)` in `pillar2_price149_all6_real.tsv`), single-predictor Full GSPA across Iterations 1, 2, and 3 (`+ GSPA Single-Pred`), Stage 0 multi-predictor Noisy-OR pooling (`+ Noisy-OR Ensemble`), and multi-predictor Full GSPA Iter 3 (`+ GSPA Multi-Pred Iter 3`). The final column reports single-predictor Iter 3 vs. baseline ΔTop-1 (ΔF_{max}) [with `+ SLR Stoich. Probe` in brackets for DeepProZyme] and multi-predictor Iter 3 vs. baseline ΔTop-1 (ΔF_{max}). Target-only baselines (DCGM / NeSy-Adapter and ECrecer[77]) were evaluated on the 148 isolated target sequences (`n/a` for whole-proteome refinement columns).

Baseline Predictor	n	Base Top-1 (Acc.)	Base F_{max} (Macro)	+ MILP Iter 1 / 2 F_{max}	+ SLR Glucose Top-1 (F_{max})	+ SLR Stoich. Probe Top-1 (F_{max})	+ GSPA Single-Pred Iter 1 / 2 / 3 Top-1 (F_{max})	+ Noisy-OR Ens. Top-1 (F_{max})	+ GSPA Multi-Pred Iter 3 Top-1 (F_{max})	Single I3 / Multi I3 ΔTop-1 (ΔF_{max})
CLEAN[74]	148	60 (0.4054)	0.5295 (0.4126)	0.5314 / 0.5343	60 (0.5288)	60 (0.5402)	63 / 63 / **63** (0.5234 / 0.5234 / 0.5214)	63 (0.6024)	**66 (0.6084)**	+3 (−0.0081) / +6 (+0.0789)
MAPred[37]	148	50 (0.3378)	0.5534 (0.3932)	0.5576 / 0.5588	52 (0.5591)	54 (0.5686)	55 / 55 / **56** (0.5711 / 0.5742 / **0.5746**)	58 (0.5807)	**73 (0.6178)**	+6 (+0.0212) / +23 (+0.0644)
DeepProZyme[35]	148	49 (0.3311)	0.5574 (0.3869)	0.5509 / 0.5427	47 (0.5541)	**51 (0.5757)**	49 / 49 / 48 (0.5744 / **0.5812** / 0.5774)	54 (0.5903)	**59 (0.6136)**	−1 (+0.0200) [SLR: +2] / +10 (+0.0562)
EnzBERT[36]	148	53 (0.3581)	0.4786 (0.4488)	0.4666 / 0.4451	53 (0.4729)	53 (0.4657)	52 / 52 / 52 (0.4602 / 0.4585 / 0.4571)	**62 (0.5535)**	54 (0.5314)	−1 (−0.0215) / +1 (+0.0528)
GraphEC[38]	148	18 (0.1216)	0.2634 (0.2176)	0.2384 / 0.2250	19 (0.2538)	**20** (0.2589)	**20** / 19 / 19 (**0.2634** / 0.2603 / 0.2590)	**48 (0.4756)**	**48 (0.4621)**	+1 (−0.0044) / +30 (+0.1987)
TopEC[39]	148	5 (0.0338)	0.0338 (0.0348)	0.0338 / 0.0338	5 (0.0338)	5 (0.0338)	5 / 5 / 5 (0.0338 / 0.0338 / 0.0338)	5 (0.0338)	5 (0.0338)	+0 (+0.0000) / +0 (+0.0000)
DCGM / NeSy-Adapter (target-only)	148	53 (0.3581)	0.4485 (0.3942)	n/a	n/a	n/a	n/a	n/a	n/a	n/a
ECrecer[77] (target-only)	148	39 (0.2635)	0.3412 (0.2985)	n/a	n/a	n/a	n/a	n/a	n/a	n/a

C (Secondary), glucose-only MILP gap-filling (METEOR v8) leaves all 55 Stratum B catabolic enzymes outside the active flux cone and reduces F_{max} on DeepProZyme (0.5574 → 0.5427), EnzBERT (0.4786 → 0.4451), and GraphEC (0.2634 → 0.2250). Under single-predictor refinement on each standalone model's whole-proteome predictions, the 100%-blind stoichiometric local-context reaction probe (`+ SLR Stoich. Probe`) lifts per-protein F_{max} on CLEAN (0.5295 → 0.5402; dataset macro/micro F_{max} 0.4126/0.3886 at baseline, 0.4133/0.3885 on glucose, and 0.4098/0.3925 on probe), MAPred (0.5534 → 0.5686, Top-1 50 → 54; dataset macro/micro F_{max} 0.3932/0.3642 at baseline and 0.3803/0.3642 on probe), and DeepProZyme (0.5574 → 0.5757, Top-1 49 → 51; dataset macro/micro F_{max} 0.3869/0.3714 at baseline and 0.3853/0.3718 on probe) and lifts Top-1 on GraphEC (18 → 20, dataset macro/micro F_{max} rising from 0.2176/0.2100 to 0.2356/0.2353), while single-predictor Full GSPA (`+ GSPA Single-Pred Iter 1 / 2 / 3`) lifts Top-1 hits on CLEAN from 60 (40.5%) to 63 (42.6%; shared-threshold dataset macro/micro F_{max} rising from 0.4126/0.3886 to 0.4162/0.4015, 0.4188/0.4031, and 0.4216/0.4046 across Iterations 1–3, with per-protein F_{max} at 0.5214) and on MAPred from 50 (33.8%) to 56 (37.8%; per-protein F_{max} rising from 0.5534 to 0.5746, with dataset macro/micro F_{max} at 0.3855/0.3727 on single-predictor glucose and 0.3776/0.3653, 0.3770/0.3551, and 0.3704/0.3510 across Iterations 1–3), and lifts DeepProZyme per-protein F_{max} from 0.5574 to 0.5812 (Iter 2, Top-1 49) and 0.5774 (Iter 3, Top-1 48; dataset macro/micro F_{max} rising from 0.3869/0.3714 to 0.3979/0.3838, 0.4029/0.3815, and 0.4061/0.3820 across Iterations 1–3), with EnzBERT single-predictor Top-1 at 52/148 (F_{max} 0.4571, dataset macro/micro F_{max} 0.4488/0.4419 → 0.4505/0.4427 on glucose and 0.4468/0.4394 at Iter 1 / 0.4433/0.4370 at Iter 3). When multi-predictor Stage 0 Noisy-OR pooling and Full GSPA Iter 3 (`+ GSPA Multi-Pred Iter 3`) are applied across the predictors' whole-genome outputs, Top-1 accuracy and F_{max} increase over every individual baseline: CLEAN rises to 66/148 (44.6%, $F_{max} = 0.6084$, dataset macro/micro F_{max} = 0.4391/0.4160 [0.4404/0.4174 at Noisy-OR], two-sided exact McNemar $p = 0.0703$), MAPred rises to 73/148 (49.3%, $F_{max} = 0.6178$, dataset macro/micro F_{max} = 0.4834/0.4625 [0.4514/0.4366 at Noisy-OR], $p = 2.9 \times 10^{-4}$), DeepProZyme rises to 59/148 (39.9%, F_{max} = 0.6136, dataset macro/micro F_{max} = 0.4379/0.4250 [0.4223/0.4051 at Noisy-OR], $p = 0.0213$), EnzBERT rises to 62/148 at Stage 0 Noisy-OR ($F_{max} = 0.5535$, dataset macro/micro F_{max} = 0.4862/0.4755, $p = 0.00391$) and 54/148 at Iter 3 (F_{max} = 0.5314, dataset macro/micro F_{max} = 0.4817/0.4656), and GraphEC rises from 18 to 48/148 (32.4%, F_{max} = 0.4756, dataset macro/micro F_{max} = 0.3856/0.3655 at Noisy-OR and F_{max} = 0.4621, dataset macro/micro $F_{max} = 0.3062/0.3047$ at Iter 3 [0.3419/0.3268 on multi-predictor glucose], $p = 1.9 \times 10^{-9}$), while TopEC (5/148, covering 820 EC classes) retains 5 Top-1 hits and increases dataset macro/micro F_{max} from 0.0348/0.0345 to 0.0402/0.0396. Stratified by role on CLEAN (`results/pillar2_price149_by_stratum.tsv`), single-predictor GSPA lifts Stratum A ($n = 91$) Top-1 from 46 to 47 (dataset macro F_{max} 0.4740 → 0.5095) and Stratum B ($n = 55$) Top-1 from 13 to 15 (23.6% → 27.3%; stoichiometric probe F_{max} 0.3580 → 0.3753), while multi-predictor GSPA lifts Stratum A Top-1 to 52 (F_{max} = 0.6336 → 0.7019, macro $F_{max} = 0.4740 \rightarrow 0.5288$) and Stratum B F_{max} from 0.3580 to 0.4546 (macro F_{max} 0.3103 → 0.3436).

2.4 Reproduction of experimental bacterial phenotypes and enzyme activities in BacDive (Suites 7 and 8)

To evaluate external biological accuracy on experimental strain characterization data, we compared GSPA against automated metabolic reconstruction pipelines (CarveMe[56], ModelSEED[54], gapseq[57], and negative-cost MILP gap-filling METEOR[63]) and supervised genome-to-phenotype classifiers (Traitar L1-SVM[81], the Random Forest phenotype baseline of Koblitz et al.[80] [`BacDive`

Table 4. Large-scale phenotypic and enzymatic validation in BacDive (Suites 7 and 8; source: `results/pillar3_bacdive_track1_summary.tsv`, `results/pillar3_bacdive_track2_summary.tsv`, `results/pillar3_bacdive_per_fold_cv.tsv`, and `results/pillar3_bacdive_enzyme_per_ec.tsv`). Track 1 separates 100% blind unsupervised prediction (zero phenotype training labels) from supervised 5-fold genus-blocked cross-validation (`[Genus-CV]`, where no genus in a test fold appears in training; pooled out-of-fold metrics shown in table, 5-fold mean ± s.d. in text). Track 2 reveals 30% of each strain's assays to calibrate strain/EC posteriors and evaluates strictly on the held-out 70% assays. Because CarveMe, ModelSEED, and gapseq emit discrete 0/1 reaction calls, their ROC-AUC equals single-point balanced accuracy ($\frac{1}{2}$(TPR + TNR)), whereas GSPA and supervised classifiers emit continuous scores evaluated by threshold-swept ROC-AUC. Runtime reports per-genome execution after base reconstruction/feature extraction (end-to-end runtime including all 3 base models in parentheses).

Evaluation Track / Panel	Method / Predictor	Assays (n)	Precision	Recall	F_1 Score	Balanced Acc.	ROC-AUC	Runtime / Genome
Track 1 Blind: Carbon Growth (529 Strains)	CarveMe[56] [Blind]	2,332	0.6935	0.4073	0.5132	0.6233	0.6233 (bin.)	45.0 s
	ModelSEED[54] [Blind]	2,332	**0.8576**	0.5200	0.6474	0.7214	0.7214 (bin.)	180.0 s
	gapseq[57] [Blind]	2,332	0.8071	**0.7836**	**0.7952**	**0.8082**	0.8082 (bin.)	420.0 s
	GSPA Track 1 [100% Blind, Zero Labels]	2,332	0.8071	**0.7836**	**0.7952**	**0.8082**	**0.8567**	**6.1 s (651.1 s)**
Track 1 Supervised: Carbon Growth [Genus-CV]	BacDive RF Baseline[79,80] [Genus-CV]	2,332	0.8433	0.8464	0.8448	0.8530	0.9222	1.8 s
	DiASPora GBDT[80] [Genus-CV]	2,332	0.8475	0.8491	0.8483	0.8564	0.9268	1.8 s
	Traitar L1-SVM[81] [Genus-CV]	2,332	0.8472	**0.8673**	0.8571	0.8638	0.9185	1.8 s
	GSPA + DiASPora Hybrid [Genus-CV]	2,332	**0.8766**	0.8655	**0.8710**	**0.8783**	**0.9567**	**6.1 s (652.9 s)**
Track 1 Blind: Enzymes (3,017 Type Strains)	CarveMe[56] [Blind]	17,496	0.7739	0.2996	0.4319	0.5774	0.5774 (bin.)	45.0 s
	ModelSEED[54] [Blind]	17,496	0.7854	0.4779	0.5943	0.6310	0.6310 (bin.)	180.0 s
	gapseq[57] [Blind]	17,496	**0.8393**	**0.7204**	**0.7753**	**0.7461**	0.7461 (bin.)	420.0 s
	GSPA Track 1 [100% Blind, Zero Labels]	17,496	**0.8393**	**0.7204**	**0.7753**	**0.7461**	**0.7520**	**6.1 s (651.1 s)**
Track 1 Supervised: Enzymes [Genus-CV]	BacDive RF Baseline[79,80] [Genus-CV]	17,496	0.8391	0.8527	0.8458	0.7911	0.8593	1.8 s
	DiASPora GBDT[80] [Genus-CV]	17,496	0.8602	0.8499	0.8550	0.8108	0.8699	1.8 s
	Traitar L1-SVM[81] [Genus-CV]	17,496	0.8402	**0.8794**	0.8593	0.8014	0.8715	1.8 s
	GSPA + DiASPora Hybrid [Genus-CV]	17,496	**0.8652**	0.8776	**0.8713**	**0.8257**	**0.9097**	**6.1 s (652.9 s)**
Track 1: 10 Diagnostic ECs (6,313 Strains)	DeepProZyme vs. METEOR v6 / a025 / v7	50,075 (50,105)	0.6442 / 0.6194	0.8105 / 0.8640	0.7178 / 0.7215	0.6792 / 0.6639	0.6792 / 0.6639	18.0 / 313.8 s
Track 1: Urease `EC 3.5.1.5` (6,313 Strains)	DPZ vs. METEOR v6 (vs. **gapseq 17k**)	5,026 (871)	0.4520 / 0.4320	0.7891 / 0.8000	0.5748 / 0.5610 (**0.7626**)	0.7294 / 0.7184	0.7294 / 0.7184	18.0 / 313.8 s
Track 1: 108-Genome Catabolic (Glucose Mismatch)	DeepProZyme vs. METEOR v8 Minimal-Glucose	1,622	**0.6254** / 0.6051	0.7240 / **0.7559**	0.6711 / 0.6722	**0.5386** / 0.5101	**0.5716** / 0.5484	18.0 / 313.8 s
Track 2: Held-Out 70% Carbon (529 Strains)	CarveMe[56]	1,641	0.6927	0.3882	0.4975	0.6165	0.6165 (bin.)	45.0 s
	ModelSEED[54]	1,641	**0.8728**	0.5116	0.6451	0.7222	0.7222 (bin.)	180.0 s
	gapseq Track 1 (Unassisted)	1,641	0.8059	0.7789	0.7922	0.8049	0.8049 (bin.)	420.0 s
	GSPA Track 2 Bayesian Anchor (30% → 70%)	1,641	0.8362	**0.9447**	**0.8871**	**0.8889**	**0.9534**	**6.4 s (651.4 s)**
	Traitar L1-SVM[81] (30% → 70%)	1,641	0.8391	0.8380	0.8386	0.8466	0.9421	1.8 s
	BacDive RF Baseline[79,80] (30% → 70%)	1,641	0.8625	0.8625	0.8625	0.8692	0.9483	1.8 s
	DiASPora GBDT[80] (30% → 70%)	1,641	0.8727	0.8985	0.8854	0.8901	0.9575	1.8 s
	GSPA Track 2 + DiASPora Hybrid (30% → 70%)	1,641	**0.9048**	**0.9409**	**0.9225**	**0.9258**	**0.9788**	**6.4 s (653.2 s)**
Track 2: Held-Out 70% Enzymes (3,017 Strains)	CarveMe[56]	12,316	0.7791	0.3009	0.4341	0.5802	0.5802 (bin.)	45.0 s
	ModelSEED[54]	12,316	0.7868	0.4786	0.5952	0.6325	0.6325 (bin.)	180.0 s
	gapseq Track 1 (Unassisted)	12,316	**0.8376**	0.7188	0.7737	0.7446	0.7446 (bin.)	420.0 s
	GSPA Track 2 Bayesian Anchor (30% → 70%)	12,316	0.8067	**0.9084**	**0.8545**	**0.7749**	**0.8738**	**6.4 s (651.4 s)**
	BacDive RF Baseline[79,80] (30% → 70%)	12,316	0.8411	0.8575	0.8492	0.7953	0.8623	1.8 s
	DiASPora GBDT[80] (30% → 70%)	12,316	0.8541	0.8473	0.8507	0.8045	0.8691	1.8 s
	Traitar L1-SVM[81] (30% → 70%)	12,316	0.8301	**0.8854**	0.8569	0.7934	0.8728	1.8 s
	GSPA Track 2 + DiASPora Hybrid (30% → 70%)	12,316	**0.8638**	0.8819	**0.8727**	**0.8264**	**0.9080**	**6.4 s (653.2 s)**

`RF Baseline`], and DiASPora Gradient-Boosted Decision Trees[80]) across experimental carbon-source growth and diagnostic enzyme assays in the DSMZ BacDive database[79,82] (Suites 7 and 8; Figure 4d, Table 4, and Supplementary Table S6; per-fold CV metrics and runtimes in `results/pillar3_bacdive_per_fold_cv.tsv`; architectural details in Section 4.8).

First, across 50,075 strain–enzyme assay pairs spanning 10 diagnostic ECs in 6,313 strains (`results/pillar3_bacdive_track1_summary.tsv` and `results/pillar3_bacdive_enzyme_per_ec.tsv`), negative-cost minimal-glucose MILP gap-filling across three parameterizations (`METEOR v6` with default addition threshold $\tau_{\text{add}} = 0.20, n = 50{,}075$; `METEOR a025` with stricter $\tau_{\text{add}} = 0.25, n = 50{,}075$; and `METEOR v7` with relaxed core threshold including 30 additional borderline strain–EC pairs, $n = 50{,}105$) trades precision for recall: precision drops from 0.6442 (DeepProZyme baseline, recall = 0.8105, $F_1 = 0.7178$, balanced accuracy = 0.6792) to 0.6194–0.6285 (recall = 0.8285–0.8640, $F_1 = 0.7148$–0.7215, balanced accuracy dropping to 0.6631–0.6670). Similarly, on the 108-genome non-glucose catabolic panel ($n = 1{,}622$ assays, evaluated with `METEOR v8` using updated MetaNetX reaction mappings), minimal-glucose MILP gap-filling reduces precision from 0.6254 to 0.6051, balanced accuracy from 0.5386 to 0.5101, and ROC-AUC from 0.5716 to 0.5484 ($F_1 = 0.6711$ vs. 0.6722). On obligate heteromeric urease (`EC 3.5.1.5`, requiring UreA, UreB, and UreC/D/F/G subunits across $n = 5{,}026$ assayed strains in the 6,313-strain panel and $n = 871$ in the 3,017 type-strain panel), negative-cost MILP reduces precision from 0.4520 to 0.4320 and F_1 from 0.5748 to 0.5610 (BAcc = binary ROC-AUC = 0.7294 → 0.7184) by boosting isolated singleton subunits, whereas complex-aware pathway reconstruction (`gapseq` / `GSPA Track 1` on the type-strain panel) achieves $F_1 = 0.7626$.

Second, in Track 1 (2,332 experimental carbon-source growth assays across 529 strains and 17,496 diagnostic enzyme assays across 3,017 type strains), we separate 100% blind unsupervised prediction (zero phenotype training labels) from supervised 5-fold genus-blocked cross-validation (Table 4 and Supplementary Table S6). When precomputed binary reaction/pathway call matrices from `gapseq`, `ModelSEED`, and `CarveMe` are available, `GSPA Track 1` pools their calibrated log-odds with empirical cross-assay pathway co-occurrence coupling $\psi(s, y)$ in 6.1 s/genome (651.1 s/genome end-to-end when running all three base reconstructions from scratch; Section 4.8). Under 100% blind evaluation, `GSPA Track 1` preserves `gapseq`'s binary decision boundary at $\theta = 0.50$, matching `gapseq` on carbon-source binary calls (Precision = 0.8071, Recall = 0.7836, $F_1 = 0.7952$, BAcc = 0.8082, vs. $F_1 = 0.6474$ for ModelSEED and 0.5132 for CarveMe) and on the 17,496-assay blind

enzyme panel (Precision = 0.8393, Recall = 0.7204, $F_1 = 0.7753$, BAcc = 0.7461, vs. 0.5943 for ModelSEED and 0.4319 for CarveMe), while providing calibrated continuous confidence scores that lift threshold-swept ROC-AUC from 0.8082 (binary) to 0.8567 (+0.0485) on carbon growth and from 0.7461 to 0.7520 on blind enzymes. Under strict 5-fold genus-blocked cross-validation (`results/pillar3_bacdive_per_fold_cv.tsv` and `pillar3_bacdive_track1_summary.tsv`), combining GSPA's continuous posteriors with the supervised classifiers (`GSPA + DiASPora Hybrid`) improves pooled out-of-genus carbon-source F_1 from 0.8448 (BacDive RF; 5-fold mean 0.8329 ± 0.0659), 0.8483 (DiASPora alone; 5-fold mean 0.8408 ± 0.0390), and 0.8571 (Traitar; 5-fold mean 0.8467 ± 0.0438) to 0.8710 (5-fold mean 0.8616 ± 0.0495, Precision = 0.8766, BAcc = 0.8783, ROC-AUC = 0.9567) and improves diagnostic enzyme F_1 from 0.8458 (BacDive RF; 5-fold mean 0.8456 ± 0.0076), 0.8550 (DiASPora; 5-fold mean 0.8550 ± 0.0075), and 0.8593 (Traitar; 5-fold mean 0.8595 ± 0.0076) to 0.8713 (5-fold mean 0.8712 ± 0.0083, Precision = 0.8652, Recall = 0.8776, BAcc = 0.8257, ROC-AUC = 0.9097).

Third, in Track 2 (where 30% of each strain's assays are revealed to calibrate strain metabolic activity and co-occurring pathway posteriors, and evaluation is performed strictly on the held-out 70% assays), `GSPA Track 2 Bayesian Anchor` improves unsupervised held-out carbon-source F_1 (n = 1,641) from 0.7922 (gapseq and unassisted GSPA Track 1 on the same 70% split) to 0.8871 (Precision = 0.8362, Recall = 0.9447, BAcc = 0.8889, continuous ROC-AUC = 0.9534) and held-out enzyme F_1 (n = 12,316) from 0.7737 (gapseq) to 0.8545 (Precision = 0.8067, Recall = 0.9084, BAcc = 0.7749, continuous ROC-AUC = 0.8738). Combining Track 2 posteriors with DiASPora (`GSPA Track 2 + DiASPora Hybrid`) reaches $F_1 = 0.9225$ (Precision = 0.9048, Recall = 0.9409, BAcc = 0.9258, ROC-AUC = 0.9788) on held-out carbon sources and F_1 = 0.8727 (Precision = 0.8638, Recall = 0.8819, BAcc = 0.8264, ROC-AUC = 0.9080) on held-out diagnostic enzymes, outperforming DiASPora (0.8854 / 0.8507), BacDive RF (0.8625 / 0.8492), and Traitar (0.8386 / 0.8569; Table 4).

2.5 Obligate heteromeric protein complex completion (Suite 5)

We next evaluated completing obligate heteromeric protein complexes when individual subunits have twilight-zone homology while suppressing false-positive singleton subunit predictions (Suite 5; Figure 5a,b and Table 5; `results/pillar3_complex_lso.tsv`). Across curated heteromeric complexes from Complex Portal and GO (including proton-transporting ATP synthase `GO:0045259`, respiratory chain complex I `GO:0045271`, cytochrome bo_3/c oxidase, DNA-directed RNA polymerase, 30S/50S ribosomal subunits, and the SecYEG translocon), we performed 175 leave-subunit-out (LSO) trials across six prokaryotic proteomes (*E. coli* K-12 [$n = 30$], *B. subtilis* 168 [$n = 30$], *M. tuberculosis* H37Rv [$n = 30$], *P. aeruginosa* PAO1 [$n = 30$], *E. coli* O157:H7 [$n = 30$], and *Synechocystis* sp. PCC 6803 [$n = 25$]). In each trial, a true complex subunit's primary evidence was attenuated into the twilight zone ($q_e \in [0.22, 0.38]$) alongside five matched decoy proteins carrying comparable raw domain scores in unrelated genomic loci (6 candidates per trial; random-chance Hit@1 = 1/6 = 0.1667; Precision@0.5, Recall@0.5, and F_1@0.5 are evaluated per trial and macro-averaged across all trials).

Under baseline Noisy-OR pooling (`M1`), all 175 attenuated subunits remain below the $\theta = 0.50$ threshold (Recall@0.5 = 0.0000, F_1@0.5 = 0.0000, genome Complex Coherence = 0.3901), and Hit@1 among the 6 candidates is 0.5600 (95% CI [0.483, 0.635]). Applying the Stage 1 complex coherence prior alone (`M2`) detects that the partner subunits of the complex are present in the genome and boosts all domain-matching candidates equally, keeping Hit@1 at 0.5600 while lifting Recall@0.5 to 0.5029, Precision@0.5 to 0.1729, and per-trial F_1@0.5 to 0.2459. Coupling complex coherence with the Stage 1 operon neighborhood prior (`M3`) disambiguates the true co-operonic subunit from non-operonic decoys: across all 175 trials, Hit@1 rises from 0.5600 to 0.9371 (95% CI [0.890, 0.968]), Recall@0.5 reaches 0.9257, Precision@0.5 rises more than three-fold to 0.5951, and F_1@0.5 reaches 0.6680 (0.9310 Hit@1, 0.9103 Recall@0.5, 0.5886 Precision@0.5, and 0.6595 F_1@0.5 across the 145 non-*E. coli* K-12 trials; Table 5).

Comparing per-genome recovery across the six species highlights the complementary roles of operon context and Stage 3 agentic reasoning (Figure 5b). In *E. coli* K-12, *B. subtilis*, *M. tuberculosis*, *P. aeruginosa*, and *E. coli* O157:H7, most heteromeric complex subunits reside in contiguous operons, allowing `M3` to achieve F_1@0.5 between 0.6862 (*P. aeruginosa*) and 0.7279 (*E. coli* O157:H7; 0.7089 in *E. coli* K-12, 0.7090 in *B. subtilis*, and 0.7244 in *M. tuberculosis*). In the cyanobacterium *Synechocystis* sp. PCC 6803, subunits of photosystem I/II, phycobilisomes, and NADH dehydrogenase are frequently split across dispersed chromosomal loci: baseline `M1` achieves 0.5200 Hit@1 (Complex Coherence = 0.3587), `M2` reaches 0.4800 Recall@0.5 and 0.2507 F_1@0.5 (Complex Coherence = 0.6125), and operon context (`M3`) reaches 0.6400 Hit@1 and 0.4080 F_1@0.5. Activating Stage 2 singleton pruning and the Stage 3 neuro-symbolic agent (`M4`) under symmetric zero-leakage candidate formatting elevates *Synechocystis* Hit@1 from 0.6400 to 0.6800 (17/25), Recall@0.5 from 0.6400 to 0.6800, Precision@0.5 from 0.3393 to 0.3793, and F_1@0.5 from 0.4080 to 0.4480 (+0.0400 gain), bringing overall 6-genome subunit recovery to 0.9429 Hit@1 (95% CI [0.898, 0.972]), 0.9314 Recall@0.5, 0.6009 Precision@0.5, and 0.6737 F_1@0.5 (0.9379 Hit@1, 0.9172 Recall@0.5, 0.5955 Precision@0.5, and 0.6664 F_1@0.5 on

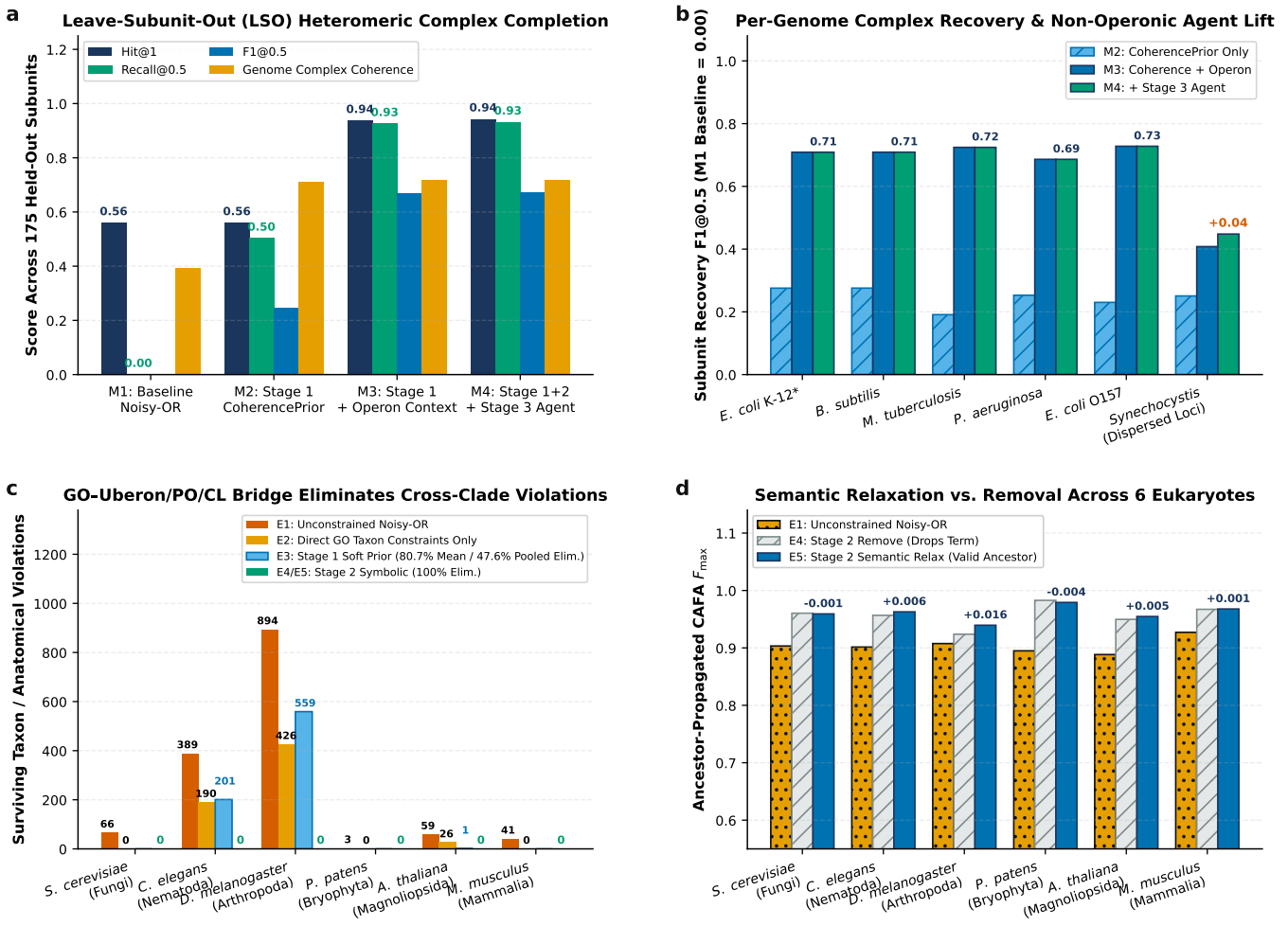


Figure 5. Obligate heteromeric protein complex completion and multicellular eukaryotic anatomical constraint enforcement (Suites 2 and 5). (**a**) Leave-subunit-out (LSO) recovery across 175 held-out heteromeric complex subunits in six prokaryotic genomes (chance Hit@1 = 0.167 over 6 candidates per case: 1 held-out subunit + 5 decoys), showing Hit@1, Recall@0.5, F_1@0.5, and Genome Complex Coherence across four configurations (`M1`–`M4`). (**b**) Per-genome subunit recovery F_1@0.5 across six prokaryotic species ($*$ marks the *E. coli* K-12 calibration reference), showing the +0.040 F_1 gain from the zero-leakage Stage 3 neuro-symbolic agent on *Synechocystis* sp. PCC 6803 where photosynthetic and respiratory complex subunits are dispersed across non-adjacent genomic loci. (**c**) Surviving cross-clade taxonomic and anatomical violations across six eukaryotic proteomes under unconstrained Noisy-OR (`E1`: 1,452 stress-test / 1,009 endogenous violations), direct GO taxon constraints only (`E2`: 55.8% stress-test elimination), Stage 1 soft anatomical prior (`E3`: 80.7% unweighted species mean / 47.6% pooled elimination), and Stage 2A deterministic enforcement (`E4`/`E5`: 100% elimination). (**d**) Ancestor-propagated transfer-evaluated micro $F_{\max}$ against unmodified Swiss-Prot ground truth across all six eukaryotes comparing unconstrained Noisy-OR (`E1`), Stage 2A naive removal (`E4`), and Stage 2A semantic ancestor relaxation (`E5`).

the 5 non-*E. coli* K-12 genomes).

2.6 Multicellular eukaryotic anatomical and developmental constraints (Suite 2)

We next evaluated anatomical and developmental constraint enforcement across six eukaryotic species spanning unicellular fungi (*Saccharomyces cerevisiae*), non-chordate invertebrates (*Caenorhabditis elegans* and *Drosophila melanogaster*), non-vascular bryophyte mosses (*Physcomitrium patens*), vascular flowering plants (*Arabidopsis thaliana*), and vertebrates (*Mus musculus*), evaluating both (i) the unperturbed endogenous homology-transfer proteome (6,155 curated proteins across the 6 species; `results/pillar3_eukaryote_endogenous_vs_injected.tsv`) and (ii) a cross-clade perturbation stress-test adding 1,546 cross-clade developmental claims across 7,655 proteins (300 in 5 species and 46 in *P. patens*), scored strictly against unmodified Swiss-Prot ground-truth annotations (Figure 5c,d and Table 6; `results/pillar3_eukaryote_anatomy.tsv` and `results/pillar3_relax_vs_remove.tsv`). On the unperturbed endogenous pro-

Table 5. Leave-subunit-out (LSO) recovery of 175 obligate heteromeric protein complex subunits across six prokaryotic reference genomes (Suite 5; source: `results/pillar3_complex_lso.tsv`). Each trial evaluates 1 true attenuated subunit ($q_e \in [0.22, 0.38]$) against 5 matched non-complex decoy proteins carrying comparable raw domain scores and symmetric zero-leakage formatting in unrelated genomic loci (6 candidates per case; random-chance Hit@1 = 1/6 = 0.1667; Precision@0.5, Recall@0.5, and F_1@0.5 are computed per trial and macro-averaged across trials). Middle rows report the 5-genome out-of-sample weighted mean ($n = 145$) excluding the hyperparameter development genome *E. coli* K-12.

Genome Scope	Method Configuration	Subunits (n)	Hit@1 (Chance = 0.167)	Recall@0.5	Precision@0.5	F_1@0.5	Complex Coh.
All 6 Genomes Combined	`M1`: Baseline Noisy-OR	175	0.5600	0.0000	0.0000	0.0000	0.3901
	`M2`: Stage 1 Complex Coherence Prior Only	175	0.5600	0.5029	0.1729	0.2459	0.7107
	`M3`: Stage 1 Coherence + Operon Context	175	0.9371	0.9257	0.5951	0.6680	**0.7170**
	`M4`: Stage 1 + Stage 2 + Stage 3 Agent	175	**0.9429**	**0.9314**	**0.6009**	**0.6737**	**0.7170**
5 Non-*E. coli* K-12 Genomes	`M3`: Stage 1 Coherence + Operon Context	145	0.9310	0.9103	0.5886	0.6595	**0.7011**
	`M4`: Stage 1 + Stage 2 + Stage 3 Agent	145	**0.9379**	**0.9172**	**0.5955**	**0.6664**	**0.7011**
Synechocystis sp. PCC 6803	`M1`: Baseline Noisy-OR	25	0.5200	0.0000	0.0000	0.0000	0.3587
	`M2`: Stage 1 Complex Coherence Prior Only	25	0.5200	0.4800	0.1813	0.2507	**0.6125**
	`M3`: Stage 1 Coherence + Operon Context	25	0.6400	0.6400	0.3393	0.4080	**0.6125**
	`M4`: Stage 1 + Stage 2 + Stage 3 Agent	25	**0.6800**	**0.6800**	**0.3793**	**0.4480**	**0.6125**

Table 6. Anatomical and developmental constraint enforcement, semantic ancestor relaxation, and active-structure developmental rescue across six diverse eukaryotes under both cross-clade perturbation stress-test (7,655 proteins) and unperturbed endogenous-only (6,155 proteins) regimes evaluated against unmodified Swiss-Prot ground truth (Suite 2; source: `results/pillar3_eukaryote_anatomy.tsv`, `results/pillar3_relax_vs_remove.tsv`, and `results/pillar3_eukaryote_endogenous_vs_injected.tsv`). Violation columns report surviving formal violations under unconstrained Noisy-OR (`E1`: stress-test violations with unperturbed endogenous violations in parentheses), direct GO taxon constraints only (`E2`), Stage 1 soft anatomical log-odds prior (`E3`), and Stage 2A deterministic enforcement (`E4` remove / `E5` relax). Ancestor-propagated transfer-evaluated micro $F_{\max}$, Recall, and Mean Information Content (IC, exact bits from `pillar3_relax_vs_remove.tsv`) compare `E1`, `E4`, and `E5`; bottom rows report the 6-species stress-test mean and the unperturbed endogenous-only 6-species CAFA $F_{\max}$ mean.

Species (Clade)	Taxon ID	E1 Viol. (Endog.)	E2 Direct TC Viol.	E3 Stage 1 Viol.	E4/E5 Viol.	E1 Noisy-OR $F_{\max}$ (Rec.)	E4 Remove $F_{\max}$ (Rec.)	E5 Relax $F_{\max}$ (Rec.)	$\Delta F_{\max}$ E5−E1 (E5−E4)	E1 / E4 / E5 Mean IC	Dev. Boost
S. cerevisiae (Fungi)	`NCBITaxon_4932`	66 (3)	0 (100.0%)	0 (100.0%)	**0** (100%)	0.9033 (0.9966)	**0.9604** (0.9249)	0.9502 (**0.9349**)	+0.0559 (−0.0012)	12.48 / 12.02 / **12.07**	2/3 (66.7%)
C. elegans (Nematoda)	`NCBITaxon_6239`	389 (289)	190 (51.2%)	201 (48.3%)	**0** (100%)	0.9015 (0.9720)	0.9567 (0.9268)	**0.9628 (0.9476)**	+0.0613 (+0.0061)	12.52 / 12.15 / 12.09	44/66 (66.7%)
D. melanogaster (Arthropoda)	`NCBITaxon_7227`	894 (717)	426 (52.3%)	559 (37.5%)	**0** (100%)	0.9076 (0.9378)	0.9238 (0.8608)	**0.9395 (0.8913)**	+0.0319 (+0.0157)	12.86 / 12.55 / 12.35	140/198 (70.7%)
P. patens (Bryophyta)	`NCBITaxon_3218`	3 (0)	0 (100.0%)	0 (100.0%)	**0** (100%)	0.8949 (0.9986)	**0.9828** (0.9697)	0.9792 (**0.9796**)	+0.0843 (−0.0036)	10.10 / 9.64 / **9.79**	0/0 (none in [0.22, 0.49])
A. thaliana (Magnoliopsida)	`NCBITaxon_3702`	59 (0)	26 (55.9%)	1 (98.3%)	**0** (100%)	0.8884 (0.9699)	0.9497 (0.9054)	**0.9547 (0.9232)**	+0.0663 (+0.0050)	11.76 / 11.13 / **11.18**	62/69 (89.9%)
M. musculus (Mammalia)	`NCBITaxon_10090`	41 (0)	0 (100.0%)	0 (100.0%)	**0** (100%)	0.9271 (0.9534)	0.9672 (0.9478)	**0.9678 (0.9545)**	+0.0407 (+0.0006)	13.01 / 12.84 / **12.85**	254/282 (90.1%)
Stress-Test 6-Eukaryote Mean (7,655 proteins; 1,452 viol.)		242.0 (1,009 endog.)	107.0 (55.8%)	126.8 (80.7% sp.)	**0** (100%)	0.9038 (0.9714)	0.9568 (0.9226)	**0.9605 (0.9385)**	+0.0567 (+0.0037)	12.12 / 11.72 / **11.72**	502/618 (81.2%)
Unperturbed Endogenous-Only 6-Eukaryote Mean (6,155 proteins)		168.2 (1,009 tot.)	n/a	n/a	**0** (100%)	0.7096 (CAFA)	0.7109 (Iter 3)	**0.7117 (Iter 1)**	+0.0021 (+0.0013 I3)	n/a	n/a

teomes alone, unconstrained Noisy-OR (`E1`) already produces 1,009 formal cross-clade anatomical and taxonomic violations transferred from mammalian or angiosperm homologs (3 in *S. cerevisiae*, 289 in *C. elegans*, and 717 in *D. melanogaster*; 0 in *P. patens*, *A. thaliana*, and *M. musculus*; mean 168.2 violations/species); GSPA eliminates 100% (1,009/1,009) of these endogenous violations and increases endogenous 6-species mean CAFA $F_{\max}$ from 0.7096 (`E1`) to 0.7117 at Iter 1 (+0.0021 mean gain at Iter 1: +0.0086 in *S. cerevisiae* [0.6644 → 0.6730], +0.0032 in *C. elegans* [0.5797 → 0.5829], +0.0010 in *D. melanogaster* [0.5863 → 0.5873], +0.0010 in *P. patens* [0.8841 → 0.8851], +0.0011 in *A. thaliana* [0.6594 → 0.6605], and −0.0020 in *M. musculus* [0.8835 → 0.8815]; 0.7109 6-species mean at Iter 3 with Stage 2A relaxation). Across the combined stress-test panel (1,452 total baseline violations at $\theta = 0.50$: 66 in *S. cerevisiae*, 389 in *C. elegans*, 894 in *D. melanogaster*, 3 in *P. patens*, 59 in *A. thaliana*, and 41 in *M. musculus*), applying direct GO taxon constraints alone (`E2`, using the 26,911 rules in `go-taxon-constraints.tsv`) eliminates 810/1,452 (55.79%) of violations while leaving 642/1,452 (44.21%) unresolved (190 in *C. elegans*, 426 in *D. melanogaster*, and 26 in *A. thaliana*), because those errors involve vertebrate, arthropod, or angiosperm organ terms whose taxonomic restrictions are defined in Uberon, PO, and CL. Applying the Stage 1 soft anatomical prior alone (`E3`, $-\alpha_{\text{anat}} = -3.5$ log-odds shift) eliminates 100% of violations in three species and 98.3% (58/59) in *A. thaliana*, while leaving 201 high-confidence violations in *C. elegans* (48.3% eliminated) and 559 in *D. melanogaster* (37.5% eliminated; 80.68% unweighted species mean elimination across the 6 species, or 47.59% [691/1,452] pooled), where raw homology scores exceed $q_e \geq 0.95$, showing why Stage 2A deterministic enforcement (`E4` and `E5`, 0 surviving violations) is required after soft Bayesian updates.

Comparing unconstrained Noisy-OR (`E1`), Stage 2A naive removal (`E4`), and Stage 2A semantic ancestor relaxation (`E5`) on transfer-evaluated micro $F_{\max}$ against unmodified Swiss-Prot ground truth clarifies the exact regime in which ancestor relaxation outperforms naive deletion (Figure 5d and Table 6; `results/pillar3_relax_vs_remove.tsv`). Naive deletion (`E4`) strips ancestral GO branches whenever an overly specific cross-clade developmental term is transferred, causing recall to drop by 0.56 to 7.70 percentage points across the six species. Under semantic ancestor relaxation (`E5`), forbidden terms are relaxed to their nearest anatomically valid GO ancestor, retaining 95.1%–98.7% of unconstrained `E1` mean annotation Information Content (12.07 vs. 12.48 bits in *S. cerevisiae*, 12.09 vs. 12.52 in *C. elegans*, 12.35 vs. 12.86 in *D. melanogaster*,

9.79 vs. 10.10 in *P. patens*, 11.18 vs. 11.76 [95.1%] in *A. thaliana*, and 12.85 vs. 13.01 [98.7%] in *M. musculus*; Table 6). Across all four multicellular animals and vascular plants (*C. elegans*, *D. melanogaster*, *A. thaliana*, and *M. musculus*), where homologous developmental regulators genuinely participate in ancestral tissue and organ morphogenesis, `E5` outperforms `E4` in both micro $F_{\max}$ ($\Delta F_{\max}(\text{E5} - \text{E4}) = +0.0006$ in *M. musculus*, +0.0050 in *A. thaliana*, +0.0061 in *C. elegans*, and +0.0157 in *D. melanogaster*; 6-species mean $F_{\max}$ rising from 0.9038 [`E1`] and 0.9568 [`E4`] to 0.9605 [`E5`]) and Recall (+0.67 to +3.05 percentage points). Conversely, in unicellular yeast (*S. cerevisiae*) and non-vascular moss (*P. patens*), where injected organ-development claims land on metabolic or housekeeping proteins annotated to core cellular processes in unmodified Swiss-Prot, `E5` recovers higher Recall (+1.00 and +0.99 percentage points) and higher Mean IC (12.07 vs. 12.02 and 9.79 vs. 9.64 bits) than `E4`, while complete removal (`E4`) achieves higher precision and $F_{\max}$ (0.9604 vs. 0.9592 in yeast; 0.9828 vs. 0.9792 in moss). In addition, for true species-appropriate developmental annotations attenuated into the borderline regime ($P^{(0)} \in [0.22, 0.49]$), the active-structure developmental boost in Stage 1 (`E3`) rescues 502/618 (81.2%) of borderline developmental claims above $\theta = 0.50$.

2.7 Evaluation of the Stage 3 neuro-symbolic agentic layer and extremophile case study (Suite 8)

Finally, we evaluated the optional Stage 3 neuro-symbolic agentic completion layer (Suite 8) against deterministic Stage 1+2, a posterior threshold baseline (≥ 0.35 + Guardrail), and supervised non-agentic tabular classifiers (Logistic Regression, Random Forest, Gradient-Boosted Decision Trees, and Multi-Layer Perceptron trained via 5-fold cross-validation; Figure 6, Table 7, Supplementary Section S5, and Supplementary Tables S7 and S8; `results/pillar4_agent_evaluation.tsv`, `results/pillar4_ml_baselines.tsv`, and `results/pillar4_multiseed_summary.tsv`). Across the 88-proteome multi-kingdom benchmark, the Stage 3 anomaly triggers select a mean of 1.84% of proteins per proteome (interquartile range 1.12%–2.38%, < 2.5% across all four domains). To evaluate anomaly resolution under strictly controlled, de-identified conditions with masked gene symbols and standardized feature formatting, we constructed a 120-anomaly benchmark (30 pathway gaps, 30 incomplete complexes, 30 borderline essentiality gaps, and 30 borderline conflicts across 7 genomes: *E. coli* K-12 [$n = 29$], *P. aeruginosa* [$n = 18$], *B. subtilis* [$n = 17$], *M. tuberculosis* [$n = 16$], *S. cerevisiae* [$n = 16$], *C. elegans* [$n = 13$], and *A. thaliana* [$n = 11$]; `results/pillar4_agent_benchmark_120.jsonl`).
Every candidate protein in the benchmark has its gene symbol masked to `"uncharacterized"`, candidate order is deterministically hash-ordered (`md5(anomaly_id:protein_id)`), and all true-positive and decoy candidates expose symmetric domain and real co-strand proteome neighbor feature formatting. The 90 completeness anomalies comprise 54 true-positive promotable gaps (18 per category) and 36 matched-score true-negative decoys (12 per category) including hard feature-overlap cases where decoys share borderline posteriors (0.24–0.36) or moderate RLGC Bayes factors (1.5–4.2) with true gaps, while the 30 `BORDERLINE_CONFLICT` anomalies serve as a symbolic-verification unit test comprising 15 real cross-clade conflicts that require ancestor relaxation or demotion and 15 adversarial cross-clade traps that must be blocked or relaxed to a valid ancestor.

Deterministic Stage 1+2 alone (Config 1) leaves post-Stage-2 borderline gaps unpromoted (0/54 promotions, accuracy = 55.00%), while lowering the posterior decision threshold to 0.35 with the symbolic guardrail (`Thresh-0.35`) promotes 29 true gaps alongside 22 false positives (Precision = 0.5686, Recall = 0.5370, $F_1 = 0.5524$, Accuracy = 0.7083). Supervised tabular classifiers trained via 5-fold CV on the 3 pre-tool summary scalars (`[current_posterior, max_claim_score, suggestion_score]`) achieve $F_1 = 0.7500$ (Precision = 0.8571, Recall = 0.6667, 3/120 [2.5%] violations accepted) for Logistic Regression (`CV-LR3`) and $F_1 = 0.8283$ (Precision = 0.9111, Recall = 0.7593, Accuracy = 0.8583) for GBDT (`CV-GB3`), while reaching 94.0% (MRR = 0.9667) and 90.0% (MRR = 0.9383) Hit@1 on the 50 ambiguous LRO cases when trained via 5-fold CV on those cases' 3 pre-tool scalars. When supervised 5-fold CV classifiers are trained directly on all 8 post-tool numerical features (`[current_posterior, likelihood_logodds, final_logodds, max_prior_contrib, rlgc_bayes_factor, suggestion_score, max_claim_score, mean_claim_score]`) and paired with the deterministic `SymbolicGuardrail` (`CV-LR8+G`, `CV-RF8+G`, `CV-GB8+G`, and `CV-MLP8+G`), they establish a strong in-distribution supervised upper bound on binary promotion ($F_1 = 0.9216$–0.9720, Accuracy = 0.9333–0.9750, and LRO-50 Hit@1 = 92.0%–96.0% in < 0.01 s). These supervised classifiers predict binary promotions when trained on in-distribution labeled anomaly splits, whereas semantic ancestor relaxation on conflict cases requires structured ontological traversal (0–2 relaxations under tabular classification).

Operating zero-shot directly from the five genomic and ontological tools across a bounded multi-turn loop (≤ 6 turns, mean 5.74 tool calls, 32.53 s single-request HTTP latency or 4.07 s wall-clock time per anomaly at 8× asynchronous concurrency) with the symbolic ontology guardrail active (Config 3, `Qwen/Qwen3.8-27B-FP8`; `results/pillar4_resolutions_c3_guarded.jsonl`), the Stage 3 agent achieves single-seed (seed 42)

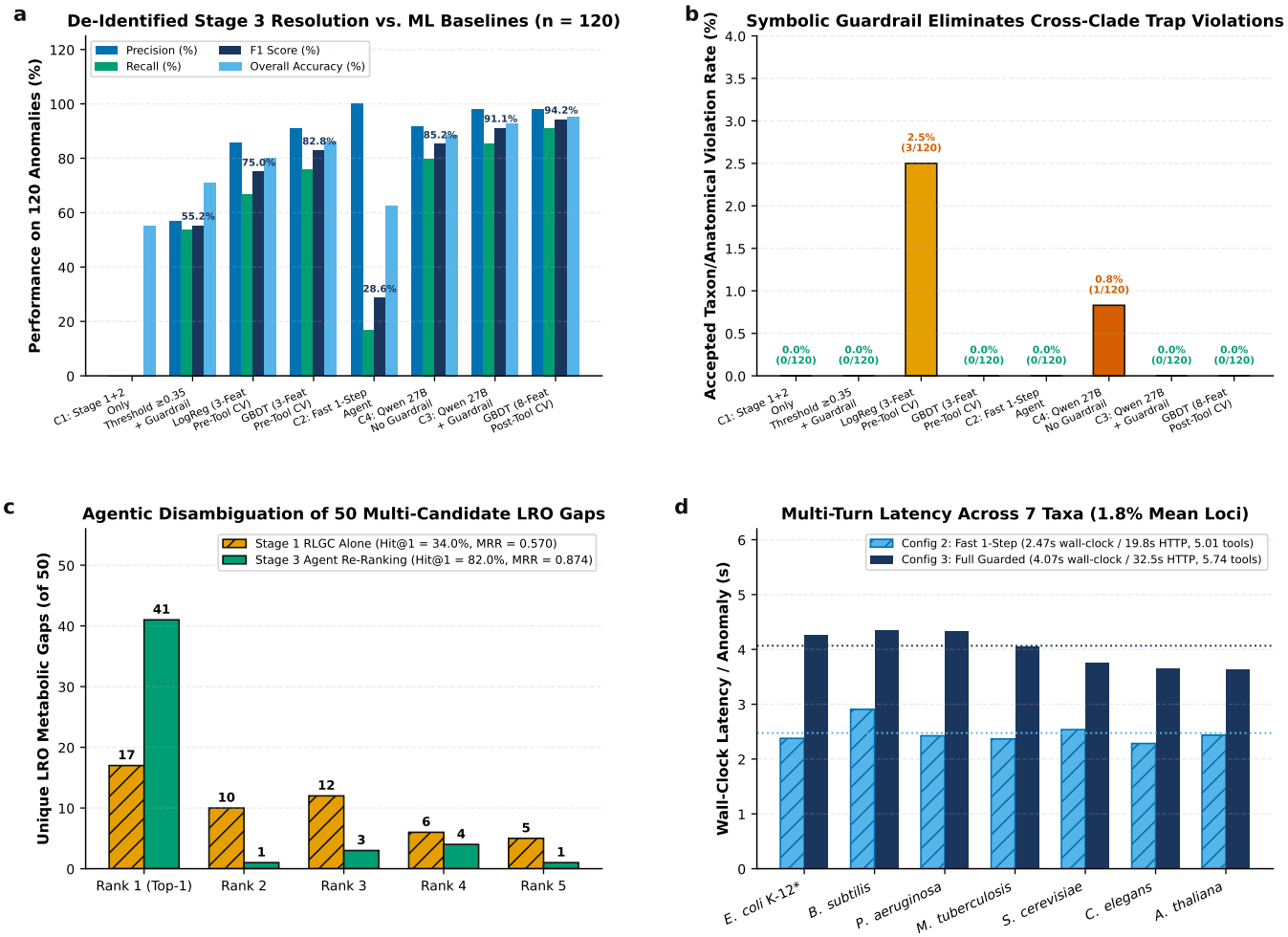


Figure 6. Evaluation and ablation of the optional Stage 3 neuro-symbolic agentic completion layer on the de-identified 120-anomaly and 50-unique-protein LRO benchmarks (Suite 8). (**a**) Precision, Recall, F_1 score, and overall resolution accuracy across 120 de-identified post-Stage-2 genome-scale anomalies (30 pathway gaps, 30 incomplete complexes, 30 borderline essentiality gaps, and 30 borderline conflicts) comparing deterministic Stage 1+2 only (Config 1), posterior threshold ≥ 0.35 + Guardrail, 3-scalar and 8-feature supervised 5-fold CV tabular baselines (LogReg, RF, GBDT, MLP), Fast 1-Step single-turn mode (Config 2), `Qwen/Qwen3.8-27B-FP8` with the symbolic guardrail ablated (Config 4), and full multi-turn `Qwen/Qwen3.8-27B-FP8` with the symbolic ontology guardrail (Config 3, single seed and 3-seed mean ± s.d.). (**b**) Accepted taxonomic and anatomical violation rate across the 120 anomalies, showing that unguarded tabular classifiers and unguarded LLMs accept 0.83%–2.50% violations whereas the symbolic guardrail guarantees 0.0% (0/120). (**c**) Rank distribution of the true enzyme across 50 unique ambiguous multi-candidate LRO metabolic gaps before and after Stage 3 agentic re-ranking (Hit@1 rises from 34.0% [17/50] to 82.0% [41/50] at seed 42 and 82.7% ± 0.9% across 3 seeds, MRR from 0.5700 to 0.8740 / 0.8792). (**d**) Mean concurrency-normalized wall-clock time per anomaly (4.07 s at 8× async concurrency; 32.53 s single-request HTTP latency) across seven prokaryotic and eukaryotic species.

Precision = 97.87% (46/47), Recall = 85.19% (46/54), $F_1 = 0.9109$, and overall resolution accuracy = 92.50% (111/120: 46/54 TP promotions [1 decoy candidate promotion counted as FP, 1 `RELAX`, and 6 `ABSTAIN` including 2 guardrail rejections], 35/36 decoy abstentions/demotions [25 `ABSTAIN`, 10 `DEMOTE`, and 1 `RELAX`], and 30/30 conflict/trap non-promotions across the 30 `BORDERLINE_CONFLICT` cases [13 valid ancestral GO relaxations (2 on real conflicts and 11 on cross-clade traps, bringing total `RELAX` actions across the 120 anomalies to 15) plus 15 `DEMOTE` and 2 `ABSTAIN`], with 0/120 = 0.0% accepted ontology violations), and across 3 independent random seeds (seed $\in \{42, 43, 44\}$; Supplementary Table S7 and `results/pillar4_multiseed_summary.tsv`) achieves Precision = 0.9722 ± 0.0092, Recall = 0.8581 ± 0.0087, F_1 = 0.9115 ± 0.0008, Accuracy = 0.9278 ± 0.0039, 13.7 ± 1.2 total `RELAX` actions (15 at seed 42, 12 at seed 43, and 14 at seed 44), and 0.00 ± 0.00% accepted ontology violation rate (Table 7). Zero-shot Config 3 outperforms the pre-tool 3-scalar CV classifiers on the 120-anomaly benchmark (F_1 = 0.9115 vs. 0.7500–0.8283) and ap-

Table 7. Ablation of the optional Stage 3 neuro-symbolic agent (`Qwen/Qwen3.8-27B-FP8`) against deterministic, threshold, and supervised 5-fold CV tabular baselines across 120 de-identified post-Stage-2 genome-scale anomalies and 50 ambiguous LRO metabolic gaps (Suite 8; source: `results/pillar4_agent_evaluation.tsv`, `results/pillar4_ml_baselines.tsv`, and `results/pillar4_multiseed_summary.tsv`). All candidate gene symbols are masked to `"uncharacterized"`, candidate order is alternated, and domain/neighbor strings expose symmetric zero-leakage InterPro/Pfam and GO/EC identifiers with real proteome accessions. The 120 anomalies (30 per category across 7 genomes) comprise 54 true-positive promotable gaps (18 per completeness category), 36 matched-score true-negative decoys (12 per category), and 30 borderline conflict cases (15 real cross-clade conflicts and 15 adversarial cross-clade traps; overall resolution accuracy credits any non-promotion [`RELAX`, `DEMOTE`, or `ABSTAIN`] on real conflicts and valid `RELAX`/`DEMOTE`/`ABSTAIN` on traps, while **TP Prom. / Relax** reports true-positive promotions [`tp_promoted`] and total valid ancestral GO relaxations [`relaxations_count`] across the 120 anomalies). Latency reports concurrency-normalized wall-clock seconds per anomaly at 8× async concurrency (with mean per-request HTTP latency in parentheses).

Config	Stage 3 Architecture / Baseline	TP Prom. / Relax	Precision	Recall	F_1 Score	Accuracy	Viol. Accepted	LRO-50 Hit@1 (MRR)	Tool Calls	Wall-Clock (HTTP)
Config 1	Stage 1 + Stage 2 Deterministic Only	0 / 0	0.0000	0.0000	0.0000	0.5500	**0 / 120 (0.0%)**	34.0% (0.5700)	0.00	0.00 s (0.00 s)
Thresh-0.35	Posterior Threshold (≥ 0.35) + Symbolic Guardrail	29 / 30	0.5686	0.5370	0.5524	0.7083	**0 / 120 (0.0%)**	34.0% (0.5700)	0.00	< 0.01 s
CV-LR3	Logistic Regression (5-fold CV, 3 Pre-Tool Scalars, Guardrail Ablated)	36 / 0	0.8571	0.6667	0.7500	0.8000	3 / 120 (2.5%)	94.0% (0.9667)	0.00	< 0.01 s
CV-GB3	GBDT (5-fold CV, 3 Pre-Tool Scalars, Guardrail Ablated)	41 / 0	0.9111	0.7593	0.8283	0.8583	**0 / 120 (0.0%)**	90.0% (0.9383)	0.00	< 0.01 s
Config 2	Fast 1-Step Agent (`--max-turns 2`, Guardrail ON)	9 / 4	**1.0000**	0.1667	0.2857	0.6250	**0 / 120 (0.0%)**	n/a	5.01	2.47 s (19.80 s)
Config 4	`Qwen/Qwen3.8-27B-FP8` (Guardrail Ablated)	43 / 0	0.9149	0.7963	0.8515	0.8833	1 / 120 (0.83%)	n/a	5.07	3.72 s (29.75 s)
Config 3	**`Qwen/Qwen3.8-27B-FP8` + Symbolic Guardrail (Seed 42)**	**46 / 15**	0.9787	0.8519	0.9109	0.9250	**0 / 120 (0.0%)**	**82.0% (0.8740)**	5.74	4.07 s (32.53 s)
Config 3 (3-Seed)	**`Qwen/Qwen3.8-27B-FP8` + Guardrail (3-Seed Mean ± s.d.)**	**46.3 / 13.7**	**0.9722 ± 0.0092**	**0.8581 ± 0.0087**	**0.9115 ± 0.0008**	**0.9278 ± 0.0039**	**0 / 120 (0.0%)**	**82.7 ± 0.9% (0.8792)**	5.74	4.07 s (32.53 s)
CV-LR8+G	Logistic Regression (5-fold CV, 8 Post-Tool Feat. + Guardrail)	47 / 1	0.9792	0.8704	0.9216	0.9333	**0 / 120 (0.0%)**	**96.0% (0.9767)**	0.00	< 0.01 s
CV-RF8+G	Random Forest (5-fold CV, 8 Post-Tool Feat. + Guardrail)	52 / 2	0.9286	0.9630	0.9455	0.9500	**0 / 120 (0.0%)**	92.0% (0.9517)	0.00	< 0.01 s
CV-GB8+G	GBDT (5-fold CV, 8 Post-Tool Feat. + Guardrail)	49 / 0	0.9800	0.9074	0.9423	0.9500	**0 / 120 (0.0%)**	92.0% (0.9500)	0.00	< 0.01 s
CV-MLP8+G	**MLP (5-fold CV, 8 Post-Tool Feat. + Guardrail, In-Dist. Upper)**	52 / 0	**0.9811**	**0.9630**	**0.9720**	**0.9750**	**0 / 120 (0.0%)**	92.0% (0.9507)	0.00	< 0.01 s

proaches the in-distribution 8-feature supervised CV upper bound ($F_1 = 0.9216$–0.9720) in a zero-shot setting. In Fast 1-Step mode (Config 2, ≤ 2 turns, 5.01 tool calls, 2.47 s wall-clock / 19.80 s HTTP latency), precision is 1.0000 (9/9) with recall at 0.1667 ($F_1 = 0.2857$). Ablating the symbolic guardrail (Config 4, 5.07 tool calls, 3.72 s wall-clock / 29.75 s HTTP latency) allows 1/120 (0.83%) cross-clade traps to be accepted and yields 0/15 ancestral GO relaxations on conflicts, lowering precision to 0.9149 (43/47), recall to 0.7963 (43/54), F_1 to 0.8515, and accuracy to 0.8833 (Figure 6b). On the 50 deduplicated unique-protein ambiguous LRO metabolic gaps in *E. coli* K-12 (`results/pillar4_lro_reranking.tsv`), where all gene symbols are masked to `"uncharacterized"`, candidate order is deterministically hash-ordered (`md5(anomaly_id:protein_id)`), and all 5 candidates per case expose symmetric domain and real proteome operon-neighbor metadata, zero-shot Stage 3 agentic re-ranking elevates Hit@1 from 34.0% (17/50, RLGC baseline) to 82.0% (41/50 at seed 42; two-sided exact binomial McNemar $p = 1.19 \times 10^{-7}$ on $b_{01} = 24, b_{10} = 0$ discordant unique proteins) and $82.67\% \pm 0.94\%$ across 3 seeds (84.0% [42/50] at seed 43 and 82.0% [41/50] at seed 44), Hit@3 from 78.0% (39/50) to 90.0% (45/50), and MRR from 0.5700 to 0.8740 (seed 42) and 0.8792 ± 0.0069 (3-seed mean; Figure 6c and Table 7). Finally, as an ecological case study on five desert extremophile isolates from the Empty Quarter (*Rub' al Khali*: *Salinicola* sp. [3,842 CDS], *Halomonas* sp. [4,118 CDS], *Pseudomonas* sp. [5,490 CDS], *Bacillus endophyticus* [4,625 CDS], and *Streptomyces* sp. [7,184 CDS]; Supplementary Table S8 and `results/pillar6_empty_quarter_vignette.tsv`), applying GSPA lifts mean core essential completeness from 0.8884 (raw homology) and 0.9190 (Stage 0) to 0.9718, increases pathway coherence from 0.4892 to 0.6226 (+13.34 percentage points), and identifies 4–8 (mean 5.6) flux-certified pFBA core repairs and 4–9 (mean 6.2) Stage 3 candidate promotions per isolate across ectoine/betaine osmoprotection and oxidative stress pathways.

3 Discussion

Protein function prediction has traditionally been formulated as a per-protein classification problem evaluated on isolated sequences or structures[1,28,31]. Our evaluation across eight benchmark suites shows that while per-protein multi-channel calibration (Stage 0) establishes bulk protein-centric accuracy on well-annotated reference proteomes, treating the whole genome as the unit of inference (Stages 1–3) restores pathway connectivity, heteromeric complex completeness, and cross-ontology taxonomic and anatomical consistency across the proteome.

Four main conclusions emerge from our study. First, our audit of 108 bacterial metabolic reconstructions clarifies why negative-cost MILP gap-filling (including our prior METEOR formulation[63]) degrades gene-level enzyme precision on diagnostic and catabolic panels: assigning negative addition costs ($c_j^{\text{add}} < 0$) to candidate reactions above an addition threshold causes 83.72% (262.34/313.35, vs. 51.01 flux-carrying additions) of added reactions to carry zero steady-state flux, while uniform reaction-to-gene boosting inflates false-positive calls across paralogs. While any positive-cost LP formulation over the sanitized universal network $\mathcal{U}_{\text{clean}}$ eliminates uncertified zero-flux additions (81.31/81.31 flux-carrying additions under threshold LP repair and 5.70 pFBA-active additions per genome with 0.00 zero-flux additions by construction under two-step lexicographic pFBA, of which 2.29 are certified FVA-essential at $\leq 1.01\,C^*$ and 1.96 are certified knockout-essential at $\leq 1.10\,C^*$), combining role-aware DNF mechanism aggregation (Top-M_{iso} isozyme OR and bottleneck-gated geometric-mean subunit AND), two-step lexicographic pFBA repair, cognate substrate flux probing, and competitive locus responsibility against an explicit

unassigned-gene hypothesis $h_{\varnothing}$ eliminates the 2.23–2.36 dead-end active reactions per genome left by unregularized LP/MILP repair, improves `Price-149` enzyme Top-1 hits under both single-predictor refinement (CLEAN 60 → 63, MAPred 50 → 56, DeepProZyme 49 → 51 with stoichiometric local-context probe, GraphEC 18 → 20) and multi-predictor refinement (+0.0528 to +0.1987 $F_{\max}$), and improves `BacDive` phenotype predictions across blind, Track 2 phenotype-anchored, and genus-blocked evaluation tracks.

Second, our ablation of the leave-reaction-out (LRO) benchmark separates the contribution of protein language model embeddings from that of local genomic synteny. On the well-studied *E. coli* K-12 proteome (Track B), per-protein ESM-2 reaction-centroid similarity alone recovers 62.11% of masked reactions at Rank 1 (87.85% given the candidate pool), with Stage 1 pathway coherence and RLGC operon neighborhood density adding +1.56 percentage points unconditionally (+2.21 percentage points given pool, to 63.67% / 90.06%). Conversely, when PLM centroids are excluded or cosine-masked (Track A, cosine > 0.50) or when multiple candidate genes in a proteome share similar domain and embedding scores ($n = 50$ unique-protein ambiguous LRO gaps and $n = 175$ heteromeric complex LSO trials), unfiltered sequence similarity alone recovers 1.95% (LRO), 34.00% (ambiguous LRO), and 56.00% (LSO), strict cosine-masked Track A reaches 11.33% (17.97% with Track A GBDT), and unfiltered RLGC operon neighborhood coupling + Stage 1 coherence (without PLM centroids) reaches 23.44% on LRO, while Stage 3 neuro-symbolic disambiguation elevates Hit@1 to 82.67% (3-seed mean on ambiguous LRO) and 94.29% (LSO across 6 candidates per trial).

Third, our multi-kingdom eukaryotic evaluation shows that enforcing ontology consistency in multicellular organisms requires bridging the Gene Ontology with external anatomy ontologies (Uberon, PO, and CL). On unperturbed 7-channel predictions across six model eukaryotes, GSPA eliminates all 1,009 endogenous homology-transferred violations while improving mean CAFA $F_{\max}$ by +0.0021 (+0.0086 in *S. cerevisiae*). Under cross-clade perturbation stress-testing evaluated against unmodified Swiss-Prot ground truth (1,452 total baseline violations across fungi, nematodes, insects, mosses, flowering plants, and mammals), direct GO taxon constraints leave 44.21% (642/1,452) of errors unresolved and soft Stage 1 log-odds penalties still leave 201–559 high-score violations in invertebrate proteomes, whereas Stage 2A semantic relaxation over our 2,715 GO–Uberon/PO/CL cross-product axioms eliminates 100% of violations, retains 95.1%–98.7% of unconstrained mean IC, and outperforms naive deletion across all four multicellular animals and vascular plants ($\Delta F_{\max}(\text{E5} - \text{E4}) = +0.0006$ to $+0.0157$; Recall +0.67 to +3.05 percentage points), while recovering higher Recall and Mean IC than deletion in unicellular yeast and non-vascular moss.

Fourth, comparing the zero-shot Stage 3 neuro-symbolic agent (`Qwen/Qwen3.8-27B-FP8`) against both pre-tool 3-scalar and post-tool 8-feature supervised 5-fold CV classifiers on the de-identified 120-anomaly and 50-LRO benchmarks clarifies when an LLM agent is warranted over a lightweight tabular classifier. When in-distribution labeled anomaly data and pre-extracted tool features are available and only binary promotion or candidate ranking is needed, an 8-feature supervised classifier paired with `SymbolicGuardrail` achieves $F_1 = 0.9216$–0.9720 on the 120 anomalies and 92.0%–96.0% Hit@1 on LRO-50 in < 0.01 s. When labeled anomaly training data are unavailable for a new clade or when multi-class semantic ancestor relaxation is required on cross-clade conflicts, zero-shot multi-turn tool use with `SymbolicGuardrail` achieves $F_1 = 0.9115 \pm 0.0008$ and accuracy $= 0.9278 \pm 0.0039$ on the 120 anomalies and $82.67\% \pm 0.94\%$ Hit@1 on LRO-50 across 3 seeds, outperforming pre-tool 3-scalar CV classifiers on anomaly resolution ($F_1 = 0.7500$–0.8283), proposing 13.7/15 valid ancestral GO relaxations, and guaranteeing a 0.0% accepted ontology violation rate.

3.1 Limitations

Several methodological boundaries of the present study define priorities for future work. First, on well-annotated reference proteomes where roughly 85% of proteins carry high-confidence Swiss-Prot homologs ($P_0 \geq 0.90$), Stage 0 multi-channel pooling accounts for the bulk of overall protein-centric $F_{\max}$, and Stages 1–3 concentrate their updates on the $< 15\%$ of twilight-zone, pathway-hole, complex-subunit, and taxon-conflicted loci; similarly, when targets are supplied as isolated sequences dispersed across species (`CAFA6` and `Holdout-125`), inference relies on Stage 0 pooling, Stage 2A ontological enforcement, and Stage 3 domain/ontology reasoning, and linking target accessions to their source genome assemblies will activate full operon and stoichiometric context on prospective sequence benchmarks. Second, in the LRO metabolic gap-filling benchmark, unconditional enzyme recovery is bounded by Stage 0 candidate pool recall ($n_{\text{cov}} = 175$–181 of 256 reactions in *E. coli* and 76 of 144 in *B. subtilis*), and Track B uses frozen pretrained ESM-2 weights that had pretraining exposure to UniRef50 sequences; expanding Stage 0 candidate generation with remote structural fold matches and training strictly time-split protein language models will further separate de novo sequence representations from homology transfer. Third, Stage 2B stoichiometric locus refinement models steady-state reaction connectivity over reactions mapped in $\mathcal{U}_{\text{clean}}$ ($\mathcal{C}_{\text{struct}} = 1$) that carry flux under the probed medium ($\mathcal{C}_{\text{task}} = 1$) on unique single-copy steps ($\mathcal{C}_{\text{unique}} = 1$), whereas in vivo growth and diagnostic enzyme phenotypes in `Price-149` and `BacDive` also depend on transcriptional regulation, transporter specificity, and unmodeled cofactors; integrating transcriptomic expression priors and automated transporter substrate

models, while continuing to pair the internal GAEF structural diagnostic with held-out experimental validation, will broaden coverage of condition-specific and secondary pathways. Fourth, our cross-ontology anatomical bridge covers the 2,715 GO terms that carry explicit Uberon, PO, or CL logical definitions in the current GO release, and semantic ancestor relaxation improves precision-recall $F_{\max}$ primarily in multicellular animals and vascular plants whose ground-truth annotations include tissue-level developmental ancestors, whereas candidate pathway fills in the 97 uncultured MAGs and five Empty Quarter extremophile isolates represent computational hypotheses awaiting heterologous expression or mutant fitness assays. Fifth, while the zero-shot Stage 3 neuro-symbolic agent supports out-of-distribution anomaly disambiguation and ancestral GO relaxation in zero-shot settings, pre-extracted 8-feature supervised tabular classifiers paired with `SymbolicGuardrail` provide a 100× faster alternative for binary promotion when labeled anomaly training splits are available.

4 Methods

4.1 Overview of the four-stage architecture and notation

Let $\mathcal{P} = \{p_1, \ldots, p_N\}$ denote the protein-coding genes of a target genome with NCBI Taxonomy identifier τ_{tax}, genomic coordinates $\mathcal{K} = \{(c_i, s_i, e_i, \zeta_i)\}_{i=1}^{N}$ (contig c_i, start s_i, end e_i, and strand $\zeta_i \in \{+, -\}$), and let $\mathcal{T}$ denote the directed acyclic graph (DAG) of the Gene Ontology with true-path relation $t \sqsubseteq a$ indicating that term t is a subtype or part of ancestor a. GSPA estimates the proteome-wide annotation posterior over $\mathbf{Y} \in \{0,1\}^{N \times |\mathcal{T}|}$ conditioned on genomic coordinates $\mathcal{K}$, taxon τ_{tax}, metabolic network $\mathcal{M}$, and cross-ontology axioms $\mathcal{O}$ through four sequential stages (Figure 1; benchmark suite organization, Stage 0 channel configuration, and active stages per suite are summarized in Table 8).

4.2 Stage 0: Multi-channel per-protein evidence calibration and Noisy-OR pooling

GSPA can refine either a single standalone predictor (such as CLEAN[34], MAPred[37], DeepProZyme[35], EnzBERT[36], GraphEC[38], or TopEC[39]) or integrate claims from up to seven per-protein evidence channels $e \in \mathcal{E}_{\text{ch}}$: (1) sequence alignment via DIAMOND[19] against Swiss-Prot GOA entries; (2) profile HMM domain architecture via HMMER 3[20] against Pfam-A/Pfam2GO; (3) domain signatures via InterProScan v5[17]; (4) ESM-2 (650M) embedding kNN transfer[21]; (5) ProtT5-XL-U50 embedding kNN transfer[22]; (6) sequence-based neural classifiers (ProteInfer[32], DeepGOWeb[83], DeepGO-SE[31]); and (7) structural similarity transfer via FoldSeek[40] and ProstT5 3Di[23].

For each evidence claim c from tool $e(c) \in \mathcal{E}_{\text{ch}}$ with raw confidence $s(c) \in [0,1]$, `CalibrationTable.groovy` (`gspa.integration.CalibrationTable`) maps $s(c)$ to a calibrated probability $\tilde{q}(c) \in [\varepsilon_{\text{cal}}, 1 - \varepsilon_{\text{cal}}]$ ($\varepsilon_{\text{cal}} = 10^{-4}$) via Platt logistic calibration $\tilde{q}(c) = \text{clip}\big(\sigma_{\log}(a_{e(c)} s(c) + b_{e(c)}), \varepsilon_{\text{cal}}, 1 - \varepsilon_{\text{cal}}\big)$ with tool-specific default parameters (a_e, b_e) (`diamond/mmseqs2`: $(6.0, -3.0)$; `hmmer/pfam`: $(4.0, -1.5)$; `interproscan/foldseek/dbcan`: $(5.0, -2.0)$; `eggnog-mapper`: $(3.5, -1.3)$; `gapseq`: $(4.0, -1.8)$; `antismash`: $(4.0, -1.5)$; `signalp/deeptmhmm`: $(3.0, -1.0)$; `operon`: $(3.0, -1.2)$; `amrfinder`: $(8.0, -4.0)$; default $(4.0, -2.0)$; or identity clipping $\text{clip}(s(c), \varepsilon_{\text{cal}}, 1-\varepsilon_{\text{cal}})$ when `calibrate = false`, overrideable via JSON/TSV calibration files). Each calibrated score is scaled by the evidence-type reliability weight $r_{e(c)} = \text{defaultReliability}() \in (0,1]$ defined in `EvidenceCombiner.groovy` (`gspa.integration.EvidenceCombiner`: `DOMAIN_SPECIFIC_AMR` 0.85; `STRUCTURE_SIMILARITY` and `DOMAIN_SPECIFIC_CAZY` 0.80; `SEQUENCE_DOMAIN`, `STRUCTURE_DEEPLEARNING`, `DOMAIN_SPECIFIC_BGC`, and `DOMAIN_SPECIFIC_VF` 0.75; `SEQUENCE_SIMILARITY` and `ORTHOLOGY` 0.70; `SEQUENCE_DEEPLEARNING` and `PROTEIN_LM_EMBEDDING` 0.65; `GENOMIC_REGION_ML`, `SEQUENCE_REGION_ML`, `DARK_MATTER`, `REACTION_LOCAL_CONTEXT`, `CROSS_GENOME_TRANSFER`, and `ML_RANKER` 0.60; `GENOMIC_LANGUAGE_MODEL` and `METABOLIC_CONTEXT` 0.55; `SEQUENCE_MOTIF` and `LOCALIZATION` 0.50; `GENOMIC_CONTEXT` 0.45), yielding $q(c) = r_{e(c)} \tilde{q}(c)$. To prevent correlated channels from over-counting redundant evidence, `EvidenceCombiner.groovy` groups claims for (p, f) by `EvidenceType.correlationGroup()` into correlation groups $g \in \mathcal{G}_{\text{corr}}(p, f)$ (`homology` for sequence similarity, sequence domains, and orthology; `ml_protein_seq` for sequence DL and PLM embeddings; `structure` for structural similarity and structural DL; `context` for genomic and metabolic context; `ml_genomic` for gLMs; `inferred_context` for dark-matter, RLGC, cross-genome transfer, and ML ranker claims; `motif` for sequence motifs; `localization` for subcellular localization; `region_features` for `SEQUENCE_REGION_ML`; `viral` for `GENOMIC_REGION_ML`; and `domain_*` for domain-specific predictors), takes the group maximum $q_{g,(1)}(p,f) = \max_{c \in g} q(c)$ (with optional second-opinion bonus $\beta_{\text{so}} q_{g,(2)}(p,f)$ from the second-highest group member $q_{g,(2)}$, default $\beta_{\text{so}} = 0.0$), clips each group score to $[10^{-3}, 0.999]$, and pools across correlation groups via Noisy-OR:

$$q_g(p,f) = \text{clip}\Big(\max_{c \in g}\big(r_{e(c)}\tilde{q}(c)\big) + \beta_{\text{so}}\, q_{g,(2)}(p,f),\ 10^{-3},\ 0.999\Big), \qquad P_0($$

(1)

Converting $P_0(p,f)$ to log-odds and clipping to $[L_{\min}, L_{\max}] = [-12.0, 12.0]$ (`lMin = -12.0`, `lMax = 12.0` in `EvidenceCombiner.groovy`) yields

Table 8. Overview of the eight benchmark suites, nomenclature mapping, Stage 0 channel configuration, active GSPA stages, and canonical data files in `results/`. Full column-level provenance is documented in `results/PROVENANCE_MAP.md`.

Suite	Internal Pillar / Scenario	Cohort & Scale	Stage 0 Channels	Active GSPA Stages	Primary Comparators	Canonical results/ File(s)
Suite 1	Pillar 1 / Scen. 1A–1C	CAFA holdout (16,125), CAFA6 (769), EC-125, 95 genomes (69,666)	7-ch GO / 3-pred EC / 1-pred EC	Stage 0 + 1 + 2A (+ 3 on CAFA6)	ProteInfer, DeepGOZero, DeepGOPlus, CLEAN, DeepProZyme, EnsBERT, METEOR	`scenario1_*.tsv`, `suite1_cafa6_prospective_summary.tsv`
Suite 2	Pillars 1, 2, 4 / Scen. 2A–2C, C	9-gen. ablation (81 runs), 13/21 panels, 1,000 RefSeq, 88 4-king., 6 Enk.	3-ch (9/13/21/88) / 7-ch (21/1000/6 Enk.)	Stage 0 + 1 + 2A (A1–A9, E1–E8)	NCBI PGAP, InterProScan 5, metagenomic-deepFRI, DeepGO-SE, SPROF-GO, gLM2	`pillar1_2_ablation_summary.tsv`, `pillar1_suite2_*.tsv`, `pillar3_eukaryote_*.tsv`
Suite 3	Pillars 2, 3 / Scen. 3A–3B (Str. A)	108 models, 6 BiGG GEMs, *E. coli* LRO (256/307), *B. subtilis* (144)	3-ch + Pfam + ESM-2 centroids	Stage 1 (RLGC/M4) + Stage 2B (SLR)	CarveMe, Reconstructor, METEOR v8, Positive-Cost LP, ESM-2 PLM Centroids	`pillar2_invariant_audit_summary.tsv`, `pillar2_curated6_gems_real.tsv`, `pillar2_lro15_summary.tsv`
Suite 4	Pillar 3 / Scen. 4 (Stratum B)	Rule-stratified LRO Alternative Catabolic ($n = 40$)	3-ch + Pfam + ESM-2 centroids	Stage 1 (RLGC/M4) + Stage 2B (Cognate)	RLGC Kernel Density, Sequence-only ranking, METEOR v8	`pillar2_lro15_summary.tsv`
Suite 5	Pillar 3 / Scen. 5 (Str. C), B. MAGs	LRO Secondary ($n = 58$), 106 BGCs, 175 LSO complexes (91–94), 97 MAGs	3-ch + Pfam + ESM-2 centroids	Stage 1 + Stage 2A + Stage 3 Agent	Baseline Noise-OR, Stage 1 Coherence/Operon, Minimal-Glucose MILP	`pillar2_3_antismash_106bgc_specificity.tsv`, `pillar3_complex_lso.tsv`, `pillar3_dark_matter_97mags.tsv`
Suite 6	Pillar 2 / Scen. 6 (Prize-149)	148 host-mapped RB-TnSeq enzymes across 22 bacterial genomes (149 seqs)	1-pred & 5-pred ensemble	Single/Multi-pred + Stage 1 + Stage 2B	CLEAN, MAProd, DeepProZyme, EnsBERT, GraphEC, TopEC, METEOR, DCGM, ECrecer	`pillar2_prize149_all5_real.tsv`, `pillar2_prize149_by_stratum.tsv`
Suite 7	Pillar 3 / Scen. 7 (BacDive Trk 1)	529 carbon strains (2,322), 3,017 type strains (17,496), 6,313 strains (50,075)	3-ch + 3 GEM reconstructions	Stage 1+2 Multi-Model + DiASPora Hybrid	gapseq, ModelSEED, CarveMe, METEOR, Traitar, BacDive RF, DiASPora GBDT	`pillar3_bacdive_track1_summary.tsv`, `pillar3_bacdive_enzyme_per_ec.tsv`, `pillar3_bacdive_per_fold_cv.tsv`
Suite 8	Pillars 3, 5, 6 / Scen. 8 (Trk 2), Agent	Held-out 70% BacDive (1,641/12,316), 120 anomalies, 50 LRO, 5 isolates	3-ch (7-ch on Enk. anomalies)	Stage 2B Anchor + Stage 3 Agent	gapseq, Traitar, BacDive RF, DiASPora, Non-Agentic LR/RF/GBDT/MLP Baselines	`pillar3_bacdive_track2_summary.tsv`, `pillar4_agent_*.tsv`, `pillar4_ml_baselines.tsv`

the anchored Stage 0 log-odds $L^{(0)}(p,f) = \mathrm{clip}\big(\log \frac{P_0(p,f)}{1-P_0(p,f)}, -12.0, 12.0\big)$, with GO annotations propagated upward across GO `is_a` and `part_of` ancestors (`IntegrationState.baselineAnnotatedGoTermsPropagated` and `goOntology.propagateAnnotations`).

4.3 Stage 1: Proteome-wide iterative Bayesian log-odds refinement

In Stage 1 (`IterativeRefiner.groovy` and `PriorEngine.groovy`), GSPA updates the log-odds matrix $\mathbf{L}^{(k)}$ across iterations $k = 1, \ldots, K_{\max}$ by computing six genome-scale prior shifts $B_m\big(p, f \mid \mathbf{P}^{(k-1)}, \mathcal{K}, \tau_{\text{tax}}, \mathcal{M}, \mathcal{O}\big)$ (weighted by w_m, default $w_m = 1.0$, and summed into $\Delta L^{(k)}_{\text{prior}}(p,f)$), adding them to the anchored Stage 0 log-odds $L^{(0)}(p,f)$ (clipped to $[-12.0, 12.0]$), and applying Jacobi under-relaxation in log-odds space with damping factor $\gamma_{\text{iter}} = 0.50$ (`damping = 0.5d` by default in `IterativeRefiner.groovy`; overrideable via `theta.json`):

$$\tilde{L}^{(k)}(p,f) = \mathrm{clip}\left(L^{(0)}(p,f) + \sum_{m=1}^{6} w_m \, B_m\big(p, f \mid \mathbf{P}^{(k-1)}, \mathcal{K}, \tau_{\text{tax}}, \mathcal{M}, \mathcal{O}\big), [-12.0, 12.0] \right)$$

$$L^{(k)}(p,f) = (1-\gamma_{\text{iter}}) L^{(k-1)}(p,f) + \gamma_{\text{iter}} \tilde{L}^{(k)}(p,f), \qquad P^{(k)}(p,f) = \sigma_{\log}\big(L^{(k)}(p,f)\big). \tag{2}$$

If a pinned floor $L_{\text{pin}}(p,f)$ is active (`state.pinnedFloors`), $L^{(k)}(p,f)$ is floored at $\min(12.0, L_{\text{pin}}(p,f))$ after damping. In `IterativeRefiner.groovy`, the inner loop runs for up to $K_{\max} = 6$ iterations (`maxIter = 6`) with convergence tolerance $\epsilon = 5 \times 10^{-3}$ on the mean absolute probability change $\frac{1}{|\Delta_{\text{chg}}|} \sum_{(p,f)\in\Delta_{\text{chg}}} |P^{(k)}(p,f) - P^{(k-1)}(p,f)| < \epsilon$ computed over changed pairs $\Delta_{\text{chg}} = \{(p,f) : |P^{(k)}(p,f) - P^{(k-1)}(p,f)| > 10^{-9}\}$, and automatic rollback to the last stable iterate if the mean change increases across two consecutive iterations. The six prior classes $B_1, \ldots, B_6$ are implemented in `gspa-core/src/main/groovy/gspa/integration/prior/` as follows:

1. **Essentiality prior ($m = 1$, $\alpha_{\text{ess}} = 1.5$, `EssentialityPrior.groovy`):** Given the domain-specific essential GO profile $\mathcal{F}_{\text{ess}}(\tau_{\text{tax}})$ (`bacteria`, `archaea`, `eukaryota`, or `viruses` in `EssentialFunctions.groovy`, `gspa.config.EssentialFunctions`), let $\mathcal{U}_{\text{ess}}(\tau_{\text{tax}}) \subseteq \mathcal{F}_{\text{ess}}(\tau_{\text{tax}})$ denote essential GO categories missing from the baseline ancestor-propagated active annotation set ($P^{(0)}(p,e) \le 0.50$ for all $p \in \mathcal{P}$), and let $\mathcal{D}_{\text{uncov}}(\tau_{\text{tax}}) = \bigcup_{e\in\mathcal{U}_{\text{ess}}(\tau_{\text{tax}})} \{t \in \mathcal{T} : t \sqsubseteq e\}$ denote the union of all GO subclasses of uncovered essential categories expanded via `goReasoner.getSubClasses`. Every candidate claim (p,f) with $f \in \mathcal{D}_{\text{uncov}}(\tau_{\text{tax}})$ receives a flat log-odds boost

$$B_1(p,f) = +\alpha_{\text{ess}} \, \mathbb{I}\big[f \in \mathcal{D}_{\text{uncov}}(\tau_{\text{tax}})\big] \qquad (\alpha_{\text{ess}} = 1.5, \texttt{maxBoostM}$$

2. **Pathway and complex coherence prior ($m = 2$, $\alpha_{\text{coh}} = 1.0$, $w_{\text{part}} = 0.85$, $w_{\text{sing}} = 0.75$, `CoherencePrior.groovy`):** First, for any GO process pair (X, Y) linked by `has_part` where X is present in the baseline active set but Y is missing, all subclasses $f \sqsubseteq Y$ receive process-coherence weight $w_{\text{proc}}(f) = \max(0, 1 - c_{\text{proc}})$, where c_{proc} is the fraction of satisfied `has_part` pairs. Second, for any reference pathway W triggered in the genome (≥ 1 baseline active step) with baseline step coverage $c_W \in (0,1)$, all GO terms $f \in \mathrm{Req}(W)$ mapping to missing steps of W receive pathway-coherence weight $w_{\text{path}}(f) = \max_{W \ni f} \max(0.25, 1 - c_W)$. Third, for an obligate heteromeric complex f ($\mathcal{G}_{\text{hgt}}(f)$ = `complexClassification[f] == 'n'` in `complex_terms.tsv`), let $N_{\text{strong}}(f) = |\{p' \in \mathcal{P} : P^{(0)}(p',f) > 0.50\}|$ and $N_{\text{cand}}(f) = |\{p' \in \mathcal{P} : \max(P^{(0)}(p',f), P^{(k-1)}(p',f)) \ge 0.08\}|$ count baseline-strong active proteins and candidate proteins across the genome; candidate partners ($\max(P^{(0)}(p,f), P^{(k-1)}(p,f)) \ge 0.08$) receive $w_{\text{cplx}}(p,f) = +w_{\text{part}} = +0.85$ when $N_{\text{strong}}(f) = 1$ (and $N_{\text{cand}}(f) \ge 2$) and $+0.65\, w_{\text{part}} = +0.5525$ when $N_{\text{strong}}(f) \ge 2$, whereas an isolated strong singleton ($P^{(0)}(p,f) > 0.50$ with $N_{\text{strong}}(f) = 1$ and $N_{\text{cand}}(f) \le 1$) receives $-w_{\text{sing}} = -0.75$:

$$B_2(p,f) = \alpha_{\text{coh}} \Big(\max\big(w_{\text{proc}}(f),\, w_{\text{path}}(f)\big) + w_{\text{cplx}}(p,f) \Big).$$

3. **Taxonomic consistency prior ($m = 3$, $\alpha_{\text{cons}} = 3.0$, `ConsistencyPrior.groovy`):** Once per iteration over the genome-wide currently active GO set $\mathcal{A}^{(k-1)} = \{t \in \mathcal{T} : \max_p P^{(k-1)}(p,t) > 0.50\}$, `ConsistencyPrior.groovy` forms the conflicting term set $\mathcal{V}^{(k-1)}_{\text{conf}} = \mathcal{T}^{\text{TC}}_{\text{forb}}(\tau_{\text{tax}}) \cup \mathrm{Core}_{\text{SAT}}(\mathcal{A}^{(k-1)})$ combining direct taxon-forbidden GO terms (`checker.getForbiddenConstrainedTerms()`) and SAT4J violation terms

(`checker.check(checkSet).violations[*].involvedGoTerms`). Any claim (p, f) where f or an ancestor $a \in \mathrm{Anc}(f)$ belongs to $\mathcal{V}_{\mathrm{conf}}^{(k-1)}$ receives $B_3(p, f) = -\alpha_{\mathrm{cons}} = -3.0$ (or -1000.0 when `hardFilter = true`).

4. **Anatomical prior ($m = 4$, $\alpha_{\mathbf{anat}} = 3.5$, $\alpha_{\mathbf{boost}} = 0.65$, `AnatomicalPrior.groovy`):** Let $\mathcal{T}_{\mathrm{forb}}^{\mathrm{anat}}(\tau_{\mathrm{tax}}) \subseteq \mathcal{T}$ be the set of GO terms (and their GO descendants via ancestor closure) anatomically forbidden in τ_{tax} (Section 4.4), and let $\mathcal{U}_{\mathrm{act}}(\tau_{\mathrm{tax}})$ be the set of non-root anatomical structures u taxonomically valid in τ_{tax} and supported by $\geq N_{\mathrm{sup}} = 2$ distinct proteins with likelihood probability $P^{(0)}(p', f') \geq 0.50$ (`minSupportProteins = 2`, `supportThreshold = 0.50d`); then

$$B_4(p, f) = \begin{cases} -\alpha_{\mathrm{anat}} = -3.5, & \text{if } (\{f\} \cup \mathrm{Anc}(f)) \cap \mathcal{T}_{\mathrm{forb}}^{\mathrm{anat}}(\tau_{\mathrm{tax}}) \neq \emptyset, \\ +\alpha_{\mathrm{boost}} = +0.65, & \text{else if } \mathrm{Anat}(f) \cap \mathcal{U}_{\mathrm{act}}(\tau_{\mathrm{tax}}) \neq \emptyset, \\ 0, & \text{otherwise.} \end{cases}$$

5. **Metabolic gap-filling prior ($m = 5$, $\alpha_{\mathbf{gap}} = 1.5$, `GapFillingPrior.groovy`) and post-refinement RLGC suggester (`ReactionLocalContextSuggester.groovy`):** Inside the Stage 1 loop, `GapFillingPrior.groovy` maps open metabolic gaps $r \in \mathcal{G}_{\mathrm{gap}}$ to gap-filler EC numbers and GO terms (using explicit EC-to-GO mappings plus prefix expansion for partial ECs), applying $B_5(p, f) = +\alpha_{\mathrm{gap}} = +1.5$ to direct gap matches and $+0.7\,\alpha_{\mathrm{gap}} = +1.05$ (`gapseqGuessedFactor = 0.7d`) when the gap reaction was flagged as a low-homology `gapseq` guess. Following Stage 1 refinement, `ReactionLocalContextSuggester.groovy` scores candidate genes p for each target gap reaction R via a $K_{\mathrm{hop}} = 2$ breadth-first search over the metabolic reaction adjacency graph $\mathcal{G}_{\mathrm{rxn}}$ (graph decay $\alpha_{\mathrm{rxn}}^{d(g)}$ with $\alpha_{\mathrm{rxn}} = 0.50$ for hop distance $d(g) \in \{1, 2\}$ of neighbor reaction GO terms $g \in \mathcal{N}_1(R) \cup \mathcal{N}_2(R)$). Each anchor gene p' on the same contig with $\max_{g \in \mathcal{N}_1(R) \cup \mathcal{N}_2(R)} P(p', g) \geq \tau_{\mathrm{anc}} = 0.30$ receives anchor weight $w_{\mathrm{anc}}(p', R) = \max_{g \in \mathcal{N}_1(R) \cup \mathcal{N}_2(R) : P(p', g) \geq \tau_{\mathrm{anc}}} \alpha_{\mathrm{rxn}}^{d(g)} P(p', g)$, and `GenomicDensityField.groovy` accumulates support within $\pm 4h_{\mathrm{bp}}$ via the Gaussian base-pair distance kernel $K_{\mathrm{bp}}(\Delta\mathrm{bp}) = \exp\!\left(-\Delta\mathrm{bp}^2/(2h_{\mathrm{bp}}^2)\right)$ ($h_{\mathrm{bp}} = 5{,}000$ bp):

$$D_{\mathrm{RLGC}}(p, R) = \sum_{p' : |\Delta\mathrm{bp}(p,p')| \leq 4h_{\mathrm{bp}}} w_{\mathrm{anc}}(p', R)\, K_{\mathrm{bp}}\big(\Delta\mathrm{bp}(p, p')\big),$$

$$S_{\mathrm{RLGC}}(p, R) = \log D_{\mathrm{RLGC}}(p, R) + \beta_{\mathrm{div}}\mathrm{Div}(p) - \gamma_{\mathrm{com}}\mathrm{Commit}(p) - \delta_{\mathrm{self}} P(p, f_R) + \varepsilon_{\mathrm{str}}\mathrm{Strand}(p) + \zeta_{\mathrm{gap}} \log_{10}\!\big(\max(1, \Delta\mathrm{bp}_{\min}(p)$$

(3)

where the top $Q = 50$ candidates with $D_{\mathrm{RLGC}}(p, R) \geq 0.01$ are evaluated using neighbor reaction diversity $\mathrm{Div}(p) = \sum_{g \in \mathcal{N}_1(R) \cup \mathcal{N}_2(R)} \alpha_{\mathrm{rxn}}^{d(g)} \big(1 - \prod_{p' \in W_8(p) \setminus \{p\}} (1 - P(p', g))\big)$ over window $W_8(p) = \{p' : |\Delta\mathrm{bp}(p, p')| \leq 8h_{\mathrm{bp}}\}$ ($2 \times$ `field.halfWidth` in `ReactionLocalContextSuggester.groovy:418`, $\beta_{\mathrm{div}} = 1.0$, `computeDiversity`), non-target commitment penalty $\mathrm{Commit}(p) = -\log \max\!\big(10^{-4}, 1 - \max_{g \in (\mathcal{N}_1(R) \cup \mathcal{N}_2(R)) \setminus \{f_R\}} P(p, g)\big)$ ($\gamma_{\mathrm{com}} = 1.0$, `computeCommitmentPenalty`), self-annotation penalty $\delta_{\mathrm{self}} = 2.0$, strand concordance fraction $\mathrm{Strand}(p) \in [0, 1]$ ($\varepsilon_{\mathrm{str}} = 0.2$), and minimum anchor base-pair distance $\Delta\mathrm{bp}_{\min}(p)$ ($\zeta_{\mathrm{gap}} = 0.1$), and normalized over the top 20 candidates via softmax $P_{\mathrm{RLGC}}(p \mid R) = \exp(S_{\mathrm{RLGC}}(p, R)) / \sum_{p''} \exp(S_{\mathrm{RLGC}}(p'', R))$ with Bayes factor $\mathrm{BF}_{\mathrm{RLGC}} = \exp(S_{\mathrm{RLGC}}(p, R))$. In the M2/M4 Track A and Track B LRO rankers (`CrossGenomeReScorer` in `ConditionalLRScorer.groovy` and `benchmark/unified/pillar2_fba_lro_price149/run_pillar`), $S_{\mathrm{RLGC}}(p, R)$ is combined with cross-genome orthogroup co-occurrence ($\lambda_{\mathrm{cg}} = 0.5$ in `CrossGenomeReScorer`), Pfam domain compatibility, and (in Track B) ESM-2 PLM reaction centroid cosine similarity.

6. **Operon neighborhood prior ($m = 6$, $\alpha_{\mathbf{ctx}} = 1.0$, $\alpha_{\mathbf{gap_ctx}} = 1.5$, $w_{\mathbf{op_cplx}} = 0.95$, `GenomicContextPrior.groovy`):** For each co-directional operon O ($|O| \geq 2$) and reference pathway W with $k_O(W) = |\{p' \in O : \max_{f' \in \mathrm{Req}(W)} P^{(k-1)}(p', f') > 0.50\}| \geq 1$ active members, let $c_O(W) = k_O(W)/|O|$ denote the operon's pathway consensus strength. Claims (p, f) with $p \in O$ and $f \in \mathrm{Req}(W)$ receive $\alpha_{\mathrm{ctx}} c_O(W)$ (multiplied by $\alpha_{\mathrm{gap_ctx}} = 1.5$ when f is also an open metabolic gap-filler GO term), and claims for an obligate heteromeric complex $f \in \mathcal{C}_{\mathrm{het}}$ (`'n'`) where another member $p' \in O \setminus \{p\}$ carries active evidence $P^{(k-1)}(p', f) > 0.50$ for the same complex f receive an additional $+w_{\mathrm{op_cplx}} = +0.95$:

$$B_6(p, f) = \max_{W \ni f : k_O(W) \geq 1} \Big(\alpha_{\mathrm{ctx}}\, c_O(W)\, \big(\alpha_{\mathrm{gap_ctx}}\big)^{\mathbb{I}[f \in \mathcal{G}_{\mathrm{gap}}]}\Big) + w_{\mathrm{op_c}}$$

All 18 configurable pipeline hyperparameters across Stage 1 (13 `theta.json` keys exposed by `gspa-cli/src/main/groovy/gspa/cli/IntegrateCommand.groovy`: `alpha_ess`, `alpha_coh`, `alpha_complex_boost`, `alpha_complex_penalty`, `alpha_cons`, `alpha_anat_penalty`, `alpha_anat_boost`, `alpha_gap`, `alpha_ctx`, `alpha_gap_ctx`, `damping`, `max_iter`, and `epsilon`) and Stage 2B (5 stoichiometric refiner parameters in `scripts/stoichiometric_locus_refiner.py`: `tau_core`, `alpha_bio`, `tau_null`, `gamma_fba`, and `gamma_damp`) were calibrated exclusively on the *E. coli* K-12 MG1655 development proteome and held strictly frozen across all non-*E. coli* benchmarks (Supplementary Section S1 and Supplementary Table S1).

4.4 Cross-ontology taxonomic and anatomical constraint bridge

To capture taxonomic restrictions entailed by multicellular eukaryotic developmental and physiological GO terms, we programmatically extracted five resource tables from the official GO, Uberon[10], Plant Ontology (PO)[66], Cell Ontology (CL)[67], and NCBI Taxonomy OBO/OWL releases via `scripts/extract_anatomical_and_taxon_constraints.py` (and validated via `AnatomicalConstraints.groovy`): (1) 26,911 direct GO taxon constraint rules (13,819 `only_in_taxon` and 13,092 `never_in_taxon`) across 18,159 distinct GO terms (`go-taxon-constraints.tsv`); (2) 2,715 equivalence and subclass cross-product axioms linking 2,674 distinct GO terms to 1,294 Uberon, PO, and CL classes across 30 object properties (`RO:0002296 results_in_development_of`, `RO:0002298 results_in_morphogenesis_of`, `RO:0002297 results_in_formation_of`, `RO:0002299 results_in_maturation_of`, `BFO:0000066 occurs_in`, and transport/organization relations; `go-anatomical-constraints.tsv`); (3) 8,331 taxon constraints (4,767 `only_in_taxon` and 3,564 `never_in_taxon`) across 1,247 Uberon, PO, and CL classes (`uberon-po-taxon-constraints.tsv`); and (4) 7,722 anatomical hierarchy edges across 2,855 Uberon/PO/CL classes (`uberon-po-hierarchy.tsv`) and 1,818 taxonomic hierarchy edges across 1,305 NCBI Taxonomy clades (`ncbi-taxon-hierarchy.tsv`).

4.5 Stage 2A: Deterministic symbolic enforcement and semantic ancestor relaxation

In `IntegrateCommand.groovy` and `ConsistencyEnforcer.groovy` (invoked when `-enforce-stage2` is active), Stage 2A executes immediately after Stage 1 (and prior to Stage 2B) as a deterministic pass over all GO provenance entries in `IntegratedAnnotationSet` (`IntegrationState`). First, `computeForbiddenTerms()` identifies all GO terms among the proteome's provenance entries that violate a direct GO taxon constraint or an entailed Uberon/PO/CL anatomical constraint in τ_{tax} (including upward GO ancestor closure via `goReasoner.getSuperClasses`; when `organismTaxon` is absent or `mode == "minimal-flip"`, `TaxonConsistencyMaxSat` / `SatConsistencyChecker` extracts the unsatisfiable or minimal-weight flip set). In `relax` mode (`-enforce-consistency-mode relax`), for each protein p and forbidden term t, `AnatomicalConstraints.findValidRelaxedAncestors()` traverses $\text{Anc}(t)\backslash\{\text{GO:0008150}, \text{GO:0003674}, \text{GO:0005575}\}$ to select the most specific valid non-root ancestors $\mathcal{A}_{\text{valid}}(t, \tau_{\text{tax}})$ (retaining only ancestors a that have no valid descendant $b \in \text{Anc}(t)$ with $b \sqsubset a$). Reconstructing the unpenalized posterior $P_{\text{unpen}}(p,t) = \sigma_{\log}\big(L^{(0)}(p,t) + \sum_{m \notin \{\text{cons,anat}\} \vee B_m(p,t)>0} B_m(p,t)\big)$ prior to negative consistency and anatomical penalties, each minimal valid ancestor $a \in \mathcal{A}_{\text{valid}}(t, \tau_{\text{tax}})$ is updated directly (and if $\mathcal{A}_{\text{valid}}(t, \tau_{\text{tax}}) = \emptyset$, t is removed):

$$P_{\text{Stage2A}}(p,a) = \max\Big(P^{(K)}(p,a),\, \gamma_{\text{relax}}\, P_{\text{unpen}}(p,t)\Big),$$
$$P_{\text{Stage2A}}(p,t) = 0, \tag{4}$$

with default relaxation factor $\gamma_{\text{relax}} = 1.0$ (`relaxFactor = 1.0d` in `ConsistencyEnforcer.groovy`), updating the provenance entry of a via `logit`$(P_{\text{Stage2A}}(p,a))$ and removing (p,t) from `state.provenance`. Second, in `IntegrateCommand.groovy`, for any obligate heteromeric complex $f \in \mathcal{C}_{\text{het}}$ (`complexClassification[f] == 'n'`) that has exactly one active protein (≥ 0.50) and zero candidate partners (≥ 0.08) across the proteome, the isolated singleton prediction is deterministically attenuated to 0.45 and logged as `PRUNED_SINGLETON` in `EnforcementReport`.

4.6 Stage 2B: Flux-certified stoichiometric locus refinement

Stage 2B (`scripts/stoichiometric_locus_refiner.py`, invoked from the CLI via `StoichiometricLocusRefiner.groovy` when `-enable-fba-refiner` is passed) operates over a sanitized universal metabolic network $\mathcal{U}_{\text{clean}}$ constructed from the ModelSEED biochemistry database (47,880 total universal reactions across 33,978 metabolites, of which 14,243 reactions carry EC mappings and 11,826 are strictly mass- and charge-balanced enzymatic reactions after flagging 2,417 pool or imbalanced reactions with `imbalance_penalty` $w_{\text{imb}}(j) = 100.0$, together with basal inorganic and cofactor exchange bounds). For each universal reaction $j \in \mathcal{U}_{\text{clean}}$ with catalytic mechanisms $\mathcal{M}_{\text{dnf}}(j) = \{m_1, \ldots, m_{K_j}\}$ and required roles or subunits $\mathcal{R}_{\text{sub}}(m) = \{u_1, \ldots, u_{D_m}\}$ (where multi-role conjunctions $D_m \geq 2$ apply when explicit multi-subunit complex rules are supplied in `rxn_mechanisms`, while single-EC mappings in the default universal metadata reduce to $D_m = 1$ Top-M_{iso} isozyme disjunctions), `SparseRoleAggregator.compute_reaction_probabilities` (and `SparseRoleAggregator.groovy`) aggregates locus support $s_{p,u} \in (0,1)$ over the top $M_{\text{iso}} = 3$ candidate genes per role with bottleneck gate $\theta_{\text{sub}} = 0.25$:

$$Q(u) = 1 - \prod_{p \in \text{Top-}M_{\text{iso}}(\mathcal{C}_{\text{cand}}(u))} \big(1 - s_{p,u}\big),$$
$$S_{\text{mech}}(m) = \begin{cases} \Big(\prod_{u \in \mathcal{R}_{\text{sub}}(m)} Q(u)\Big)^{1/D_m}, & \min_u Q(u) \geq \theta_{\text{sub}}, \\ \min_{u \in \mathcal{R}_{\text{sub}}(m)} Q(u), & \text{otherwise}, \end{cases}$$
$$S_{\text{rxn}}(j) = \max_{m \in \mathcal{M}_{\text{dnf}}(j)} S_{\text{mech}}(m). \tag{5}$$

Second, reactions are partitioned into the draft core $\mathcal{R}_{\text{core}} = \{j : S_{\text{rxn}}(j) \geq \tau_{\text{core}} = 0.50\}$ (plus open medium

exchanges and spontaneous reactions) and candidate repair reactions $j \notin \mathcal{R}_{\text{core}}$ restricted to the active solver domain $\mathcal{D}_{\text{dom}}$ comprising reactions with $S_{\text{rxn}}(j) \geq 10^{-3}$ or belonging to the 108-genome universal flux support skeleton `univ_sup`, with strictly positive addition penalties $c_j^{\text{add}} = \lambda_{\text{add}}(1 - S_{\text{rxn}}(j)) + w_{\text{imb}}(j) + \epsilon_{\text{par}} > 0$ ($\lambda_{\text{add}} = 1.0$, $\epsilon_{\text{par}} = 10^{-3}$, $w_{\text{imb}}(j) = 100.0$ if j is imbalanced and 0 otherwise). Splitting reversible fluxes into non-negative variables $v_j = v_j^+ - v_j^-$ and setting the biomass target $v_{\text{bio}}^{\text{target}}(\mathcal{E}_{\text{med}}) = \max\big(\alpha_{\text{bio}}\, v_{\text{bio}}^{\max}(\mathcal{E}_{\text{med}}),\, 0.1\big)$ ($\alpha_{\text{bio}} = 0.10$), Stage 2B solves a two-step lexicographic pFBA linear program[73] in HiGHS (feasibility tolerance 10^{-9}): Step 1 minimizes $\sum_{j \in \mathcal{D}_{\text{dom}} \setminus \mathcal{R}_{\text{core}}} c_j^{\text{add}}(v_j^+ + v_j^-)$ subject to $\mathbf{S}\mathbf{v} = \mathbf{0}$ and $v_{\text{bio}} \geq v_{\text{bio}}^{\text{target}}(\mathcal{E}_{\text{med}})$ (first constraining all imbalanced candidate bounds to $(0, 0)$ via `balanced_bounds_split` so that imbalanced reactions are strictly excluded whenever a balanced route exists, and falling back to the penalized bounds $w_{\text{imb}} = 100.0$ only if the strictly balanced LP is infeasible) to obtain minimum repair cost $C^*(\mathcal{E}_{\text{med}})$; Step 2 minimizes total non-biomass network flux $\sum_{j \in \mathcal{D}_{\text{dom}} \setminus \{\text{bio}\}}(v_j^+ + v_j^-)$ subject to $\sum_{j \notin \mathcal{R}_{\text{core}}} c_j^{\text{add}}(v_j^+ + v_j^-) \leq (1 + \delta_{\text{opt}})C^*(\mathcal{E}_{\text{med}}) + 10^{-9}$ ($\delta_{\text{opt}} = 10^{-6}$). By construction, the repaired reaction set $\mathcal{J}_{\text{rep}}(\mathcal{E}_{\text{med}}) = \{j \notin \mathcal{R}_{\text{core}} : |v_j^{\text{pFBA}}(\mathcal{E}_{\text{med}})| > \epsilon_{\text{flux}} = 10^{-6}\}$ retains only non-core reactions carrying positive pFBA flux (yielding 0.00 zero-flux additions by construction). When full certificate auditing is active (`compute_certificates=True`), each pFBA-active addition in $\mathcal{J}_{\text{rep}}(\mathcal{E}_{\text{med}})$ is further tested by three essentiality and stability certificates to form $\pi_j^{\text{FBA}} = 0.5\,\mathbb{I}[j \in \mathcal{R}_{\text{FVA}}] + 0.3\,\mathbb{I}[j \in \mathcal{R}_{\text{KO}}] + 0.2\,\omega_{\text{ens}}(j) \in [0, 1]$, where $j \in \mathcal{R}_{\text{FVA}}$ iff $\min(v_j^+ + v_j^-) > \epsilon_{\text{flux}}$ subject to $\sum c_{j'}^{\text{add}}(v_{j'}^+ + v_{j'}^-) \leq 1.01\,C^* + 10^{-7}$, $j \in \mathcal{R}_{\text{KO}}$ iff constraining $v_j = 0$ either renders $\mathcal{R}_{\text{core}} \cup \mathcal{J}_{\text{rep}}$ growth-infeasible or increases the minimum repair cost on $\mathcal{D}_{\text{dom}}$ above $1.10\,C^* + 10^{-6}$, and $\omega_{\text{ens}}(j) \in \{1.0, 0.65, 0.0\}$ is the perturbation stability score across $N_{\text{ens}} = 5$ repair LPs with $\pm 5\%$ uniform multiplicative cost noise (1.0 if active in $\geq 80\%$ of perturbed solves, 0.65 if active in $> 0\%$, and 0.0 otherwise; when `compute_certificates=False`, $\pi_j^{\text{FBA}} = 1.0$ for all pFBA-active repairs).

Third, `attribute_locus_responsibility` in `scripts/stoichiometric_locus_refiner.py` (and `LocusResponsibilityUpdater.groovy`) attributes pFBA-active reaction repairs back to candidate gene loci against an explicit unassigned-gene hypothesis $h_\varnothing$ with prior mass $\eta_{\text{null}} = 0.05$, using each locus's sequence odds $o_{p,u} = \frac{s_{p,u}}{1 - s_{p,u}}$:

$$
\begin{aligned}
w_{j,u}(p) &= \big(o_{p,u}\big)^{\beta_1} \text{BF}_{\text{operon}}(p,u)^{\beta_2} \\
&\quad \times \exp\big(\beta_3 C_{\text{syn}}(p,u) + \beta_4 S_{\text{ESM2}}(p,u)\big), \\
r_{j,u}(p) &= \frac{w_{j,u}(p)}{\eta_{\text{null}} + \sum_{p' \in \mathcal{C}_{\text{cand}}(u)} w_{j,u}(p')},
\end{aligned} \tag{6}
$$

with $\beta_1 = 1.0, \beta_2 = 1.0, \beta_3 = 1.5, \beta_4 = 1.5$, yielding the stoichiometric log-odds shift $\Delta L_{\text{FBA}}(p,u) = \gamma_{\text{FBA}} \sum_{j \in \mathcal{J}_{\text{rep}}(u)} \pi_j^{\text{FBA}}\, r_{j,u}(p)\,(1 - S_{\text{rxn}}(j))$ ($\gamma_{\text{FBA}} = 2.2$), applied with step weight $\lambda_t = \lambda_0 \gamma_{\text{damp}}^{t-1}$ ($\lambda_0 = 0.70, \gamma_{\text{damp}} = 0.60$) in `refine_iterative` (in the CLI bridge `StoichiometricLocusRefiner.groovy`, the three auxiliary covariates passed to Python are extracted from `IntegratedAnnotationSet` as $\text{BF}_{\text{operon}} = \exp(B_{\text{gap}}(p,u)/1.5)$, $C_{\text{syn}} = \text{clip}(B_{\text{ctx}}(p,u) + B_{\text{coh}}(p,u), 0, 1)$, and S_{ESM2} equal to the maximum raw claim score `sc.rawScore` over ESM/ProtT5/PLM/CLEAN supporting claims on (p,u)). In the standalone `Price-149` evaluation script (`scripts/eval_price149_real_all6.py:348-440`), which evaluates both target-level and whole-genome predictions across 6 neural EC predictors, chromosomal neighbor ECs within ± 10 genes (`extract_ibex_compact_matrices.py:172`) are weighted by distance decay $w_{\text{dist}}(\Delta g) = 1/(1 + 0.25\,|\Delta g|)$ to form neighbor pathway support $S_{\text{pwy}}(u)$, 3-digit EC sub-subclass support $S_{\text{ec3}}(u)$, $S_{\text{operon}}(u) = \min(1.0,\, 0.55\,S_{\text{pwy}}(u) + 0.45\,S_{\text{ec3}}(u))$, and 1-hop and 2-hop non-currency metabolite-sharing stoichiometric neighbor supports $S_{1\text{hop}}(u), S_{2\text{hop}}(u)$ derived from $\mathbf{S}$, and locus responsibility is computed as $r_j(p) = w_h/(0.05 + w_h)$ where $w_h = \text{odds}_{\text{seq}}^{0.5} \exp(0.85\,S_{1\text{hop}} + 0.40\,S_{2\text{hop}}) \exp(1.2\,S_{\text{operon}})\, g_{\text{need}}$ and $g_{\text{need}} = \max(0.10,\, 1 - 0.85\,s_{\text{other,max}})$. Using each protein's top-1 baseline score $s_{\max}$, **+ GSPA SLR (Minimal Glucose)** solves the glucose pFBA repair LP on each of the 22 host genomes and boosts candidate ECs in repaired ($\mathcal{J}_{\text{rep}}$, $s \geq 0.03\,s_{\max}$, $\pi_j = 0.85$) or active ($\mathcal{J}_{\min}$, $s \geq 0.05\,s_{\max}$, $\pi_j = 0.45$) reactions via $\Delta L_{\text{FBA}} = 2.2\,\pi_j\,r_j(p)\,(1 - s)$; **+ GSPA SLR (Cognate Substrate)** (`+ SLR Stoich. Probe` in Table 3) applies $\Delta L_{\text{FBA}}^{\text{probe}} = 1.10\,r_j(p)(1 - s)\,\mathbb{I}[u \in \mathcal{J}_{\text{rep}} \cup \mathcal{J}_{\min}] + 2.20\,S_{1\text{hop}}(u)\,r_j(p)(1 - s)\,\mathbb{I}[S_{1\text{hop}}(u) > 0 \wedge s \geq 0.02\,s_{\max}] + 1.25\,S_{2\text{hop}}(u)\,r_j(p)(1 - s)\,\mathbb{I}[S_{1\text{hop}}(u) = 0 \wedge S_{2\text{hop}}(u) > 0 \wedge s \geq 0.03\,s_{\max}]$ (falling back to the minimal-glucose score when $\Delta L_{\text{FBA}}^{\text{probe}} = 0$); and **+ Full GSPA Iter** t ($t \in \{1, 2, 3\}$) evaluates the recursive damped log-odds outer loop combining the Stage 1 operon/coherence shift $\Delta L_{\text{Stage1}}(p,u) = (1.35\,S_{\text{operon}}(u) + 0.95\,S_{\text{coh}}(u))\,\mathbb{I}[s \geq 0.03\,s_{\max}] - 0.35\,\mathbb{I}[S_{1\text{hop}}(u) = 0 \wedge S_{2\text{hop}}(u) = 0 \wedge S_{\text{operon}}(u) < 0.08 \wedge S_{\text{coh}}(u) < 0.15 \wedge u \notin \mathcal{J}_{\min}]$ (including the -0.35 isolated paralog penalty) and stoichiometric probe shift $\Delta L_{\text{FBA}}^{\text{rxn}}(p,u) = 1.20\,r_j(p)(1 - s)\,\mathbb{I}[u \in \mathcal{J}_{\min} \wedge s \geq 0.05\,s_{\max}] + 2.20\,S_{1\text{hop}}(u)\,r_j(p)(1 - s)\,\mathbb{I}[S_{1\text{hop}}(u) > 0 \wedge s \geq 0.02\,s_{\max}] + 1.25\,S_{2\text{hop}}(u)\,r_j(p)(1 - s)\,\mathbb{I}[S_{1\text{hop}}(u) = 0 \wedge S_{2\text{hop}}(u) > 0 \wedge s \geq 0.03\,s_{\max}]$ with $\Delta_{\text{tot}}(p,u) = \Delta L_{\text{Stage1}}(p,u) + \Delta L_{\text{FBA}}^{\text{rxn}}(p,u)$ and

$L_{\text{target}}(p,u) = L^{(0)}(p,u) + \Delta_{\text{tot}}(p,u)$:

$$
\begin{aligned}
L^{(1)}_{\text{outer}}(p,u) &= (1-0.70)\, L^{(0)}(p,u) + 0.70\, L_{\text{target}}(p,u),\\
L^{(2)}_{\text{outer}}(p,u) &= (1-0.42)\, L^{(1)}_{\text{outer}}(p,u) + 0.42\, \big(L_{\text{target}}(p,u) + 0.25\, \Delta_{\text{tot}}(p,u)\big),\\
L^{(3)}_{\text{outer}}(p,u) &= (1-0.252)\, L^{(2)}_{\text{outer}}(p,u) + 0.252\, \big(L_{\text{target}}(p,u) + 0.35\, \Delta_{\text{tot}}(p,u)\big).
\end{aligned}
\tag{7}
$$

where multi-predictor refinement blends the single-predictor outer-pass probability $P^{(t)}_{\text{single}}(p,u) = \sigma_{\log}\big(L^{(t)}_{\text{outer}}(p,u)\big)$ with the 5-predictor Noisy-OR ensemble probability $P_{\text{ens}}(p,u)$ via $P^{(t)}_{\text{multi}}(p,u) = 0.55\, P^{(t)}_{\text{single}}(p,u) + 0.45\, P_{\text{ens}}(p,u)$.

To formalize the theoretical limits of metabolic gene recovery across Strata A, B, and C, we define five binary recoverability indicators for each held-out reaction j and true gene p^*: (1) $\mathcal{C}_{\text{struct}}(j) \in \{0,1\}$, equal to 1 iff reaction j is present and mass-balanced in $\mathcal{U}_{\text{clean}}$; (2) $\mathcal{C}_{\text{task}}(j, \mathcal{E}_{\text{med}}) \in \{0,1\}$, equal to 1 iff blocking j requires non-zero repair flux under medium $\mathcal{E}_{\text{med}}$ ($\mathcal{C}_{\text{task}} = 1$ on minimal glucose for Stratum A, $\mathcal{C}_{\text{task}} = 1$ on cognate substrate for Stratum B, and $\mathcal{C}_{\text{task}} = 0$ for secondary metabolism Stratum C); (3) $\mathcal{C}_{\text{unique}}(j) \in \{0,1\}$, equal to 1 iff j has no unmasked isozyme sufficient to carry full flux; (4) $\mathcal{C}_{\text{operon}}(p^*, j) = \mathbb{I}[\text{BF}_{\text{operon}}(p^*, j) > 1]$; and (5) $\mathcal{C}_{\text{complex}}(p^*, j) \in \{0,1\}$, equal to 1 iff j is catalyzed by a multi-subunit complex with partner subunits present.

4.7 Stage 3: Selective neuro-symbolic agentic annotation completion

Stage 3 (`AnomalyExtractor.groovy` and `benchmark/agent/run_gspa_agent.py`) operates selectively on post-Stage-2 genome-scale anomalies across four categories: (1) `PATHWAY_GAP` (a pathway with completeness $\geq \theta_{\text{path_anom}}$ missing a step f with sub-threshold candidates in $P_{\text{Stage2}} \in [\theta_{\text{floor}}, 0.50)$, where `AnomalyExtractor.groovy` defaults to $\theta_{\text{path_anom}} = 0.50$, $\theta_{\text{floor}} = 0.05$, up to 8 candidates per anomaly and 150 anomalies per genome, and the Suite 8 benchmark focuses on high-completeness pathways ≥ 0.65 with top 5 candidates in $[0.15, 0.49]$); (2) `INCOMPLETE_COMPLEX` (an obligate heteromeric complex `'n'` where `AnomalyExtractor.groovy` triggers when exactly one active anchor protein has posterior ≥ 0.50 and ≥ 1 sub-threshold candidate lies in $[\theta_{\text{floor}}, 0.50)$, and the Suite 8 benchmark samples incomplete complexes with $\geq 50\%$ partner subunit coverage); (3) `BORDERLINE_ESSENTIAL` (an unfulfilled core essential category $f \in \mathcal{F}_{\text{ess}}(\tau_{\text{tax}})$ with sub-threshold candidates); and (4) `BORDERLINE_CONFLICT` (a claim in $[0.35, 0.62]$ subject to a taxonomic or anatomical conflict where domain inspection can refine the ancestral target).

For each anomaly, Stage 3 strips all evaluation metadata (`_strip_ground_truth`) and runs a bounded multi-turn tool-calling loop (≤ 6 turns in full mode, ≤ 2 turns in fast 1-step mode) using our locally hosted FP8-quantized 27B open-weight instruction checkpoint `Qwen/Qwen3.8-27B-FP8` (derived from the Qwen2.5/Qwen3 open-weight model family[84] and served as `qwen3.8-27b` via vLLM on `unimatrix01` with a 32,768-token context window, temperature 0.6, maximum 450 output tokens per turn; evaluated across 3 independent random seeds seed $\in \{42, 43, 44\}$, `results/pillar4_multiseed_summary.tsv`) equipped with five deterministic read-only tools: `inspect_operon_neighborhood`, `inspect_pathway_or_complex_gap`, `query_protein_evidence`, `check_symbolic_constraints`, and `propose_resolution`. Every proposed resolution is verified by the fail-closed deterministic `SymbolicGuardrail` against all 48,101 GO term stanzas (40,122 active and 7,979 obsolete) in `go-basic.obo`, 1,305 NCBI Taxonomy clades, 26,911 direct GO taxon constraints, and 2,715 GO–Uberon/PO/CL anatomical constraints before acceptance.

4.8 BacDive Track 1, Track 2 Bayesian Anchor, and GSPA + DiASPora Hybrid protocols

In Suites 7 and 8 (`benchmark/unified/pillar3_bacdive/run_bacdi`
we evaluate experimental carbon-source growth ($n = 2{,}332$ strain–substrate assays across 529 ProTraits-matched bacterial genomes [791 total distinct `genome_id` entries in `bacdive_carbon_2332_real.tsv`]) and diagnostic enzyme activity ($n = 17{,}496$ strain–EC assays across 3,017 BacDive type-strain assemblies [3,170 total distinct `strain_id` entries in `bacdive_enzyme_17496_real.tsv`]). For each strain s and assay y ($M = 37$ carbon substrates or $M = 10$ diagnostic enzymes), given binary reaction/pathway calls $g(s,y), m(s,y), c(s,y) \in \{0,1\}$ from `gapseq` v1.2[57], `ModelSEED` v2.6.1[54], and `CarveMe` v1.5.2[56], `build_cooccurrence_coupling` forms the blended call matrix $B(s,y) = 0.55\, g(s,y) + 0.30\, m(s,y) + 0.15\, c(s,y)$ on carbon and $B(s,y) = 0.55\, g(s,y) + 0.28\, m(s,y) + 0.17\, c(s,y)$ on enzymes, column-centers $\tilde{B}_{s,y} = B(s,y) - \bar{B}_y$, computes the nonnegative cosine similarity $\Psi_0(y,y') = \max\big(0, \frac{\tilde{B}_{:,y}^{\top} \tilde{B}_{:,y'}}{\|\tilde{B}_{:,y}\|_2 \|\tilde{B}_{:,y'}\|_2 + 10^{-9}}\big)$ with $\Psi_0(y,y) = 0$, row-normalizes $\Psi(y,y') = \Psi_0(y,y') / (\sum_{y''} \Psi_0(y,y'') + 10^{-9})$, and defines the unsupervised cross-assay coupling score $\psi(s,y) = \sum_{y'} \Psi(y,y') \big(B(s,y') - 0.5\big)$. In **GSPA Track 1 (100% Blind)**, binary calls are mapped to calibrated tool probabilities $p_{\text{gap}}(s,y) \in \{0.12, 0.88\}$, $p_{\text{ms}}(s,y) \in \{0.22, 0.80\}$, and $p_{\text{cm}}(s,y) \in \{0.30, 0.72\}$ on carbon ($p_{\text{gap}} \in \{0.14, 0.86\}$, $p_{\text{ms}} \in \{0.24, 0.78\}$, and $p_{\text{cm}} \in \{0.28, 0.74\}$ on enzymes) and pooled in log-odds space:

$$
P_{\text{Track1}}(s,y) = \sigma_{\log}\Big(w_{\text{gap}} \operatorname{logit}\big(p_{\text{gap}}(s,y)\big) + w_{\text{ms}} \operatorname{logit}\big(p_{\text{ms}}(s,y)\big) + w_{\text{cm}}
\tag{8}
$$

with $(w_{\text{gap}}, w_{\text{ms}}, w_{\text{cm}}, w_{\psi}) = (0.56, 0.28, 0.16, 0.65)$ on carbon and $(0.55, 0.27, 0.18, 0.55)$ on enzymes, evaluated at $\theta = 0.50$.

For the supervised classifiers in Track 1 (`Traitar` L1-regularized `LinearSVC` wrapped in `CalibratedClassifierCV` with $C = 0.3$ on carbon and $C = 0.2$ on enzymes[81], `BacDive RF Baseline` Random Forest with 200 trees on carbon and 150 trees on enzymes at max depth 8[79,80], and `DiASPora GradientBoostingClassifier` with 150 trees, max depth 4, and learning rate 0.05 on carbon and `HistGradientBoostingClassifier` with 100 iterations, max depth 4, and learning rate 0.08 on enzymes[80]), all models are trained and evaluated under deterministic 5-fold genus-blocked cross-validation (`fold(s) = int(md5(genus)[:8], 16) mod 5`; per-fold metrics in `results/pillar3_bacdive_per_fold_cv.tsv`). Baseline classifiers use the $(3 + M)$-dimensional feature vector $\mathbf{x}_{\text{base}}(s, y) = [g(s, y), m(s, y), c(s, y), \mathbf{e}_y] \in \mathbb{R}^{3+M}$, where $\mathbf{e}_y \in \{0, 1\}^M$ is the assay one-hot indicator. Using the Laplace-smoothed training-fold calibrated `gapseq` and `ModelSEED` posteriors $(p_g^{\text{tr}}(s, y), p_m^{\text{tr}}(s, y))$, training-fold assay prevalence $\pi_{\text{assay}}^{\text{tr}}(y)$, and Track 1 blind score $P_{\text{Track1}}(s, y)$, GSPA forms the fold-local pooled score $p_{\text{pool}}(s, y) = \sigma_{\log}\big(w_g^{\text{tr}} \operatorname{logit}(p_g^{\text{tr}}(s, y)) + w_m^{\text{tr}} \operatorname{logit}(p_m^{\text{tr}}(s, y)) + w_{\pi}^{\text{tr}} \operatorname{logit}(\pi_{\text{assay}}^{\text{tr}}(y)) + w_{\text{t1}}^{\text{tr}} \operatorname{logit}(P_{\text{Track1}}(s, y)) + w_{\psi}^{\text{tr}} \psi(s, y)\big)$ (with $(0.45, 0.25, 0.15, 0.15, 0.50)$ on carbon and $(0.48, 0.24, 0.14, 0.14, 0.45)$ on enzymes), constructs the $(10 + 4M)$-dimensional feature vector $\mathbf{x}_{\text{GSPA}}(s, y) = [g(s, y), m(s, y), c(s, y), \pi_{\text{assay}}^{\text{tr}}(y), p_g^{\text{tr}}(s, y), p_m^{\text{tr}}(s, y), \bar{b}_{\text{strain}}(s), \psi(s, y), P_{\text{Track1}}(s, y), p_{\text{pool}}(s, y), \operatorname{cap}_{20}(s), \mathbf{e}_y] \in \mathbb{R}^{10+4M}$ (where $\bar{b}_{\text{strain}}(s)$ is the strain's mean blended call across assays and $\mathbf{s}_{\text{vec}}(s) \in \{0, 1\}^{3M}$ is the strain's $3M$-dimensional vector of (`gapseq`, `ModelSEED`, `CarveMe`) binary calls across all M assays; `run_bacdive_tracks1_2.py:305-309, 391-395`), trains the hybrid classifier (`GradientBoostingClassifier` with 180 trees, max depth 4, learning rate 0.05 on carbon; `HistGradientBoostingClassifier` with 160 iterations, max depth 5, learning rate 0.08 on enzymes), and blends the out-of-fold classifier probability $P_{\text{ML}}(s, y)$ with $p_{\text{pool}}(s, y)$ as $P_{\text{Hyb1}}(s, y) = 0.88\, P_{\text{ML}}(s, y) + 0.12\, p_{\text{pool}}(s, y)$ (evaluated at $\theta = 0.50$ on carbon and at the training-fold F_1-optimal threshold on the enzyme panel). In Track 2, 30% of assay rows (`int(md5(f"{id}_{assay}_{idx}")[:8], 16) mod 100 < 30`) are revealed ($\mathcal{D}_{\text{rev}}$) and evaluation is restricted to the held-out 70% ($\mathcal{D}_{\text{hold}}$). **GSPA Track 2 Bayesian Anchor** combines $p_{\text{pool}}(s, y)$ with the revealed substrate prevalence $s_{b,\text{emp}}(y)$ and revealed-assay co-occurrence shift $\psi_{\text{rev}}(s, y)$ on carbon via $P_{\text{Track2}}(s, y) = \sigma_{\log}\big(0.72 \operatorname{logit}(p_{\text{pool}}(s, y)) + 0.28 \operatorname{logit}(s_{b,\text{emp}}(y)) + 0.85\, \psi_{\text{rev}}(s, y)\big)$, and combines $p_{\text{pool}}(s, y)$ with the revealed genus–EC posterior $g_{e,\text{emp}}(s, y)$ on enzymes via $P_{\text{Track2}}(s, y) = \sigma_{\log}\big(0.70 \operatorname{logit}(p_{\text{pool}}(s, y)) + 0.30 \operatorname{logit}(g_{e,\text{emp}}(s, y))\big)$ (with decision thresholds tuned strictly on the revealed 30% split), while **GSPA Track 2 + DiASPora Hybrid** blends the Track 2 classifier $P_{\text{ML2}}(s, y)$ with $P_{\text{Track2}}(s, y)$ as $P_{\text{Hyb2}}(s, y) = 0.65\, P_{\text{ML2}}(s, y) + 0.35\, P_{\text{Track2}}(s, y)$ evaluated at $\theta = 0.50$. All wall-clock runtimes and peak heap memory were measured on dual-socket AMD EPYC 7763 compute nodes (8 CPU threads per genome; NVIDIA A100-SXM4-80GB GPU for vLLM and PLM embedding extraction).

4.9 Evaluation metrics, donor-database homology masking, and internal GAEF score

Across Suites 1–8, functional predictions are evaluated using standard protein-centric CAFA metrics ($F_{\max}$, information-accretion-weighted $WF_{\max}$, pair-level micro $F_{\max}$, and minimum semantic distance $S_{\min}$ across thresholds $\theta \in \{0.01, \ldots, 0.99\}$[1,24,85]) with 95% bootstrap or exact Clopper–Pearson confidence intervals and two-sided paired Wilcoxon or McNemar tests. Under donor-database homology masking (`Protocol H` at sequence-identity cutoffs $\eta_{\text{id}} \in \{100\%, 70\%, 50\%, 30\%\}$ and taxonomic Genus/Family masking `Protocol D` in `apply_protocol_H.py`), all reference donor proteins r with pairwise sequence identity $\text{pident}(g, r) > \eta_{\text{id}}$ (or sharing the same genus/family; when explicit donor `pident` metadata is absent on a pre-merged claim, raw homology confidence is conservatively capped at $\eta_{\text{id}}/100$) to any test protein g are excluded from the DIAMOND database, ESM-2/ProtT5 kNN donor indices, FoldSeek templates, and probability calibrators; for the neural enzyme predictors in Suite 1C (`CLEAN`, `DeepProZyme`, and `EnzBERT`), reference lookup databases and training-split homologs above η_{id} are filtered per each tool's homology-split evaluation protocol, while the underlying pretrained protein language model backbone weights (ESM-2 and ProtT5) remain frozen as released (which may retain pretraining exposure to UniRef50 sequences and therefore makes strict Track A cosine-masked LRO a more conservative test of de novo context recovery).

Complementing external ground-truth evaluation, the internal Genome-Scale Annotation Evaluation Framework (GAEF; `QualityReport.groovy`) structural audit score summarizes genome-wide invariants at $\theta_{\text{eval}} = 0.50$ across four weighted dimensions: Essential Function Completeness (C_{ess}, default weight $w_{\text{ess}} = 0.30$), Functional Coherence (C_{func}, default weight $w_{\text{func}} = 0.40$, computed as the mean of non-null Pathway Coherence C_{path}, Complex Coherence C_{cplx}, and Process Coherence C_{proc}), Taxonomic/Anatomical Consistency ($C_{\text{cons}} = \max(0, 1 - 0.01\, V(\mathcal{P}))$, default weight $w_{\text{cons}} = 0.30$, where $V(\mathcal{P})$ is the count of formal constraint violations), and Functional Resolution ($C_{\text{res}} = \min(1.0, \text{meanIC}/10.0)$, default weight $w_{\text{res}} = 0.00$), renormalized over all non-null evaluated di-

mensions:

$$\mathrm{GAEF}(\mathcal{P}) = \frac{\sum_{d \in \{\mathrm{ess,func,cons,res}\}: C_d \neq \mathrm{null}} w_d \, C_d}{\sum_{d: C_d \neq \mathrm{null}} w_d}, \quad (9)$$

which reduces to $0.30\,C_{\mathrm{ess}} + 0.40\,C_{\mathrm{func}} + 0.30\,C_{\mathrm{cons}}$ under default weights when completeness, coherence, and consistency are all evaluated. Because $C_{\mathrm{ess}}, C_{\mathrm{func}}$, and C_{cons} measure the structural invariants targeted by Stages 1 and 2, GAEF is reported strictly as an internal structural diagnostic alongside held-out experimental validation.

Data availability

All reference ontologies (Gene Ontology, Uberon, Plant Ontology, Cell Ontology, and NCBI Taxonomy), curated cross-ontology constraint tables (`go-taxon-constraints.tsv`, `go-anatomical-constraints.tsv`, `uberon-po-taxon-constraints.tsv`, `uberon-po-hierarchy.tsv`, `ncbi-taxon-hierarchy.tsv`, and `complex_terms.tsv`, derived from CC-BY-4.0 OBO Foundry ontologies and released under CC-BY-4.0), SHA-256-verified benchmark manifests (`benchmark/manifests/`), and frozen empirical result tables (`results/*.tsv` and `results/PROVENANCE_MAP.md`) are included in the repository at https://github.com/bio-ontology-research-group/gspa (release tag `v2.1.0`).

Code availability

The GSPA v2.1.0 source code, standalone command-line executable (`gspa-2.1.0.jar`), Stage 2B flux-certified stoichiometric locus refiner (`scripts/stoichiometric_locus_refiner.py`), Stage 3 neuro-symbolic agent and tool schemas (`benchmark/agent/run_gspa_agent.py`), figure generation scripts (`paper/generate_figures.py` and `paper/generate_figure6.py`), and benchmark reproduction scripts are openly available under the GPL-3.0-or-later license at https://github.com/bio-ontology-research-group/gspa (release tag `v2.1.0`).

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Author contributions

T.A.-S., M.K., and R.H. conceived and designed the study. T.A.-S. and R.H. implemented the GSPA v2.1.0 software, cross-ontology constraint bridge, stoichiometric locus refiner, and neuro-symbolic agent, executed the benchmarks, and analyzed the data. T.A.-S., M.K., and R.H. wrote and revised the manuscript.

Competing interests

The authors declare no competing interests.

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